Cheatsheet AZ-104 — 01 · Microsoft Entra ID: users, groups, licensing, guests and SSPR

1. Tenant, editions and domains

Feature	Free	P1	P2
Users, groups, B2B, Security Defaults, device registration, basic SSO	✓	✓	✓
Smart Lockout and global banned password list	✓	✓	✓
SSPR: password **change** (user knows the password)	✓	✓	✓
SSPR: password **reset** for cloud users	✗ (yes with M365 Business Standard or higher)	✓	✓
SSPR with **writeback** to AD DS	✗	✓	✓
Dynamic groups (users or devices)	✗	✓	✓
Group-based licensing	✗ (yes with Office 365 / M365 E3 or higher)	✓	✓
Role-assignable groups (Entra roles to groups)	✗	✓	✓
Administrative units (license for each delegated admin)	✗	✓	✓
Conditional Access, Application Proxy, Entra Connect Health	✗	✓	✓
Self-service group management, **expiration** and **naming policy** for M365 groups	✗	✓	✓
Password Protection: **custom** banned list and on-prem agent	✗	✓	✓
PIM, **Identity Protection**, **Access Reviews**	✗	✗	✓

Rule: P1 is included in Microsoft 365 E3 / EMS E3; P2 in Microsoft 365 E5 / EMS E5. If the wording says "minimize licensing costs" and asks for dynamic groups, CA or writeback, the minimum answer is **P1**; if it asks for PIM or risk detection, **P2**.

- **Custom domain:** verified with a **TXT** (or MX) record in public DNS; the initial `<tenant>.onmicrosoft.com` cannot be deleted or renamed. A domain can be verified in **one** tenant only; subdomains of a verified domain are verified automatically. "Make primary" sets the default UPN suffix.
- **User settings:** "Restrict non-admin users from creating tenants" (recommended Yes), "Users can register applications", "Restrict access to Microsoft Entra administration portal". A user can be a **member** of their home tenant and a **guest** in others.
- **Global Administrator ≠ subscription Owner.** A GA does not see Azure resources by default: it needs **elevate access** (User Access Administrator at root scope; see cheatsheet 02). Microsoft recommends fewer than 5 GAs.

2. Users: types, lifecycle and roles

	Cloud-only	Synchronized (hybrid)	Guest (B2B)
Created	Portal, bulk CSV, PowerShell / CLI	In AD DS; brought in by Entra Connect / Cloud Sync	Invitation (portal, CSV, `New-MgInvitation`)
Edited	In Entra ID	**Only in AD DS** (attributes greyed out in the portal); cloud exceptions: licenses, roles, usage location	In Entra ID (local attributes) and in their home tenant
userType	Member	Member	**Guest** (UPN `name_domain.com#EXT#@tenant.onmicrosoft.com`)
Authenticates in	Entra ID	Entra ID (PHS) / on-prem (PTA, federation)	Their provider: Entra, MSA, Google, SAML/WS-Fed, email code
Signal	"cloud identity", "no on-premises"	"synchronized from AD DS", "attribute is read-only"	"partner", "external user", "collaborate"

- **Bulk operations** (Users → Bulk operations): CSV template for **create**, **invite**, **delete** and **download**. The template columns cannot be changed.
- **Deletion:** the user stays in **Deleted users for 30 days** and can be **restored** with properties and memberships; afterwards (or with "Delete permanently") it is gone. "Block sign in" when the requirement is to block access without deleting.
- **Usage location** is mandatory to assign a license and **is not synchronized** from AD DS → "license cannot be assigned" error on a synced user.
- Admin reset → temporary password + "change password at next sign-in". If the wording asks for the "least privileged role", use the table:

Role	Can reset the password of	Also
Password Administrator	Users with no role, Directory Readers, Guest Inviter, Password Admin	Nothing else: it is the **minimum** for "reset passwords of users"
Helpdesk Administrator	The above + Helpdesk Admin, Message Center Reader, Reports Reader	Invalidates refresh tokens, support tickets, reads BitLocker keys
Authentication Administrator	Users with no role, readers, Password Admin, Auth Admin	Manages **MFA methods** of non-admins (re-register, delete methods)
User Administrator	Non-admins + Groups Admin, Helpdesk Admin, Password Admin, User Admin, readers	**Creates and manages users and groups**, assigns **licenses**, restores deleted users
Privileged Authentication Administrator	**Anyone**, including Global Admins and members of role-assignable groups	MFA methods of any user

Role	Can reset the password of	Also
Global Administrator	Anyone	Everything; can also elevate access over Azure

Other roles that show up: **Guest Inviter** (invite only), **License Administrator** (licenses and usage location only), **Groups Administrator** (create and manage groups and their settings), **Privileged Role Administrator** (assigns Entra roles, creates role-assignable groups and administrative units), **Global Reader** (read everything), **Cloud Device Administrator** (devices).

3. Groups

	Security	Microsoft 365
Members	Users, **groups**, **devices**, service principals	**Users only** (no groups, no devices)
Purpose	Resource access, Azure RBAC, licensing, CA, SSPR	Collaboration: mailbox, SharePoint, Teams, Planner
Dynamic	Users **or** devices (never mixed)	Users only
Role-assignable	Yes (**assigned** membership)	Yes (**assigned** membership)
Expiration / naming policy	No	Yes (P1)
Restore after deletion	Cloud groups: 30 days (preview); Exchange ones (distribution, mail-enabled security) **cannot be restored**	**30 days** (Deleted groups)
Azure RBAC	Yes (transitive nesting)	Yes

Membership type	License	Rules
Assigned	None	Manual or by owners. The only valid type for **role-assignable** groups
Dynamic user	**P1 for each unique user** who is a member of any dynamic group	Rule on `user.*`; members cannot be added manually; rule builder up to 5 expressions; text up to 3,072 characters
Dynamic device	None (devices do not consume P1)	Rule on `device.*`, only in security groups and only via the text box

```
user.department -eq "Sales"
(user.department -eq "Sales") -and (user.country -eq "ES")
user.userType -eq "Guest"                                  → all guests
(user.objectId -ne null) -and (user.userType -eq "Member")   → all members
user.department -in ["Sales","Marketing"]
user.displayName -match "^Da.*"
(user.proxyAddresses -any (_ -startsWith "SMTP:sales"))
(device.deviceOSType -eq "Windows") -and (device.deviceOSVersion -startsWith "10.0.1")
device.deviceOwnership -eq "Company"
device.deviceTrustType -eq "ServerAD"                      → hybrid joined
Direct Reports for "<manager objectId>"                    → cannot be combined with other rules
```

Operators: `-eq -ne -startsWith -notStartsWith -endsWith -notEndsWith -contains -notContains -match -notMatch -in -notIn`, logical `-and -or -not`, collections `-any -all`. Maximum **15,000 dynamic groups** per tenant. Switching assigned ↔ dynamic is possible and recalculates the membership.

- **Role-assignable** ("Microsoft Entra roles can be assigned to the group" = Yes): only **at creation**, immutable, requires **P1**, **assigned** membership, maximum **500** per tenant, no nesting, created by a **Privileged Role Administrator** and its members can only be changed by PRA / GA or owners. For a **temporary** role on the group → PIM (P2).
- **Nesting:** Azure RBAC and Conditional Access evaluate nested groups; **group-based licensing does not** (first level only); Entra roles **do not**; dynamic groups **do not contain groups** (except `memberOf`, preview); M365 groups cannot be nested.
- **Group settings** (Groups → General): "Users can create security groups / Microsoft 365 groups in Azure portals", "Owners can manage group membership requests in My Groups" (self-service, P1), **expiration** of M365 groups (180 / 365 / custom ≥ 30 days; owners renew by email; auto-renewed with activity in Teams, SharePoint or Outlook) and **naming policy** (prefix / suffix + blocked words).

4. Licensing

	Direct assignment	Group-based
Requirement	Any edition	**P1/P2** or an Office 365 / Microsoft 365 E3 or higher plan
Usage location	**Mandatory** (error if missing)	If the user has none, inherits the **tenant** location
Nesting	—	**No**: only first-level members receive the license
Removing it	From the user	From the **group** or by removing the user; an **inherited license cannot be removed on the user**
When leaving the group	Kept	Removed, unless the user also has it **directly** (then only the inheritance is lost)
Errors	Not enough licenses, Usage location not allowed	Same + **Conflicting service plans**, **Other products depend on this license**, **Proxy address conflict** → fix and **Reprocess**
Who	License Administrator, User Administrator, Global Admin	Same + Groups Administrator

- **Service plans can be disabled** (e.g. Yammer) on the group assignment: "Turn apps and services on or off".

- Move a user between licensed groups **without interruption**: add to the destination → confirm the license → remove from the source.
- PowerShell: `Update-MgUser -UserId u -UsageLocation ES` → `Set-MgUserLicense -UserId u -AddLicenses @{SkuId="..."} -RemoveLicenses @()`; for groups `Set-MgGroupLicense`; available SKUs with `Get-MgSubscribedSku`.

5. External users (B2B)

Setting (External Identities → External collaboration settings)	Options	Signal
Guest user access	Same as members · **Limited** (default: cannot enumerate users or groups) · **Restricted** (own profile only)	"guests must not see other users" → Restricted
Guest invite settings	Anyone incl. guests (default) · Members + admin roles · **Only specific admin roles** (User Administrator, **Guest Inviter**) · No one	"only the help desk can invite" → Only admin roles + Guest Inviter
Collaboration restrictions	Any domain · **Deny** list · **Allow** list	"only invite users from fabrikam.com" → Allow list
External user leave settings	Yes / No (requires the tenant's privacy information)	"guests can remove themselves"
Self-service sign-up via user flows	Yes / No	"users sign up to our app and get a guest account"
Cross-tenant access settings	Inbound / outbound per tenant, users, groups and apps; **trust** MFA, compliant and hybrid joined from the partner tenant	"guests prompted twice for MFA" → trust the partner's MFA
Identity providers	Entra ID, Microsoft account, Google, Facebook (self-service only), **SAML/WS-Fed** (domains without Entra), **email one-time passcode** (fallback)	"partner has no Microsoft account" → email OTP

- Flow: invite → status **Pending acceptance** → the guest **redeems** with their provider → **Accepted**. If they do not redeem: "Resend invitation"; if they change identity: "Reset redemption status".
- A guest is a normal object: it joins groups (also dynamic ones with `user.userType -eq "Guest"`), receives Azure RBAC and Entra roles, and Conditional Access targets it with "All guest and external users". Convert guest → member: change **User type** (no need to re-invite).
- External ID billing by **MAU**: the first **50,000** monthly active users are free.
- **B2B direct connect** (Teams shared channels) creates no guest object. **External ID for customers (formerly B2C)** is for consumer apps, not for business-to-business collaboration. **Lighthouse** manages resources in other tenants without guest accounts.

6. SSPR and password protection

Blade	Setting	Exam detail
Properties	None · **Selected** · All	Selected accepts **a single group** in the portal; **nested** groups count. Typical use: pilot
Authentication methods	Number of methods: **1 or 2** · App (notification / code), email, mobile phone (SMS / call), office phone, **security questions** (3-5 to register, 3-5 to answer)	The app counts as **one** method even if you enable notification and code. Questions **not available to admins**. Since Sept. 2025 methods are managed in the **Authentication methods policy** (migration from the legacy policy)
Registration	Require users to register when signing in (Yes / No) · Days before re-confirm: **0-730**, default **180** (0 = never)	"users must provide contact info at next sign-in" → Yes
Notifications	Notify users on password resets (default **Yes**) · Notify **all admins** when other admins reset (default **No**)	"all admins alerted when an admin resets" → second toggle
Customization	Custom helpdesk link (email or URL)	"Contact your administrator"
On-premises integration	**Write back passwords** to on-prem · Allow users to **unlock without resetting**	Requires Entra Connect (or Cloud Sync) with password writeback enabled and permissions in AD DS

- **Administrators:** always enabled for SSPR, with the **two-gate** policy (2 methods) and no security questions, even if Properties is None or Selected.
- **Licensing:** password change → anyone; cloud **reset** → M365 Business Standard or higher, P1/P2; **writeback** → **P1/P2 or M365 Business Premium**. Without writeback, a synced user cannot reset from the cloud.
- **Combined registration** (`aka.ms/mysecurityinfo`): MFA and SSPR are registered together. **Smart Lockout** (Free): locks after **10** failed attempts for **60 s** by default. **Password Protection**: global banned list in Free; custom list and agent for AD DS with **P1**.

7. Hybrid identity, devices and administrative units

	Entra Connect Sync	Entra Cloud Sync
Where it is configured	On-prem server (single active server + staging)	**In the cloud**; lightweight agents, **several active** (automatic HA)
Forests	Connected	Connected and **disconnected** (mergers)
PHS · password writeback · device sync (hybrid join) · Exchange hybrid	✓	✓
PTA and **ADFS** configuration	✓	✗ (configured separately)
Device writeback, advanced sync rules, full attribute filtering, cross-forest references	✓	✗ (OU-based / basic filtering only)
Group writeback v1	✓	✓

	Entra Connect Sync	Entra Cloud Sync
Group provisioning **to** AD DS, on-demand provisioning	✗	✓
Limits	Unlimited; groups up to 250,000 members	**150,000 objects per domain**; groups up to 50,000 members

- Synced objects are edited **in AD DS**; delta cycle every **30 min** (`Start-ADSyncSyncCycle -PolicyType Delta`, `Initial` for a full one); password hashes travel every **2 min**. Non-routable UPN suffixes (`.local`) → add a public suffix before syncing. **Entra Connect Health** (P1) alerts on sync failures.

	Entra registered	Entra joined	Hybrid joined
Device	**Personal / BYOD**	Corporate, **cloud-only**	Corporate, **joined to AD DS**
OS	Windows 10+, iOS, Android, macOS	Windows 10/11, Windows Server 2019+ (VMs), macOS, Linux	Domain-joined Windows
Sign-in	Local account or MSA + "Add work account"	**Entra ID** account	**AD DS** account
How	Settings → Access work or school; MDM enrollment	OOBE, Settings, Autopilot, bulk enrollment	GPO / SCP + Entra Connect or Cloud Sync (device sync)
trustType	Workplace	AzureAD	ServerAD

- **Device settings:** "Users may join devices" (default **All**; does not affect hybrid join or Azure VMs), "Users may register their devices" (All / None; MDM forces All), "Require MFA to register or join" (default **No**; prefer the CA user action), **Maximum number of devices per user** (default **50**, up to 100 or Unlimited; does not apply to hybrid), local admins (**Microsoft Entra Joined Device Local Administrator** role), **Enterprise State Roaming** (P1).
- **Disable** revokes the PRT and blocks access; **Delete** is irreversible and removes the **BitLocker keys**. Done by **Cloud Device Administrator** (or Intune Administrator). BitLocker keys: the device owner or Cloud Device / Helpdesk / Intune / Security Admin / Security Reader.
- **Administrative units (AU):** container of **users, groups and devices** to delegate an Entra role over them only (e.g. Helpdesk Administrator for the Bilbao office). Cannot be nested; assigned or **dynamic** membership (P1 per member); **P1 for each admin** holding a role scoped to the AU; adding a **group** to the AU does not bring in its **members**. **Restricted management AU**: its members can only be managed by admins holding a role scoped to the AU, not by tenant-level admins.

8. Commands that show up

Task	PowerShell (Microsoft Graph / Az)	Azure CLI
Create user	`New-MgUser -DisplayName -UserPrincipalName -MailNickname -AccountEnabled -PasswordProfile @{Password="..."; ForceChangePasswordNextSignIn=$true}` · `New-AzADUser`	`az ad user create --display-name --user-principal-name --password`
Create group	`New-MgGroup -DisplayName -MailNickname -SecurityEnabled -MailEnabled:$false` · `New-AzADGroup`	`az ad group create --display-name --mail-nickname`
Add member	`New-MgGroupMember -GroupId -DirectoryObjectId` · `Add-AzADGroupMember`	`az ad group member add --group --member-id`
Invite guest	`New-MgInvitation -InvitedUserEmailAddress -InviteRedirectUrl`	`az rest --method post --url https://graph.microsoft.com/v1.0/invitations --body '{...}'`
License	`Update-MgUser -UsageLocation ES` → `Set-MgUserLicense -AddLicenses @{SkuId=...} -RemoveLicenses @()` · `Get-MgSubscribedSku`	No native command (via `az rest`)
Bulk create	`Import-Csv .\users.csv` piped into `ForEach-Object { New-MgUser ... }`	Portal → Bulk operations (CSV)
Restore deleted	`Restore-MgDirectoryDeletedItem -DirectoryObjectId`	Portal → Deleted users / Deleted groups
Sync now	`Start-ADSyncSyncCycle -PolicyType Delta` (on the Connect server)	—

The **AzureAD** and **MSOnline** modules (`New-AzureADUser`, `New-MsolUser`) are retired: the modern answer is **Microsoft Graph PowerShell**, although older questions still list them as a valid option.

9. Decision tree

```
What does the wording ask for?
├─ A user from another company accesses our resources ..... B2B invitation (guest)
├─ Invitations only to certain domains ..... Collaboration restrictions (allow list)
├─ Guests must not see the directory ..... Guest user access: Restricted
├─ Automatic membership by attribute ..... Dynamic group (P1)
├─ Automatic license when joining a group ..... Group-based licensing (+ usage location)
├─ Entra ID role to a group ..... Role-assignable group (at creation, assigned)
├─ Helpdesk for one office / department only ..... Administrative unit + role at that scope
├─ Reset passwords of regular users (least privilege) ..... Password Administrator
├─ Create users, groups and assign licenses ..... User Administrator
├─ SSPR for a pilot ..... Properties: Selected + one group
├─ SSPR that also changes the password in AD DS ..... Password writeback (P1 + Connect / Cloud Sync)
├─ Change the department of a synced user ..... In AD DS, then wait for or force the delta sync
└─ Device: personal / corporate cloud-only / joined to AD DS ..... Entra registered / Entra joined / Hybrid joined
```

10. Signals → answer

Wording	Answer
"minimize administrative effort", "users move between departments"	Dynamic group with `user.department`
"automatically assign licenses to members of the group"	Group-based licensing
"license cannot be assigned", "usage location"	Fill in **Usage location** (not synced from AD DS)
"remove license from user" fails	It is inherited: remove it on the group or remove the user from the group
"assign the Entra role to the group" and the group already exists	Impossible: create a new **role-assignable** group
"partner needs access", "no new accounts in their tenant"	B2B invitation
"only users from contoso.com can be invited"	Collaboration restrictions → Allow list
"guests must not enumerate users"	Limited (the default) already blocks it; Restricted if they must not see anyone's profile or memberships either
"least privileged role to reset passwords"	Password Administrator (Helpdesk if sessions must be invalidated)
"least privileged role to create users and groups"	User Administrator
"least privileged role to invite guests"	Guest Inviter
"least privileged role to manage devices"	Cloud Device Administrator
"regional helpdesk", "only users in the Paris office"	Administrative unit (dynamic by `city`)
"SSPR for a pilot group"	Selected + that group (only one)
"administrators must use two methods for SSPR"	Already the default (two-gate); nothing to configure
"users prompted to re-register every 90 days"	Registration → days before re-confirm = 90
"deleted a user 20 days ago"	Restore from Deleted users (30 days)
"attribute is grayed out" in the portal	Synced user: edit in AD DS
"Global Administrator cannot see the subscription"	Elevate access → User Access Administrator at root (cheatsheet 02)
"device limit reached"	Device settings → Maximum number of devices per user (default 50)
"BitLocker recovery key"	Entra ID → Devices → device → Show Recovery Key
"dynamic group for Windows devices"	Security group, dynamic device, rule `device.deviceOSType -eq "Windows"` via text box

11. Mistakes to avoid

1. Assigning a license without **Usage location**, or expecting the attribute to come from AD DS.
2. Creating the group and trying to mark it role-assignable afterwards: only **at creation**, and never dynamic.
3. Nesting groups for licensing: group-based licensing **does not inherit** through nested groups.
4. Assuming a Global Administrator sees Azure subscriptions and resources by default.
5. Editing attributes of a synced user in the portal (they are greyed out): they are changed in AD DS.
6. Selecting **several groups** in SSPR Selected: it accepts only one (nested ones do count).
7. Granting **User Administrator** when Password Administrator is enough, or Global Admin when Guest Inviter is enough.
8. Choosing External ID for customers (B2C) or Lighthouse to collaborate with **users** from another company: that is **B2B**.
9. Expecting Cloud Sync to configure **PTA** or perform **device writeback**: that is Connect Sync.
10. Adding a group to an administrative unit believing its members fall within the scope.

Cheatsheet AZ-104 — 02 · RBAC: built-in roles, scopes, inheritance and interpreting assignments

1. The three pieces of an assignment

Piece	What it is	Exam details
Security principal	User, **group** (transitive nesting), service principal or managed identity	Prefer assigning to groups. Guests also receive roles. A principal can hold several assignments at once
Role definition	Collection of permissions in JSON: `Actions`, `NotActions`, `DataActions`, `NotDataActions`, `AssignableScopes`	`NotActions` **subtracts** permissions within the same role; **it is not a deny**: another role can grant the action again
Scope	Management group → subscription → resource group → resource	**Inherited** downwards. The effective result is the **union** of all assignments (the most permissive wins). Only **deny assignments** (deployment stacks / managed apps; not created by hand) subtract

- Limits: **5,000** assignments per subscription, **500** per management group, **5,000** custom roles per tenant.
- A new assignment can take **a few minutes** to apply (up to 10, more if the token is cached). Every change is recorded in the **Activity log** (Microsoft.Authorization/roleAssignments/write).
- RBAC governs the **control plane** (`management.azure.com`). The data plane (blobs, queues, Key Vault in RBAC mode) requires roles with **DataActions**, or else keys / SAS.
- Optional: **ABAC condition** (blob and queue data roles, and to delegate role assignments with the RBAC Administrator), description, and **eligible** assignments with PIM for Azure resources (P2).

2. Built-in roles you must tell apart

Role	Can	Cannot	Signal
Owner	Everything, including **assigning roles** and managing **locks**	—	"full access including access management"
Contributor	Create and manage any resource	Assign roles, create or delete locks, elevate access (NotActions `Microsoft.Authorization/*/Write`)	"manage everything except access"
Reader	View resources	Change anything; cannot **list storage keys** either (it is a POST)	"view only"
User Access Administrator	Manage access (assignments) and locks	Manage resources	"manage user access to resources"
Role Based Access Control Administrator	Assign roles only (ideal for delegating with conditions)	Nothing else; neither locks nor resources	"assign roles, nothing else, least privilege"
Virtual Machine Contributor	Create, start, stop, resize and delete VMs and disks; create NICs and join them to subnets	Create VNets or NSGs, manage the storage account, assign access, sign in to the VM	"manage VMs but not the network they connect to"
Network Contributor	All networking (VNet, NSG, LB, PIP)	Nothing outside Microsoft.Network	"manage networking only"
Storage Account Contributor	Manage accounts and **list keys**	Read data with an Entra identity	"manage storage accounts"
Storage Blob Data Owner / Contributor / Reader	Blob data via Entra (data plane)	Manage the account	"access blobs with Entra ID"
Storage Account Key Operator Service Role	List and **regenerate** keys	Nothing else	"rotate keys, least privilege"
Security Admin / Security Reader	Defender for Cloud, security policies	Manage resources	"security recommendations"
Monitoring Contributor / Reader · Log Analytics Contributor / Reader	Alerts, diagnostic settings, workspaces and queries	Other resources	"configure alerts", "query logs"
Backup Contributor / Operator / Reader	Manage backups in the vault (Operator: no deleting)	Create the vault (Contributor can)	"manage backups"
Website Contributor · Web Plan Contributor	Web apps · App Service plans	The other one	"manage the web app only"
Key Vault Administrator · Key Vault Secrets User	The entire Key Vault data plane (RBAC mode) · read secrets	Manage the vault (control plane)	"read secrets, least privilege" → Secrets User
Managed Identity Operator / Contributor	Assign a user-assigned MI to resources · create it	—	"assign the identity to the VM"
Resource Policy Contributor	Create and assign Azure Policy	Assign roles	"manage policies"
Management Group Contributor / Reader	Manage or read management groups	Resources	"create management groups"
Cost Management Reader / Contributor · Billing Reader	View costs and budgets · create budgets · invoices	Resources	"auditors review costs"
Tag Contributor	Tag any resource (via PowerShell / API)	Modify the resource	"add tags without modifying resources"
Virtual Machine Administrator Login / User Login	Sign in to the VM with Entra ID (admin / user)	Manage the VM (that is VM Contributor)	"sign in to the VM with Entra credentials"

Classic administrators (Service Administrator, Co-Administrator) have been **retired** since August 2024: if they appear in a question, the correct answer is always an RBAC role.

3. Interpreting assignments

Assignments	Result	Why
Reader on the subscription + Contributor on RG1	Contributor on RG1, Reader on the rest	Union; the more specific scope does not "downgrade", it adds
Owner on the management group (via group) + explicit Reader on the subscription	**Owner** on the whole subscription	There is no "most restrictive assignment" concept; only deny assignments subtract
Contributor on the subscription	Creates everything, but does **not** assign roles or create locks	NotActions on Microsoft.Authorization
Reader on a storage account	Sees the account; sees neither keys nor blobs	listKeys is an action; blobs are data plane
Owner on the subscription, portal with "Switch to Entra ID user account"	Cannot read blobs; can with "Access key"	Owner has no DataActions
Virtual Machine Contributor on RG1	Starts, stops, deletes and resizes VMs; does not create VNets or NSGs	Microsoft.Network only partially (NIC, subnet join, LB)
Custom role with AssignableScopes = RG1	Only assignable on RG1 or its resources	Outside the scope it does not appear in the list
Resource moved to another RG or subscription	Assignments **on the resource** are lost (orphaned)	They must be recreated after the move
Deny assignment from a deployment stack	Blocks even if you are Owner	The only way to subtract
User with an assigned role but "Authorization failed"	Wait for propagation, check scope/NotActions/DataActions, PIM not activated, deny assignment	Exam checklist

- **Check access:** Access control (IAM) → **Check access** (what a user / group / MI has at that scope, including inheritance) and **View my access**. In CLI: `az role assignment list --assignee ana@contoso.com --all --include-inherited`; in PowerShell: `Get-AzRoleAssignment -SignInName ana@contoso.com -ExpandPrincipalGroups`.
- **Who can assign roles:** Owner, User Access Administrator, Role Based Access Control Administrator; and a Global Administrator **only after elevating** (section 5).

4. Custom roles

```
{
  "Name": "VM Operator",
  "IsCustom": true,
  "Description": "Start, stop and restart VMs without being able to delete them",
  "Actions": [
    "Microsoft.Compute/virtualMachines/read",
    "Microsoft.Compute/virtualMachines/start/action",
    "Microsoft.Compute/virtualMachines/restart/action",
    "Microsoft.Compute/virtualMachines/deallocate/action",
    "Microsoft.Resources/subscriptions/resourceGroups/read"
  ],
  "NotActions": [],
  "DataActions": [],
  "NotDataActions": [],
  "AssignableScopes": [ "/subscriptions/<sub-id>", "/subscriptions/<sub-id>/resourceGroups/RG1" ]
}
```

- Created in the portal (IAM → Add → **Add custom role**: from scratch, by **cloning** a role or pasting JSON), with `New-AzRoleDefinition -InputFile role.json` or `az role definition create --role-definition role.json`. Update: `Set-AzRoleDefinition` / `az role definition update`. It cannot be deleted while it has assignments.
- **AssignableScopes:** subscriptions, resource groups or **a single** management group; never the root `/`. A role with `DataActions` cannot be assigned at a management group. Wildcards (`Microsoft.Compute/*/read`) are valid.
- To discover operations: `Get-AzProviderOperation "Microsoft.Compute/virtualMachines/*"` or `az provider operation show --namespace Microsoft.Compute`. To start from a built-in role: `Get-AzRoleDefinition "Virtual Machine Contributor"` and export to JSON.
- Typical case: "the team must be able to restart VMs but not delete or create them" → custom role (no built-in fits). "Minimal effort" + a built-in that fits → built-in.

5. Entra ID roles, Azure roles and elevation

	Microsoft Entra ID roles	Azure roles (RBAC)
Scope	Tenant: users, groups, apps, licenses	Resources: management group, subscription, RG, resource
Examples	Global Administrator, User Administrator, Groups Administrator	Owner, Contributor, Reader, UAA
Where	Entra ID → Roles and administrators	Access control (IAM) at each scope
Relationship	**A Global Administrator is not Owner** of anything in Azure	A subscription Owner does not manage users in Entra

- **Elevate access:** Entra ID → Properties → "Access management for Azure resources" = **Yes** (only a Global Administrator can). It gets **User Access Administrator at the root scope** `/`, which lets it assign itself Owner on the Tenant Root Group, see all subscriptions or recover an orphaned subscription. Best practice: **remove the elevation afterwards** (toggle to No and delete the assignment at `/`). It is recorded in the Activity log as `Microsoft.Authorization/elevateAccess/action`.

- Entra roles and Azure roles can be combined in the same person, but they are assigned in different places and neither implies the other.

6. Commands that show up

Task	PowerShell (Az)	Azure CLI
Assign role	`New-AzRoleAssignment -SignInName ana@contoso.com -RoleDefinitionName Contributor -ResourceGroupName RG1` · to a group: `-ObjectId <id> -Scope /subscriptions/<id>`	`az role assignment create --assignee ana@contoso.com --role Contributor --resource-group RG1` · `--scope /subscriptions/<id>`
List	`Get-AzRoleAssignment -ResourceGroupName RG1` · `-SignInName ana@contoso.com -ExpandPrincipalGroups`	`az role assignment list --resource-group RG1 --include-inherited` · `--assignee ana@contoso.com --all`
Remove	`Remove-AzRoleAssignment -SignInName ... -RoleDefinitionName ... -ResourceGroupName RG1`	`az role assignment delete --assignee ... --role ... --resource-group RG1`
View a role	`Get-AzRoleDefinition "Virtual Machine Contributor"` · `(Get-AzRoleDefinition Contributor).NotActions`	`az role definition list --name "Virtual Machine Contributor"`
Custom role	`New-AzRoleDefinition -InputFile role.json` · `Set-AzRoleDefinition -InputFile role.json`	`az role definition create --role-definition role.json` · `az role definition update ...`
Provider operations	`Get-AzProviderOperation "Microsoft.Storage/*"`	`az provider operation show --namespace Microsoft.Storage`
Elevate access (GA)	Portal: Entra ID → Properties → toggle	`az rest --method post --url "/providers/Microsoft.Authorization/elevateAccess?api-version=2016-07-01"`
Remove the elevation	`Remove-AzRoleAssignment -SignInName ga@contoso.com -RoleDefinitionName "User Access Administrator" -Scope "/"`	`az role assignment delete --assignee ga@contoso.com --role "User Access Administrator" --scope "/"`

7. Decision tree

```
What does the wording ask for?
├─ Manage everything except access ..... Contributor
├─ Manage access without touching resources ..... User Access Administrator
├─ Assign roles only, least privilege ..... Role Based Access Control Administrator
├─ Manage VMs without touching the network ..... Virtual Machine Contributor
├─ Read blobs with an Entra identity ..... Storage Blob Data Reader
├─ Rotate storage keys and nothing else ..... Storage Account Key Operator Service Role
├─ View costs without seeing resources ..... Cost Management Reader (or Billing Reader)
├─ Restart VMs but not delete them (no built-in fits) ..... Custom role
├─ Same role across many subscriptions ..... Assign at the management group
├─ Global Admin who cannot see the subscriptions ..... Elevate access → UAA at "/"
├─ User with a role and still "Authorization failed" ..... Propagation / scope / DataActions / deny assignment
└─ Find out what permissions someone has on a resource ..... IAM → Check access
```

8. Signals → answer

Wording	Answer
"least privilege", "manage VMs, cannot modify the virtual network"	Virtual Machine Contributor
"create and manage all resources, cannot grant access"	Contributor
"grant access to others, must not manage resources"	User Access Administrator
"delegate role assignments only"	Role Based Access Control Administrator
"read blob data using Microsoft Entra ID"	Storage Blob Data Reader
"Owner cannot see blobs in the portal"	Switch to key authentication, or add Storage Blob Data Reader
"Reader cannot view the storage account keys"	Correct: listKeys requires Contributor or Storage Account Key Operator
"same permissions on 20 subscriptions, minimal effort"	One assignment at the management group
"user has Reader on subscription and Contributor on RG"	Contributor on the RG, Reader elsewhere
"role assignment missing after moving the resource"	Resource-level assignments do not move: recreate them
"custom role cannot be assigned at management group"	It contains DataActions, or the MG is not in AssignableScopes
"cannot delete the custom role"	Remove its assignments first
"Global Administrator must manage all subscriptions"	Elevate access (toggle) → User Access Administrator at root
"Co-Administrator", "Service Administrator"	Retired classic roles: use Owner / Contributor
"which role assignments does the user have"	IAM → Check access / `az role assignment list --assignee --all`
"sign in to Windows VM with Entra credentials as admin"	Virtual Machine Administrator Login (+ AADLoginForWindows extension)
"temporary / just-in-time Owner on a subscription"	PIM for Azure resources (eligible assignment, P2)

9. Mistakes to avoid

1. Believing that a more restrictive assignment at a lower scope "downgrades" permissions: permissions **add up**; only deny assignments subtract.
2. Confusing Global Administrator with Owner: without elevate access, a GA does not see subscriptions.

3. Granting Owner when Contributor is enough, or Contributor when a service role is enough (VM Contributor, Network Contributor).
4. Expecting Reader or Owner to read blobs with Entra ID: those are DataActions (Storage Blob Data *).
5. Putting the root / or several management groups in a custom role's AssignableScopes.
6. Forgetting that moving a resource loses its resource-level role assignments.
7. Choosing Co-Administrator or Service Administrator: they no longer exist.
8. Trying to delete a custom role with active assignments, or assigning a role with DataActions at a management group.
9. Leaving elevated access enabled after using it.

Cheatsheet AZ-104 — 03 · Governance: subscriptions, management groups, Policy, locks, tags, costs and Advisor

1. Hierarchy: tenant, management groups, subscriptions and resource groups

Level	What it is	Exam facts
Tenant (Entra ID)	Directory that the subscriptions trust	A subscription trusts **a single** tenant. Changing its tenant = transfer and **redo RBAC** (assignments do not travel)
Tenant Root Group	Root management group; ID = tenant ID; cannot be moved or deleted	Nobody has access by default: a Global Admin **elevates**. Anything assigned here (RBAC or Policy) affects **everything**
Management groups	Containers of subscriptions and other MGs to apply RBAC and Policy by inheritance	**10,000** per tenant, **6 levels** (not counting root or subscriptions), **a single parent**. New subscriptions land in the root (or in the configured "default management group"). "Require authorization" prevents anyone from creating MGs
Subscription	Unit of billing, quotas and trust boundary	**980** resource groups. Types: Free, Pay-As-You-Go, EA, MCA, CSP, **Dev/Test** (reduced rates, no Windows / SQL license charges, non-production only). Cancel → resources disabled, reactivable for **90 days**. You must **register resource providers** (`Microsoft.Compute...`) before using a new type
Resource group	Logical container; its region only stores metadata; resources from several regions	A resource lives in **one** RG; deleting the RG deletes everything; **800** resources per type. Moving resources: same tenant, source and destination RGs **locked for up to 4 h**, the resource's RBAC assignments **are lost**, validate first, a ReadOnly lock on the source or destination RG blocks the move. Changing subscription: VM + disks + NIC + VNet **together**

- **Moving a subscription between MGs:** write on the source and destination MG + `roleAssignments/write` and `delete` on the subscription (that is, subscription **Owner**; if you inherit it from the source MG, you lose it when moving). If the source or destination is the root, no permissions on it are needed.
- **Billing:** EA = Enrollment → Departments → Accounts → Subscriptions (roles: Enterprise Administrator, Department Administrator, Account Owner). MCA = Billing account → Billing profiles → Invoice sections → Subscriptions. These roles are **not** RBAC: a Billing account owner is not Owner of the resources.
- **Subscription policies** (portal → Subscriptions → Manage Policies; only a Global Administrator with elevate access; since May 2026 both default to "Allow no users"): prevent subscriptions from **leaving** the tenant or **entering** from another; list of exempted users.
- **Quotas** per region and VM family (vCPU): "QuotaExceeded" → Usage + quotas → **Request increase** (do not switch subscriptions).

2. Azure Policy

Piece	What it is	Detail
Definition	JSON with `if` (condition) and `then` (effect); parameters; mode **All** or **Indexed** (only resources with tags/location)	Built-in or custom; created in an MG or subscription (definition scope)
Initiative (policy set)	Group of definitions assigned as a single one	E.g.: Microsoft Cloud Security Benchmark, ISO 27001, "Enable Azure Monitor for VMs"
Assignment	Definition or initiative + **scope** (MG, subscription, RG) + **exclusions** + parameter values + enforcement (Enabled / **DoNotEnforce** for testing)	If the effect is Modify or DeployIfNotExists it needs a **managed identity** with permissions (Contributor) at the scope
Exemption	Excludes a resource or scope from an assignment with reason **Waiver** or **Mitigated** and an expiration date	Different from exclusion: the exemption is done after assigning and is audited
Remediation task	Applies DeployIfNotExists / Modify to **already existing** resources	Policy **does not fix** existing resources on its own: it only flags them non-compliant
Compliance	Evaluation: when a resource is **created or updated**, on **assignment** (≈ 30 min), every **24 h**, or on demand (`Start-AzPolicyComplianceScan`, `az policy state trigger-scan`)	States: Compliant, Non-compliant, Exempt, Conflicting, Not started

Effect	What it does	Built-in example
Deny	Blocks create or update; error `RequestDisallowedByPolicy`	Allowed locations · Allowed resource types · Not allowed resource types · Allowed virtual machine size SKUs · Require a tag and its value
Audit	Allows, but flags non-compliant	Audit VMs that do not use managed disks
AuditIfNotExists	Audits if a **related** resource is missing (extension, diagnostic setting)	Audit VMs without the antimalware extension
DeployIfNotExists	Deploys the missing related resource (MI + remediation)	Deploy diagnostic settings to Log Analytics · Deploy Azure Monitor Agent
Modify	Adds, replaces or removes **tags** or properties on create/update and in remediation	**Inherit a tag from the resource group** · Add a tag to resources
Append	Adds fields to the request (e.g. a network rule)	Append IP rules to storage accounts
DenyAction	Blocks an action, today **DELETE** (complement to locks)	Do not allow deletion of resource types
Manual	Compliance attested by hand	Regulatory compliance controls
Disabled	To switch off a definition without removing the assignment	—

- **Policy ≠ RBAC:** Policy decides **what can be done with resources** (properties, locations, SKUs) and applies **even to an Owner**; RBAC decides **who** can do what. "Users must not create VMs larger than D4" is not solved with RBAC.
- Order: disabled → append / modify → deny → audit → manual → auditIfNotExists → denyAction; AINE / DINE are evaluated after the resource is created (≈ 10 min). Several assignments accumulate: the **most restrictive** wins.

- **Allowed locations** uses mode Indexed and usually excludes global resources; **assigning at an MG** applies to all subscriptions under it. Policy for **tags** = Modify (add / inherit) or Deny (require). Role: **Resource Policy Contributor**.

3. Resource locks

Lock	Portal	Effect
CanNotDelete	Delete	Read and modify yes; **delete no**
ReadOnly	Read-only	Read-only for **all roles**, including Owner (equivalent to Reader)

- Inherited (subscription → RG → resource); the **most restrictive** wins. Cannot be placed on management groups. They affect the **control plane**: a lock on the storage account does **not** prevent deleting blobs.
- Who manages them: **Owner** and **User Access Administrator** (`Microsoft.Authorization/locks/*`). **Contributor cannot**.
- Surprise effects: ReadOnly on a storage account → cannot **list keys** or create containers from the portal · ReadOnly on an RG with VMs → **they cannot be started or restarted** (POST) · ReadOnly on an RG with an App Service plan → cannot **scale** · ReadOnly on an RG → cannot **move** resources in or out · ReadOnly on the subscription → Advisor stops working · CanNotDelete on the Azure Backup RG (`AzureBackupRG_*`) → **backups fail** (it cannot clean up recovery points) · CanNotDelete prevents deleting RBAC assignments · a locked resource makes the deletion of its **entire** RG fail.
- Commands: `New-AzResourceLock -LockLevel CanNotDelete -LockName NoDelete -ResourceGroupName RG1` (or `-ResourceName vm1 -ResourceType Microsoft.Compute/virtualMachines`), `Get-AzResourceLock`, `Remove-AzResourceLock -LockId <id>` · `az lock create --name NoDelete --lock-type ReadOnly --resource-group RG1 [--resource-name app1 --resource-type Microsoft.Web/sites]`, `az lock list`, `az lock delete --name NoDelete --resource-group RG1`.

4. Tags

Rule	Value
Maximum	**50** pairs per resource, RG or subscription; name **512** characters (**128** on storage accounts), value **256**
Case	Name **insensitive**, value **sensitive**. Not allowed in the name: `< > % & \ ? /`
Where	Resources, RGs and subscriptions; **not** on management groups or classic resources. Some types only allow 15 (Automation, CDN, DNS zones, saved searches)
Inheritance	**Not inherited** from the RG or the subscription → Policy "Inherit a tag from the resource group" (Modify) or "Require a tag" (Deny)
Permissions	**Tag Contributor** (`Microsoft.Resources/tags/write`, even without access to the resource) or Contributor on the resource. Not Reader
Billing	Tags column in the usage file; Cost analysis **group by tag**; they only count from the moment they exist (**not retroactive**) and in services that support them

- PowerShell: `Update-AzTag -ResourceId <id> -Tag @{Owner="Ana"} -Operation Merge` (Merge adds, Replace replaces, Delete removes) · `New-AzTag -ResourceId <id> -Tag @{Env="Prod"}` (replaces everything) · `Set-AzResourceGroup -Name RG1 -Tag @{Env="Prod"}` (replaces) · `(Get-AzResource -Name vm1).Tags`.
- CLI: `az tag update --resource-id <id> --operation Merge --tags Owner=Ana` · `az tag create --resource-id <id> --tags Env=Prod` · `az group update -n RG1 --set tags.Env=Prod` · `az resource tag --tags Env=Prod --ids <id>` (replaces unless `--is-incremental`) · `az resource list --tag Env=Prod`.

5. Costs: Cost Management + Billing and Advisor

Tool	What it does	Exam details
Cost analysis	Explore actual or amortized costs; **group by** resource, RG, service, location, **tag**; filters; saved views; forecast	Needs Reader or **Cost Management Reader** at the scope. Data lags **8-24 h**; a new subscription takes up to 48 h
Budgets	Amount per period (monthly, quarterly, annual; "billing month" on PAYG) with alerts	Up to **5 thresholds** (% of the budget) and 5 emails; alerts on **actual** or **forecasted** cost; **action group** only at subscription or RG; evaluated every **24 h**; **does not stop** spending or resources; Owner creates, Contributor / Cost Management Contributor create their own, Reader only views
Cost alerts	Budget, **credit** (EA / MCA), department quota (EA) and **anomaly** alerts	Shown in Cost Management → Cost alerts
Exports	Scheduled dump (daily, weekly, monthly) to a storage account	"analyze in Power BI", "retain cost data"
Pricing calculator / **TCO calculator**	Estimate the cost of an architecture / compare it with on-prem	"estimate before deploying" / "compare with datacenter"
Savings	**Reservations** (1 or 3 years, up to ~72 %) · **Savings plan** (hourly compute spend, flexible) · **Spot** (up to ~90 %, evictable) · **Azure Hybrid Benefit** (Windows / SQL licenses with Software Assurance) · Dev/Test pricing · auto-shutdown · right-size	"predictable 3 years" → reservation; "existing licenses" → Hybrid Benefit; "interruptible" → Spot
Advisor	Recommendations in 5 categories: **Cost, Security, Reliability, Operational Excellence, Performance** + Advisor score	Cost: **shut down or resize underutilized VMs** (looks at **7 days**, configurable 7-90; shut down if P95 CPU < 3 % and outbound network < 2 %), unattached disks, unassociated public IPs, idle gateways, recommended reservations. Can be **Postponed / Dismissed**; **alerts** for new recommendations with an action group; weekly digest. Reader to view, Contributor to postpone. **Does not act on its own**

- "Stop spending when the budget is reached" **does not exist**: a budget only notifies; to act, action group → Automation runbook / Logic App that shuts down VMs.

- "Group costs by department" → **tags** + Cost analysis; if the tags did not exist before, earlier costs do **not** show up grouped.
- "Prevent users from creating expensive VMs" → Policy (Allowed virtual machine size SKUs), not a budget or RBAC.

6. Commands that show up

Task	PowerShell (Az)	Azure CLI
Management groups	`New-AzManagementGroup -GroupName mg-prod -DisplayName "Prod" -ParentId <id>` · `New-AzManagementGroupSubscription -GroupName mg-prod -SubscriptionId <id>`	`az account management-group create --name mg-prod --display-name Prod --parent <id>` · `az account management-group subscription add --name mg-prod --subscription <id>`
Subscriptions and providers	`Get-AzSubscription` · `Set-AzContext -Subscription "Prod"` · `Register-AzResourceProvider -ProviderNamespace Microsoft.Compute`	`az account list -o table` · `az account set -s "Prod"` · `az provider register --namespace Microsoft.Compute` · `az provider list --query "[?registrationState=='Registered']"`
Resource groups	`New-AzResourceGroup -Name RG1 -Location westeurope` · `Remove-AzResourceGroup -Name RG1 -Force` · `Get-AzResourceGroup -Tag @{Env="Prod"}`	`az group create -n RG1 -l westeurope` · `az group delete -n RG1 --yes --no-wait` · `az group list --tag Env=Prod`
Move resources	`Move-AzResource -DestinationResourceGroupName RG2 -ResourceId <id> [-DestinationSubscriptionId <id>]` · validate: `Invoke-AzResourceAction -Action validateMoveResources`	`az resource move --destination-group RG2 --ids <id> [--destination-subscription-id <id>]` · `az resource invoke-action --action validateMoveResources`
Policy	`New-AzPolicyAssignment -Name loc -Scope <id> -PolicyDefinition (Get-AzPolicyDefinition -BuiltIn ...) -PolicyParameterObject @{listOfAllowedLocations=@("westeurope")}` · `Get-AzPolicyState -Filter "ComplianceState eq 'NonCompliant'"` · `Start-AzPolicyRemediation`	`az policy assignment create --name loc --scope <id> --policy <def> --params '{"listOfAllowedLocations": {"value":["westeurope"]}}'` · `az policy state list --filter "complianceState eq 'NonCompliant'"` · `az policy remediation create`
Locks and tags	See sections 3 and 4	See sections 3 and 4
Budgets	`New-AzConsumptionBudget -Amount 500 -Name b1 -Category Cost -TimeGrain Monthly -StartDate ... -NotificationKey Key1 -NotificationEnabled -NotificationThreshold 80 -ContactEmail ...`	`az consumption budget create --amount 500 --budget-name b1 --category cost --time-grain monthly --start-date ... --end-date ...`

7. Decision tree

```
What does the wording ask for?
├─ Same RBAC or Policy across many subscriptions ..... Management group
├─ Prevent creating resources outside a region ..... Policy: Allowed locations (Deny)
├─ Block expensive VM SKUs ..... Policy: Allowed virtual machine size SKUs
├─ Require a tag on creation ..... Policy: Require a tag and its value (Deny)
├─ Add the RG's tag to its resources ..... Policy: Inherit a tag from the RG (Modify) + remediation
├─ Fix already existing resources ..... Remediation task (DeployIfNotExists / Modify)
├─ Prevent accidental deletions ..... Lock CanNotDelete (Owner / UAA)
├─ Freeze a resource completely ..... Lock ReadOnly (beware VM start/restart)
├─ Notify when spending exceeds X ..... Budget (+ action group)
├─ View costs by department ..... Tags + Cost analysis (group by tag)
├─ Find underutilized VMs ..... Advisor (Cost)
├─ Move resources to another RG or subscription ..... Move (same tenant; redo RBAC)
├─ Move resources to another region ..... Azure Resource Mover (not a move)
└─ "Subscription not registered for resource type" ..... Register the resource provider
```

8. Signals → answer

Wording	Answer
"apply the policy to all current and future subscriptions"	Assign at the **management group** (or at the root)
"resources can only be created in West Europe"	Allowed locations with effect Deny
"existing non-compliant resources remain", "no resources were changed"	Normal Policy behavior; to fix them, remediation
"users should be unable to delete the resource group", "even Owners"	Lock CanNotDelete
"VM cannot be started after applying a lock"	ReadOnly on the RG: switch to CanNotDelete
"cannot list storage keys in the portal"	ReadOnly lock on the account (or Reader role)
"cannot move the resource"	ReadOnly lock on source or destination; dependencies not included; different tenant; provider not registered
"role assignments missing after moving"	Recreate them: they do not move
"notify when cost exceeds 80 %", "forecasted"	Budget with actual or forecasted alert
"stop the VMs when the budget is reached"	Budget → action group → Automation runbook (the budget only notifies)
"charge back costs to departments"	Mandatory tags (Policy) + Cost analysis
"tags not applied to new resources in the resource group"	RG tags are not inherited: Policy Modify
"identify underutilized VMs", "resize recommendations"	Advisor → Cost
"the subscription must trust a different tenant"	Transfer subscription → change directory; RBAC is lost
"move subscription to another management group" fails	Missing write on source/destination MG or Owner on the subscription
"QuotaExceeded when creating a VM"	Request quota increase for that family and region

Wording	Answer
"minimize cost for a 3-year predictable VM"	Reserved instance (+ Hybrid Benefit if licenses exist)

9. Mistakes to avoid

1. Using RBAC to limit regions, SKUs or tags: that is **Azure Policy**.
2. Expecting Policy to fix or delete existing resources: it only flags them; fixing = remediation.
3. Believing a ReadOnly lock is harmless: it blocks starting VMs, scaling plans, listing keys and moving resources.
4. Thinking Contributor can create locks: Owner or User Access Administrator is needed.
5. Assuming resource group tags flow down to the resources.
6. Believing a budget stops spending.
7. Moving resources between subscriptions in different tenants, or leaving out the dependencies (disks, NIC, VNet).
8. Forgetting to register the resource provider in the destination subscription.
9. Creating more than 6 levels of management groups or a second parent.
10. Confusing billing roles (EA / MCA) with RBAC roles.

Cheatsheet AZ-104 — 04 · Storage accounts: types, redundancy, encryption, object replication, firewall, Storage Explorer and AzCopy

1. Account types

Type	Services	Redundancy	When
Standard general-purpose v2 (StorageV2)	Blob (incl. Data Lake Gen2), Files, Queue, Table	LRS, ZRS, GRS, RA-GRS, GZRS, RA-GZRS	The default option for almost everything; the only one with **geo-redundancy** and **access tiers**
Premium block blobs (BlockBlobStorage)	Block and append blobs	LRS, ZRS	Many transactions, small objects, consistently low latency. **No tiers**
Premium file shares (FileStorage)	Azure Files only: SSD provisioned v1 / v2 (and HDD provisioned v2)	LRS, ZRS (HDD v2 also GRS / GZRS)	Files with high IOPS; the only type with **NFS 4.1**
Premium page blobs	Page blobs only	LRS (ZRS in some regions)	Unmanaged disks / VHD; rare today

- **Legacy:** General-purpose v1 and Blob Storage are being retired → **upgrade to GPv2** (no downtime, **irreversible**). Classic (ASM) accounts were retired in August 2024. The type cannot be changed after creation: create another account and copy.
- **Name:** 3-24 characters, lowercase letters and numbers only, **unique across all of Azure**. Endpoints: `https://<account>.blob.core.windows.net`, `.file.`, `.queue.`, `.table.`, `.dfs.` (Data Lake) and `.z<n>.web.` (static website, `$web` container).
- Default limits: **250** accounts per region and subscription (500 on request), **5 PiB** per account, 20,000-40,000 requests/s, 400 IP rules and 400 VNet rules per account.
- At creation: performance (Standard / Premium), redundancy, default **access tier** (Hot / Cool / Cold), **hierarchical namespace** (Data Lake: real folders, POSIX ACLs; can be enabled later but not disabled), secure transfer (HTTPS) on, minimum TLS 1.2, anonymous blob access **disabled**, key access allowed, **infrastructure encryption** (only at creation), **large file shares**.

2. Redundancy

Option	Copies	Durability	Survives	Read from secondary
LRS	3 in one datacenter	11 nines	Disk, node or rack failure	—
ZRS	3 across **3 zones** in the region	12 nines	Datacenter / zone outage	—
GRS	LRS + 3 **asynchronous** LRS copies in the paired region	16 nines	Regional outage (with failover)	No
RA-GRS	GRS + `-secondary` endpoint	16 nines	Same, and also **reads** without failover	**Yes**
GZRS	ZRS + 3 LRS copies in the paired region	16 nines	Zone and region	No
RA-GZRS	GZRS + read	16 nines	Zone and region, read always	**Yes**

- **Restrictions:** Premium only LRS / ZRS. **Azure Files does not support RA-GRS or RA-GZRS** (it never reads from the secondary). The **Archive** tier does not exist in ZRS / GZRS / RA-GZRS. Geo copies are **asynchronous**: RPO ≈ 15 min (**Last Sync Time** shows how far replication has reached).
- **Changing redundancy:** most changes are made **on the account itself** (Settings → Redundancy): LRS ↔ GRS / RA-GRS are immediate; LRS ↔ ZRS, GRS ↔ GZRS are a **conversion** (customer-initiated, can take days; or open a ticket). If there are blobs in Archive they must be rehydrated before moving to ZRS / GZRS. There is no Standard ↔ Premium conversion: new account + copy (AzCopy).
- **Failover:** with GRS / GZRS you can force a **customer-managed failover** (unplanned) when the primary region goes down: the secondary becomes primary, the account ends up as **LRS** and **GRS must be reconfigured**; unreplicated data may be lost. There is also the **planned failover** (no loss, for drills). With RA-GRS it is enough to read from the `-secondary` endpoint without failover. Failover is not possible with object replication enabled, nor with Azure File Sync without support.

3. Encryption

Mechanism	What it is	Detail
SSE with Microsoft-managed keys	AES-256 encryption at rest **always on**, no cost or configuration	Cannot be disabled
Customer-managed keys (CMK)	Your key in **Key Vault** (or Managed HSM) + the account's **managed identity**	The vault needs **soft delete and purge protection**; automatic rotation if you reference the key without a version; revoking the key = data inaccessible
Customer-provided keys	The client sends the key in each request (blobs only)	For apps that manage their own keys; not for the general exam
Infrastructure (double) encryption	Second encryption layer at the infrastructure level	Only enabled **at account creation**
Encryption scopes	Different keys per container or blob within the same account	Isolate data from several customers in one account
In transit	**Secure transfer required** (HTTPS and encrypted SMB 3.x only), minimum TLS	Disable it only for legacy SMB 2.1 (and only within the region)

- Access keys, SAS and RBAC are **authorization**, not encryption. Azure Disk Encryption and encryption at host belong to **VM disks**, not to the account (cheatsheet 08).

4. Object replication (blobs)

Aspect	Rule
What it copies	**Block blobs** (not append, not page, not snapshots) from a source container to a destination one, **asynchronously**, with their versions and metadata
Requirements	**Blob versioning** on source and destination, **change feed** on source; GPv2 or Premium block blob accounts; **no** hierarchical namespace
Topology	Same or different region, same or different subscription, even another tenant (if allowed). One source → maximum **2** destination accounts; **1,000 rules** per policy
Filters	Name prefix and minimum creation date; by default only **new** blobs (existing ones can be included)
Destination	Becomes **read-only** while a policy exists (writing returns 409); an already replicated blob can be archived at the destination, but its later changes no longer replicate
Purpose	Reduce latency per region, process in several regions, distribute results, **compliance** / data copy; **does not replace GRS** (it is per container and you control it)
Breaks if	A source or destination blob moves to **Archive**, there is an immutability policy on the destination, or account failover is used

5. Account firewall and networking

Setting	Options	Detail
Public network access	Enabled from all networks · Enabled from **selected virtual networks and IP addresses** · **Disabled**	Disabled = private endpoints only
Virtual network rules	Subnets with the `Microsoft.Storage` **service endpoint** (same region) or `Microsoft.Storage.Global` (any region)	Traffic arrives with a **private** source IP; up to 400 rules; VNets from any subscription or tenant
IP network rules	**Public** IPv4 addresses: a single IP or a CIDR range of **/30 or larger** (not /31 or /32, no private ranges); they do not work for clients in the same Azure region (those use VNet rules)	For on-prem: the public egress IP (or the ExpressRoute Microsoft peering NAT IPs); up to 400
Resource instance rules	Allow specific Azure resources (Synapse, Backup vaults, etc.) by identity	For services that cannot be isolated by IP / VNet
Exceptions	"Allow Azure services on the **trusted services list**" · read access to **resource logs** · read access to **metrics**	Backup, Monitor, Defender, Site Recovery... keep working with the network closed
Private endpoint	NIC with a private IP in your VNet per sub-resource (blob, file, dfs, queue, table, web) + private zone `privatelink.<service>.core.windows.net`	Access from on-prem via VPN / ExpressRoute and from peered VNets; it is the answer to "no public endpoint" (cheatsheet 13)

- With the firewall enabled, the **portal, Storage Explorer and AzCopy from your PC** fail ("not authorized") until you add your **public IP** to the rules (or use "Add your client IP address").
- **Service endpoint** = still the public endpoint, but it only accepts those subnets; free; does not work from on-prem. **Private endpoint** = private IP, private DNS, cost per hour and per GB; works from on-prem.
- Other account switches: **Allow Blob anonymous access** (if Disabled no container can be public), **Allow storage account key access** (Disabled = Entra ID and user delegation SAS only), **Default to Microsoft Entra authorization in the portal**, **Allow cross-tenant replication**.

6. Storage Explorer, AzCopy and bulk transfer

Tool	How it authenticates	Purpose
Azure Storage Explorer (desktop app)	Subscription (Entra ID), name + key, **SAS URI**, connection string, emulator	Browse and manage blobs, files, queues and tables; upload / download; change tier; create SAS and stored access policies; Data Lake ACLs; view soft-deleted items; share snapshots. To view data with Entra ID it needs the **Storage Blob Data Reader** role (plain Reader does not see data)
AzCopy (command line)	`azcopy login` (Entra ID), `--identity` (managed identity) or **SAS in the URL**; Files: SAS (or Entra with the Storage File Data Privileged Contributor role)	Bulk, scriptable copies: local ↔ blob / Files, **blob ↔ blob** between accounts (server to server), **S3 / Google Cloud → Azure**, incremental sync, resume jobs
Azure Import/Export	Your own BitLocker-encrypted SATA/SSD drives + the WAImportExport tool	Import to Blob or Files, export Blob only; when you do not want Microsoft hardware
Data Box Disk / Data Box 120 / Data Box 525	Microsoft devices (up to 40 TB on SSD drives · **120 TB** usable · **525 TB** usable; the 80 TB Data Box and Data Box Heavy are retired or being retired)	Tens or hundreds of TB with little bandwidth; Data Box Gateway = **online** virtual appliance
Azure Storage Mover	Managed service	Migrate on-prem SMB / NFS shares to Azure Files or Blob over the network

```
azcopy login                                  # Entra ID (or --identity on a VM with MI)
azcopy copy "C:\data" "https://st1.blob.core.windows.net/cont?<SAS>" --recursive
azcopy copy "https://src.blob.core.windows.net/c1?<SAS>" "https://dst.blob.core.windows.net/c2?<SAS>" --recursive
azcopy sync "C:\data" "https://st1.blob.core.windows.net/cont?<SAS>" --recursive --delete-destination=true
azcopy copy ... --block-blob-tier Cool --include-pattern "*.pdf" --put-md5 --cap-mbps 100
azcopy make "https://st1.blob.core.windows.net/new?<SAS>"     · azcopy list ... · azcopy remove ...
azcopy jobs list · azcopy jobs resume <jobId>                # resume an interrupted copy
```

- `copy` overwrites by default; `sync` is **one-way** and incremental (compares last-modified time) and only deletes at the destination with `--delete-destination`. **Account-to-account** copies do not go through your PC. For Azure Files, AzCopy requires SAS except in recent versions with Entra ID.

- Choosing: a few GB or TB with good bandwidth → AzCopy / Storage Explorer; tens of TB and little network → Data Box Disk / Data Box 120; hundreds of TB → Data Box 525; "no third-party hardware", "use our own drives" → Import/Export; continuous migration of a file server → Azure File Sync.

7. Commands that show up

Task	PowerShell (Az.Storage)	Azure CLI
Create account	`New-AzStorageAccount -ResourceGroupName RG1 -Name st1 -Location westeurope -SkuName Standard_GRS -Kind StorageV2 -AccessTier Hot -MinimumTlsVersion TLS1_2 -AllowBlobPublicAccess $false`	`az storage account create -n st1 -g RG1 -l westeurope --sku Standard_GRS --kind StorageV2 --access-tier Hot --min-tls-version TLS1_2 --allow-blob-public-access false`
Change redundancy or tier	`Set-AzStorageAccount -ResourceGroupName RG1 -Name st1 -SkuName Standard_GRS` · `-AccessTier Cool` · to ZRS / GZRS (conversion): `Start-AzStorageAccountMigration -ResourceGroupName RG1 -AccountName st1 -TargetSku Standard_ZRS`	`az storage account update -n st1 -g RG1 --sku Standard_RAGRS` · `--access-tier Cool` · to ZRS / GZRS: `az storage account migration start -n st1 -g RG1 --sku Standard_ZRS`
Upgrade GPv1 → GPv2	`Set-AzStorageAccount -ResourceGroupName RG1 -Name st1 -UpgradeToStorageV2`	`az storage account update -n st1 -g RG1 --set kind=StorageV2`
Firewall	`Update-AzStorageAccountNetworkRuleSet -ResourceGroupName RG1 -Name st1 -DefaultAction Deny -Bypass AzureServices` · `Add-AzStorageAccountNetworkRule ... -IPAddressOrRange 203.0.113.0/24` · `-VirtualNetworkResourceId <subnet id>`	`az storage account update -n st1 --default-action Deny --bypass AzureServices` · `az storage account network-rule add -n st1 --ip-address 203.0.113.0/24` · `--vnet-name vnet1 --subnet sub1`
Failover	`Invoke-AzStorageAccountFailover -ResourceGroupName RG1 -Name st1`	`az storage account failover -n st1 -g RG1`
Endpoints and keys	`(Get-AzStorageAccount -ResourceGroupName RG1 -Name st1).PrimaryEndpoints` · `Get-AzStorageAccountKey`	`az storage account show -n st1 --query primaryEndpoints` · `az storage account keys list -n st1`
Object replication	`New-AzStorageObjectReplicationPolicy -ResourceGroupName RG1 -StorageAccountName dst -SourceAccount src -Rule <rule>`	`az storage account or-policy create --account-name dst --source-account src --source-container c1 --destination-container c2`
PowerShell context	`$ctx = New-AzStorageContext -StorageAccountName st1 -UseConnectedAccount` (Entra) · `-StorageAccountKey <key>` · `-SasToken <sas>`	CLI: `--auth-mode login` (Entra) · `--account-key` · `--sas-token` · `--connection-string`

8. Decision tree

```
What does the wording ask for?
├─ Blobs, queues, tables and files, geo-redundancy, tiers ..... Standard general-purpose v2
├─ NFS, high IOPS on files ..... Premium file shares (FileStorage)
├─ Many small transactions, minimal latency on blobs ..... Premium block blobs
├─ Survive a datacenter outage within the region ..... ZRS
├─ Survive a regional outage and be able to READ meanwhile ..... RA-GRS / RA-GZRS
├─ Survive region + zone ..... GZRS
├─ Data that cannot leave the region ..... LRS or ZRS
├─ Copy specific containers to another account or region ..... Object replication (versioning + change feed)
├─ Encryption with our own key ..... CMK in Key Vault + managed identity
├─ Double encryption required by compliance ..... Infrastructure encryption (at creation)
├─ Only reachable from our VNet ..... Firewall: selected networks + service endpoint
├─ No public endpoint, access from on-prem ..... Private endpoint + private DNS
├─ Backup / Monitor must keep working with the firewall closed ..... Exception: trusted Microsoft services
├─ Copy TB over the network in a scriptable way ..... AzCopy
└─ Hundreds of TB with little bandwidth ..... Data Box 120 / 525, or Import-Export with your own drives
```

9. Signals → answer

Wording	Answer
"minimize cost", "data can be reconstructed"	LRS
"survive a datacenter outage", "zone failure"	ZRS (or GZRS if the region too)
"read access during a regional outage"	RA-GRS / RA-GZRS
"Azure Files", "read from secondary region"	Impossible: Files does not support RA-*; use GRS + failover
"archive tier" + "ZRS"	Incompatible: Archive only LRS / GRS / RA-GRS
"change from LRS to GRS"	Settings → Redundancy (no recreation)
"change from Standard to Premium"	New account + AzCopy
"after failover the account is LRS"	Normal: reconfigure GRS afterwards
"NFS file share"	Premium file shares (FileStorage)
"hierarchical namespace", "ACLs on folders", "analytics"	Data Lake Storage Gen2 (GPv2 + HNS)
"replicate blobs from container A to container B in another region"	Object replication
"object replication fails", "destination 409"	Missing versioning / change feed, blob in Archive, or writing to the destination
"encrypt with keys we control", "rotate keys"	Customer-managed keys in Key Vault
"storage not accessible from the portal after enabling firewall"	Add the client's public IP (or trusted services for services)
"VM in a VNet must reach storage, no public access from internet"	Service endpoint + VNet rule (or private endpoint)
"on-premises access through private IP"	Private endpoint + VPN / ExpressRoute + DNS

Wording	Answer
"Azure Backup fails after restricting network"	Exception "trusted Microsoft services"
"copy 50 TB, 10 Mbps link"	Data Box
"sync a local folder to a container daily"	`azcopy sync` (scheduled)
"copy between two storage accounts without downloading"	`azcopy copy` account to account (with SAS on both sides)
"Storage Explorer: Reader cannot see blobs"	Add Storage Blob Data Reader or use the key

10. Mistakes to avoid

1. Choosing RA-GRS for Azure Files or Archive on a ZRS account.
2. Believing GRS allows reading from the secondary without failover: that is **RA**-GRS.
3. Thinking failover leaves the account as GRS: it ends up as LRS and must be reconfigured.
4. Trying to convert a Standard account to Premium or change its type: new account + copy.
5. Configuring object replication without versioning on both accounts and change feed on the source, or writing to the destination container.
6. Enabling infrastructure encryption or hierarchical namespace without knowing that one can only be enabled at creation and the other cannot be removed.
7. Closing the firewall and forgetting the trusted services exception (Backup, Monitor) or the client IP.
8. Using `azcopy sync` expecting it to be bidirectional, or `azcopy copy` expecting it to delete what no longer exists at the source.
9. Sending 500 TB over a slow line instead of using Data Box.

Cheatsheet AZ-104 — 05 · Storage access: keys, SAS, stored access policies and identity for Azure Files

1. The four ways to authorize

Method	What it is	Scope	Revoke	Signal
Shared Key (access keys)	Two 512-bit keys with **full access** to the whole account (all services and data)	Everything	Regenerate the key	"full access", "connection string", "mount with account key"
SAS (shared access signature)	Token in the URL with permissions, resources, time window, IP and protocol	Whatever the token says	Depends on the type (section 3)	"temporary access", "specific container", "partner without an account"
Microsoft Entra ID (data RBAC)	Roles with DataActions on blob, queue, table or Files	Per role and scope (account, container)	Remove the assignment	"most secure", "identity-based", "no keys in code"
Anonymous access	Container with Blob or Container level, if the account allows it	Public read	Change the level / disable on the account	"publicly readable images on a website"

Golden rule of the exam: when asked for "the most secure" or "least privilege" → **Entra ID** (managed identity for apps, user delegation SAS for sharing); keys and key-signed SAS only if the wording requires them.

2. Access keys

- Two keys (**key1 / key2**) to rotate without interruption: switch the apps to key2, regenerate key1, and vice versa. Regenerating a key **invalidates every SAS signed with it** (account and service SAS). A **rotation reminder** can be set (e.g. 90 days) and **Key Vault** can manage and rotate the keys for you.
- Who can see them: requires the `Microsoft.Storage/storageAccounts/listKeys/action` action → Owner, Contributor, Storage Account Contributor, **Storage Account Key Operator Service Role** (list and regenerate only). **Not Reader** (and a ReadOnly lock blocks it too).
- **"Allow storage account key access" = Disabled** → keys and key-signed SAS are blocked; only Entra ID and user delegation SAS work. Azure Files over SMB with a key stops mounting. Azure Policy can enforce this setting. Apps using a connection string fail with 403.
- Connection string = `DefaultEndpointsProtocol=https;AccountName=st1;AccountKey=...` → equivalent to handing over the key.

3. SAS: three types

	Account SAS	Service SAS	User delegation SAS
Signed with	Account key	Account key	**Entra ID credentials** (user delegation key)
Scope	One or several **services** (blob, file, queue, table) and service-level operations (create containers, properties)	**One** resource of **one** service: container, blob, share, file, queue, table	Blob (container or blob; also Data Lake) and, recently, queue, table and Files
Stored access policy	No	**Yes** (up to **5** per container / share / queue / table)	No
Duration	Whatever you set (expiration policy recommended)	Whatever you set	Maximum **7 days**
Revoke	Regenerate the key	Delete or edit the **stored access policy**; if ad hoc, regenerate the key	Revoke the user delegation key or remove the user's role
Permission to create it	Having the key	Having the key	Role with `generateUserDelegationKey` (e.g. **Storage Blob Delegator** or Storage Blob Data Contributor)
Signal	"access to multiple services", "create containers"	"share one blob / container", "revoke without regenerating keys" → + policy	"most secure", "no account key"

Parameter	Meaning
`sv`	Service version
`ss`	Services (account SAS): **b**lob, **f**ile, **q**ueue, **t**able
`srt`	Resource types (account SAS): **s**ervice, **c**ontainer, **o**bject
`sr`	Resource (service SAS): b = blob, c = container, s = share, f = file, d = directory
`sp`	Permissions: **r**ead, **w**rite, **d**elete, **l**ist, **a**dd, **c**reate, **u**pdate, **p**rocess, **t**ags, **x** delete versions...
`st` / `se`	Start / **expiry** (UTC, ISO 8601 format)
`sip`	Allowed IP or range
`spr`	Protocol: `https` (recommended) or `https,http`
`si`	**Stored access policy** identifier
`sig`	HMAC signature

- **Stored access policy:** defined on the container (or share, queue, table) with permissions, start and end; the service SAS that reference it **inherit** those limits. Changing the expiry or deleting the policy revokes all associated SAS at once **without regenerating keys**. Account SAS and user delegation SAS cannot use it.

- A SAS **cannot be audited or counted** (it is generated on the client). Best practices: HTTPS, short expiry, start 15 min in the past (clock skew), least privilege, **SAS expiration policy** on the account (warns if someone creates a longer SAS), revocation plan.
- "SAS stopped working" → the key was regenerated, it expired, the client clock is off, IP outside `sip`, HTTP with `spr=https`, or key access was disabled.

4. Entra ID and data roles

Service	Roles	Notes
Blob	**Storage Blob Data Owner** (everything + Data Lake ACLs) · **Data Contributor** (read, write, delete) · **Data Reader** · **Storage Blob Delegator** (create user delegation key)	Scope: account, container; ABAC conditions possible
Queue	Storage Queue Data Contributor · Reader · **Message Processor** (read and delete) · **Message Sender** (add)	For Functions / apps with MI
Table	Storage Table Data Contributor · Reader	—
Files (SMB)	**Storage File Data SMB Share Reader** · **Contributor** · **Elevated Contributor** (modifies NTFS ACLs)	**Share**-level permission; inside, the Windows ACLs rule
Files (REST / portal)	**Storage File Data Privileged Contributor / Reader**	Bypass the ACLs; to operate from the portal, PowerShell or AzCopy with Entra ID

- **Owner / Contributor have no data access** with Entra ID: the portal shows "You do not have permissions" when opening a container with Entra authentication; switch to "Access key" or add Storage Blob Data Reader. The **"Default to Microsoft Entra authorization in the Azure portal"** toggle sets the default method.
- For an **app or VM** → managed identity + data role (never keys in code). For a **user** → data role on just the right container.

5. Anonymous blob access

- Two levels: the **account** must have "Allow Blob anonymous access" = Enabled (new accounts come with it **Disabled**) and the **container** a level: **Private** (default), **Blob** (read blobs if you know the URL, no listing) or **Container** (read and list). "Public website images, no listing" → Blob. For security, Azure Policy can forbid anonymous access.

6. Identity for Azure Files (SMB)

Identity source	Requirements	Who it is for
On-premises AD DS	Join the storage account to the domain (computer object or service account with `Join-AzStorageAccount`), users **synchronized** to Entra ID, clients with line of sight to the **DC**	File server migration; AD DS-joined machines
Microsoft Entra Domain Services	Enable on the account; the managed domain issues the Kerberos tickets	Lift-and-shift without your own DCs; VMs joined to Entra DS
Microsoft Entra Kerberos	**Hybrid** users (synchronized from AD DS); **Entra joined or hybrid joined** clients; no DC line of sight needed to mount (needed to configure ACLs)	Remote work, Windows 365 / AVD, no VPN
Account key	Nothing, but it is **superuser**: ignores the ACLs	Initial mount to set ACLs with `icacls`, scripts; not recommended for users

- Two permission levels: **share** (Storage File Data SMB Share * RBAC role or a **default permission** for all authenticated users) and **directory / file** (regular NTFS ACLs with icacls or Explorer, after mounting with the key).
- Network: SMB uses port **445** with SMB 3.x encryption; many ISPs block it → mounting from on-prem requires **VPN / ExpressRoute + private endpoint**, or caching with **Azure File Sync**. **NFS 4.1** (Premium only) does not use identities: network control only (VNet / private endpoint); optional encryption in transit with TLS (AZNFS client) and its own "Require Encryption in Transit for NFS" setting.
- Mounting: Windows `net use Z: \\st1.file.core.windows.net\share1 /user:Azure\st1 <key>` (or the portal's "Connect" script); Linux `mount -t cifs //st1.file.core.windows.net/share1 /mnt/share1 -o username=st1,password=<key>,serverino`. With identity, a plain `net use` without credentials from a joined machine is enough.

7. Commands that show up

Task	PowerShell (Az.Storage)	Azure CLI
Regenerate key	`New-AzStorageAccountKey -ResourceGroupName RG1 -Name st1 -KeyName key1`	`az storage account keys renew -n st1 -g RG1 --key primary`
Disable keys	`Set-AzStorageAccount -ResourceGroupName RG1 -Name st1 -AllowSharedKeyAccess $false`	`az storage account update -n st1 --allow-shared-key-access false`
Account SAS	`New-AzStorageAccountSASToken -Service Blob,File -ResourceType Service,Container,Object -Permission "rl" -ExpiryTime (Get-Date).AddDays(7) -Protocol HttpsOnly -Context $ctx`	`az storage account generate-sas --account-name st1 --services bf --resource-types sco --permissions rl --expiry 2026-12-31T00:00Z --https-only`
Service SAS	`New-AzStorageContainerSASToken -Name c1 -Permission rl -ExpiryTime ... -Context $ctx` · `New-AzStorageBlobSASToken -Container c1 -Blob f.txt -Permission r ...` · with policy: `-Policy p1`	`az storage container generate-sas -n c1 --permissions rl --expiry ... --account-name st1` · `az storage blob generate-sas` · with policy: `--policy-name p1`
User delegation SAS	`$ctx = New-AzStorageContext -StorageAccountName st1 -UseConnectedAccount` → `New-AzStorageContainerSASToken ... -Context $ctx` (generates a user delegation SAS)	`az storage container generate-sas -n c1 --permissions r --expiry ... --auth-mode login --as-user`
Stored access policy	`New-AzStorageContainerStoredAccessPolicy -Container c1 -Policy p1 -Permission rl -ExpiryTime ...` · `Set-AzStorageContainerStoredAccessPolicy` · `Remove-...`	`az storage container policy create -c c1 -n p1 --permissions rl --expiry ...` · `update` · `delete`

Task	PowerShell (Az.Storage)	Azure CLI
Data role	`New-AzRoleAssignment -SignInName ana@contoso.com -RoleDefinitionName "Storage Blob Data Reader" -Scope <account or container id>`	`az role assignment create --assignee ana@contoso.com --role "Storage Blob Data Reader" --scope <id>`
Anonymous access	`Set-AzStorageAccount ... -AllowBlobPublicAccess $true` · `Set-AzStorageContainerAcl -Name c1 -Permission Blob`	`az storage account update --allow-blob-public-access true` · `az storage container set-permission -n c1 --public-access blob`
Files with AD DS	`Join-AzStorageAccount -ResourceGroupName RG1 -StorageAccountName st1 -DomainAccountType ComputerAccount`	Entra Kerberos: `az storage account update -n st1 --enable-files-aadkerb true --domain-name contoso.com --domain-guid <guid>`

8. Decision tree

```
What does the wording ask for?
├─ App in Azure accesses blobs without secrets ..... Managed identity + Storage Blob Data Contributor
├─ User views blobs in the portal with their account ..... Storage Blob Data Reader (Owner is not enough)
├─ Share a blob with an external party for a few days ..... Service SAS (user delegation SAS if "secure")
├─ Be able to revoke without regenerating keys ..... Service SAS + stored access policy
├─ Access to several services or create containers ..... Account SAS
├─ The most secure SAS, no key ..... User delegation SAS (max. 7 days)
├─ Forbid the use of keys and key-signed SAS ..... Allow storage account key access = Disabled
├─ Rotate keys without interruption ..... Alternate key1 / key2 (or Key Vault)
├─ Public images for a website ..... Account with anonymous access + Blob-level container
├─ SMB share with AD permissions for domain machines ..... Join to AD DS + SMB Share roles + NTFS ACLs
├─ SMB share for Entra joined machines without VPN or DC ..... Entra Kerberos
├─ SMB share without your own DCs (lift and shift) ..... Entra Domain Services
├─ Users cannot mount from the office ..... Port 445 blocked → VPN/ER + private endpoint or File Sync
└─ NFS ..... Premium + VNet / private endpoint (no identity)
```

9. Signals → answer

Wording	Answer
"most secure way for the app to access the storage account"	Managed identity + data role
"Owner cannot browse the container in the portal"	Missing Storage Blob Data Reader (or use the key)
"generate a SAS that can be revoked without regenerating keys"	Service SAS + stored access policy
"SAS valid for 7 days, no storage key, Microsoft Entra"	User delegation SAS
"allow listing blobs in the container for a SAS"	`l` (list) permission in `sp`
"SAS must only work over HTTPS from the office IP"	`spr=https` + `sip`
"SAS works on one blob but fails to create containers"	It is a service SAS: an account SAS with `srt` = s,c is needed
"after regenerating key1 the application fails"	Its SAS / connection strings used key1 → rotate to key2 first
"prevent the use of shared keys"	Allow storage account key access = Disabled (and Policy to audit it)
"images must be publicly readable but not listable"	Container public access = Blob
"anonymous access option is greyed out"	The account has "Allow Blob anonymous access" set to Disabled
"file share with NTFS permissions for domain users"	AD DS identity + Storage File Data SMB Share Contributor + icacls
"users must access the share from Entra joined devices over the internet"	Entra Kerberos
"mount fails with error 53 / 67 from on-premises"	Port 445 blocked
"set NTFS ACLs on the share"	Mount with the **key** (superuser) or the Elevated Contributor role
"copy files to the share with AzCopy from the portal/REST using identity"	Storage File Data Privileged Contributor
"Storage Account Key Operator Service Role"	List and regenerate keys only (least privilege for rotation)

10. Mistakes to avoid

1. Handing over the account key (or the connection string) when temporary or scoped access is requested: that is a SAS.
2. Believing an ad hoc key-signed SAS can be revoked without regenerating the key.
3. Attaching a stored access policy to an account SAS or a user delegation SAS: service SAS only.
4. Requesting a 30-day user delegation SAS: the maximum is 7.
5. Thinking Owner or Contributor read blobs with Entra ID: **data** roles are needed.
6. Forgetting that regenerating a key breaks the SAS signed with it and the apps using a connection string.
7. Setting anonymous access on the container without enabling it on the account first.
8. Choosing Entra Domain Services when users and machines are already in AD DS (and vice versa), or Entra Kerberos for **cloud-only** users (it needs hybrid identities).
9. Trying to mount SMB from home without VPN: 445 is usually blocked.

Cheatsheet AZ-104 — 06 · Blob and Files: tiers, lifecycle, versioning, soft delete, snapshots and Azure File Sync

1. Blobs and containers

Blob type	Purpose	Detail
Block blob	Files in general (images, backups, videos)	Up to ≈ 190 TiB; **the only one with access tiers**; uploaded in blocks
Append blob	Logs, append-only writes	No tiers
Page blob	VHD / disks, random access	Up to 8 TiB; no tiers

- A container is a **flat** namespace ("folders" are name prefixes) except with **hierarchical namespace** (Data Lake Gen2: real directories and ACLs). Container names: lowercase, 3-63 characters.
- Each blob has **metadata** (key-value pairs), **index tags** (up to 10, queryable with `Find blobs by tags` and usable in lifecycle), properties (Content-Type, tier, lease) and can have **snapshots** and **versions**.
- **Static website:** enabled on the account, publishes the `$web` container at `https://<account>.z<n>.web.core.windows.net` (static content only; HTTPS with a custom domain → Azure CDN / Front Door).

2. Access tiers (block blobs, GPv2 accounts)

	Hot	Cool	Cold	Archive
Use	Frequent access	Infrequent	Rare, but immediate read	Almost never; **offline**
Cost	Expensive to store, cheap to access	Cheaper to store, more expensive to access	Cheaper still	Cheapest to store; most expensive to read
Minimum retention	—	**30 days**	**90 days**	**180 days** (otherwise early deletion charge, prorated)
Latency	ms	ms	ms	**Hours**: standard rehydration up to **15 h**, high priority under 1 h (blobs < 10 GB)
Availability	99.9 %	99 %	99 %	99 %
Redundancies	All	All	All	Only **LRS, GRS, RA-GRS**
Account default tier	Yes	Yes	Yes	**No** (blob level only)

- A blob with no explicit tier shows "**Hot (inferred)**" (inherits the account's). Changing the account default tier recalculates all inferred ones (and charges operations).
- Moving to a colder tier is instant and billed as a **write**; moving to a warmer one is billed as a **read** (+ early deletion if the minimum retention was not met). To rehydrate from Archive: **Set Blob Tier** (in place) or **Copy Blob** to another blob in an online tier (avoids the penalty and keeps the original). A blob in Archive only allows reading **metadata**, not the content.
- **Premium** block blobs **have no tiers**; to move to tiers you must copy to a Standard account. Changing the tier only applies to **block blobs**.

3. Lifecycle management

```
{ "rules": [ { "enabled": true, "name": "logs", "type": "Lifecycle",
  "definition": {
    "filters": { "blobTypes": [ "blockBlob" ], "prefixMatch": [ "logs/2026" ] },
    "actions": {
      "baseBlob": {
        "tierToCool":    { "daysAfterModificationGreaterThan": 30 },
        "tierToArchive": { "daysAfterModificationGreaterThan": 90 },
        "delete":        { "daysAfterModificationGreaterThan": 365 } },
      "snapshot": { "delete": { "daysAfterCreationGreaterThan": 90 } },
      "version":  { "delete": { "daysAfterCreationGreaterThan": 90 } } } } } ] }
```

- **Filters:** `blobTypes` (blockBlob, appendBlob), `prefixMatch` (`container/prefix`), `blobIndexMatch` (tags). **Actions:** tierToCool, tierToCold, tierToArchive, delete, and `enableAutoTierToHotFromCool` (back to Hot if accessed). **Conditions:** days since modification, since creation, since **last access** (*last access time tracking* must be enabled) or since the last tier change.
- Runs **once a day**; the first run can take up to 24 h and policy changes up to 48 h. Up to **100 rules** per policy. **Does not rehydrate** from Archive. On Premium block blob accounts only delete actions (no tiering). "Minimize cost", "automatically move to archive after 180 days" → lifecycle, not a script.

4. Data protection for blobs

Feature	What it protects	Exam details
Soft delete (blobs)	Deleted or overwritten blobs, snapshots and versions	Retention **1-365 days** (default 7 when enabled from the portal); **Undelete** restores; does not protect if the account is deleted
Soft delete (containers)	Deleted containers	Retention 1-365 days; the whole container is restored; independent from the blob one

Feature	What it protects	Exam details
Versioning	Every overwrite or deletion creates an automatic **version**	Restore = **copy** (promote) the previous version; required for object replication and point-in-time restore; versions cost → combine with lifecycle
Snapshots	**Manual** read-only copy of a blob at a point in time	Snapshots must be deleted before the blob (or use "delete snapshots"); not in Archive
Point-in-time restore	Return a **range of containers** to an earlier point in time	Block blobs only; requires **versioning + change feed + soft delete**; PITR retention must be **shorter** than the soft delete one; "ransomware, restore to yesterday 10:00"
Change feed	Ordered log of changes	Prerequisite for PITR and object replication; auditing
Immutable storage (WORM)	Nobody (not even admins) can delete or modify	**Time-based retention** (unlocked for testing, **locked** irreversible) and **legal hold** (until removed); at container or **version** level; "regulatory compliance", "SEC 17a-4"
Blob inventory	Daily / weekly report of blobs and properties	"report of all blobs with their tier"
Recover the account	Deleted storage account	Up to **14 days** if the name was not reused (portal → Storage accounts → Restore)

- For "**the blob was overwritten** and we need the previous content" → versioning (or snapshots if they were being taken). For "**deleted** by mistake" → soft delete + undelete. For "**deleted container**" → container soft delete. For "**compliance, no deletions for 7 years**" → locked immutability policy.

5. Azure Files

Aspect	SSD (Premium, FileStorage)	HDD (Standard: GPv2 pay-as-you-go or FileStorage provisioned v2)
Protocols	**SMB and NFS 4.1**	SMB only
Billing	**Provisioned** (v1 / v2): you pay for what is provisioned (GiB, IOPS, MiB/s)	Provisioned v2 or **pay-as-you-go** with **Transaction optimized, Hot, Cool** tiers (more transactions → optimized; more storage → cool)
Redundancy	LRS, ZRS	LRS, ZRS, GRS, GZRS (**never RA-***)
Size	**100 TiB** (provisioned v1) or **256 TiB** (provisioned v2) per share	**100 TiB** on pay-as-you-go with any redundancy (the **large file shares** setting is only needed on very old accounts) or **256 TiB** on provisioned v2
Latency	Single-digit milliseconds	Higher
When	Databases, IOPS, NFS	Team shares, lift-and-shift, File Sync

- **Quota** per share (maximum size). Share **snapshots**: up to **200** per share, retention up to 10 years, incremental, read-only; files are restored with "Previous versions" in Windows or from the portal. **Share soft delete**: enabled by default on new accounts, 1-365 days (default 7), recovers the whole share.
- **Azure Backup for Files:** snapshot-based, in a Recovery Services vault in the **same region**; daily or hourly schedule and GFS retention (daily, weekly, monthly, yearly); restores the whole share or **individual files** (item-level) in place or to another location; enables soft delete and places a lock on the account.
- Connecting: encrypted SMB 3.x over **port 445**; "Connect" script from the portal; Linux with `cifs`. If users cannot mount from the office → the ISP blocks 445: VPN / ExpressRoute + private endpoint or Azure File Sync (cheatsheet 05 for identities).

6. Azure File Sync

Piece	What it is	Rules
Storage Sync Service	Root resource in Azure	A server registers with only **one** Storage Sync Service; the storage account must be in the **same region**
Registered server	Windows Server **2016-2025** (Full or Core) with the **agent** installed and registered	Up to **30** synced shares per server; do not clone / sysprep after installing the agent; failover cluster supported (File Server for general use)
Sync group	Sync relationship	**A single cloud endpoint** per sync group; a share can only be in **one** sync group
Cloud endpoint	The Azure file share (**SMB** only)	Changes made **directly in Azure** are detected by a job every **24 h** (server changes are uploaded within minutes)
Server endpoint	Folder on a registered server	**One per server** in each sync group; paths cannot overlap; not on the system volume for tiering
Cloud tiering	Keeps only hot files locally; cold ones remain as **pointers** (grey icon with an X, offline attribute) and are recalled on open	Policies: **volume free space** (% free to keep) and **date policy** (not accessed in N days); files > 64 KiB only; ideal for "reduce on-premises storage"

- Scenarios: **cache** a share across several offices (multi-site, all servers see the same), **replace** a file server while keeping local access, **migrate** to Azure Files without downtime, **DR** of a file server (install the agent on a new server and add it to the sync group). NTFS ACLs are preserved.
- Backup: Azure Backup of the share in the cloud, or local backup on a server **without cloud tiering** (with tiering only pointers would be copied). Simultaneous edit conflicts → the losing file is saved renamed with "Conflict". Role to register servers and create cloud endpoints: **Azure File Sync Administrator** (or Owner / Contributor).

7. Commands that show up

Task	PowerShell (Az.Storage)	Azure CLI
Container and blob	`New-AzStorageContainer -Name c1 -Context $ctx` · `Set-AzStorageBlobContent -File .\a.pdf -Container c1 -Blob a.pdf -StandardBlobTier Cool -Context $ctx` · `Get-AzStorageBlob -Container c1 -Context $ctx`	`az storage container create -n c1 --account-name st1 --auth-mode login` · `az storage blob upload -f a.pdf -c c1 -n a.pdf --tier Cool --account-name st1 --auth-mode login` · `az storage blob list -c c1`
Change tier	`$b = Get-AzStorageBlob -Container c1 -Blob a.pdf -Context $ctx; $b.BlobClient.SetAccessTier("Archive")` · rehydrate: `SetAccessTier("Hot", $null, "High")`	`az storage blob set-tier -c c1 -n a.pdf --tier Archive` · `--tier Hot --rehydrate-priority High`
Soft delete and versioning	`Enable-AzStorageBlobDeleteRetentionPolicy -ResourceGroupName RG1 -StorageAccountName st1 -RetentionDays 14` · `Enable-AzStorageContainerDeleteRetentionPolicy ... -RetentionDays 14` · `Update-AzStorageBlobServiceProperty ... -IsVersioningEnabled $true`	`az storage account blob-service-properties update --account-name st1 --enable-delete-retention true --delete-retention-days 14 --enable-container-delete-retention true --container-delete-retention-days 14 --enable-versioning true`
Restore	`Get-AzStorageBlob -Container c1 -IncludeDeleted -Context $ctx` → `$b.BlobBaseClient.Undelete()` · snapshot: `$b.BlobClient.CreateSnapshot()`	`az storage blob undelete -c c1 -n a.pdf` · `az storage blob snapshot -c c1 -n a.pdf` · `az storage container restore -n c1 --deleted-version <v>`
Lifecycle	`Set-AzStorageAccountManagementPolicy -ResourceGroupName RG1 -StorageAccountName st1 -Policy (New-AzStorageAccountManagementPolicyRule ...)` or `-Policy` with a policy object	`az storage account management-policy create --account-name st1 -g RG1 --policy @policy.json`
File share	`New-AzRmStorageShare -ResourceGroupName RG1 -StorageAccountName st1 -Name share1 -QuotaGiB 1024 -EnabledProtocol SMB` · snapshot: `New-AzRmStorageShare ... -Snapshot` · soft delete: `Update-AzStorageFileServiceProperty ... -EnableShareDeleteRetentionPolicy $true -ShareRetentionDays 14`	`az storage share-rm create --storage-account st1 -g RG1 -n share1 --quota 1024 --enabled-protocols SMB` · `az storage share-rm snapshot` · `az storage account file-service-properties update --account-name st1 --enable-delete-retention true --delete-retention-days 14`
Azure File Sync	`New-AzStorageSyncService` · `Register-AzStorageSyncServer` · `New-AzStorageSyncGroup` · `New-AzStorageSyncCloudEndpoint` · `New-AzStorageSyncServerEndpoint -ServerLocalPath D:\data -CloudTiering -VolumeFreeSpacePercent 20 -TierFilesOlderThanDays 30` · `Invoke-AzStorageSyncChangeDetection` (force detection on the cloud endpoint)	`az storagesync` (extension): `az storagesync create`, `az storagesync sync-group create`, `az storagesync sync-group cloud-endpoint create`, `az storagesync sync-group server-endpoint create --cloud-tiering on --volume-free-space-percent 20 --tier-files-older-than-days 30`

8. Decision tree

```
What does the wording ask for?
├─ Infrequently used data, immediate access, minimum cost ..... Cool (≥ 30 days) or Cold (≥ 90 days)
├─ Long retention, hours of wait acceptable ..... Archive (≥ 180 days; LRS / GRS / RA-GRS only)
├─ Read an archived blob ..... Rehydrate (Set tier or Copy Blob; high priority < 1 h)
├─ Automatically move by age / delete ..... Lifecycle management (rules by prefix)
├─ Recover a deleted blob ..... Soft delete → Undelete
├─ Recover the previous version of an overwritten blob ..... Versioning (or snapshot if one existed)
├─ Recover a deleted container ..... Container soft delete
├─ Return a whole container to a point in time (ransomware) ..... Point-in-time restore
├─ Nobody can delete or modify for N years ..... Immutable storage (time-based, locked) / legal hold
├─ Share with high IOPS or NFS ..... Azure Files Premium (SSD)
├─ Cheap share for documents ..... Azure Files Standard (HDD) Hot / Cool / Trans. optimized
├─ Share backup with per-file recovery ..... Azure Backup (snapshots) or manual snapshots
├─ Several offices share files with a local cache ..... Azure File Sync (multi-site) + cloud tiering
└─ Free up file server disk without losing files ..... Cloud tiering (volume free space / date policy)
```

9. Signals → answer

Wording	Answer
"accessed once a month, must be available immediately"	Cool
"accessed rarely, retrieval within hours is acceptable, lowest cost"	Archive
"blob in archive, need it in 1 hour"	Rehydrate high priority
"deleted blobs before 30 days incur charges"	Early deletion for Cool (30), Cold (90), Archive (180)
"move blobs to cool after 60 days, delete after 2 years, minimal effort"	Lifecycle management rule
"lifecycle rule based on last read"	Enable last access time tracking + daysAfterLastAccessTimeGreaterThan
"accidentally deleted a blob", "recover"	Soft delete (retention) → Undelete
"restore previous version after overwrite"	Blob versioning (copy the version)
"point-in-time restore prerequisites"	Versioning + change feed + soft delete (longer retention)
"WORM", "cannot be deleted even by administrators"	Immutability policy (locked)
"storage account deleted 5 days ago"	Recover (up to 14 days)
"premium file share", "NFS"	FileStorage account (SSD)
"file share larger than 5 TiB"	Large file shares (legacy setting: current accounts already reach 100 TiB)
"restore a single file from the share"	Share snapshot (manual or Azure Backup)

Wording	Answer
"branch offices need low latency access to the same files"	Azure File Sync, server endpoint in each office
"files appear with a grey X icon / offline"	Cloud tiering: they are in Azure, recalled on open
"files changed in the Azure portal take a day to appear on the server"	Change detection on the cloud endpoint every 24 h (or `Invoke-AzStorageSyncChangeDetection`)
"second server endpoint on the same server for the same sync group" fails	One server endpoint per server and sync group
"sync group with two Azure file shares"	Impossible: one cloud endpoint per sync group

10. Mistakes to avoid

1. Setting Archive as the account default tier, or on a ZRS account.
2. Expecting to read a blob in Archive without rehydrating, or expecting lifecycle to rehydrate it.
3. Enabling point-in-time restore without versioning, change feed and soft delete.
4. Confusing soft delete (deletions) with versioning (overwrites).
5. Assuming a locked immutability policy can be removed.
6. Choosing Azure Files Standard when NFS or high IOPS are required (that is Premium), or RA-GRS for Files.
7. Creating a sync group with two cloud endpoints or two server endpoints on the same server.
8. Running a local backup of a server with cloud tiering (it copies pointers).
9. Installing the File Sync agent and then cloning or running sysprep on the server.

Cheatsheet AZ-104 — 07 · ARM templates and Bicep: structure, deployment, export and conversion

1. Anatomy of an ARM template (JSON)

Section	Required	What it contains
`$schema`	Yes	Schema URL according to the scope: `deploymentTemplate.json#` (resource group), `subscriptionDeploymentTemplate.json#`, `managementGroupDeploymentTemplate.json#`, `tenantDeploymentTemplate.json#`
`contentVersion`	Yes	Template version (`1.0.0.0`); informational only
`parameters`	No	Values requested at deployment: `type` (string, int, bool, object, array, **secureString**, secureObject), `defaultValue`, `allowedValues`, `minLength`, `maxLength`, `minValue`, `maxValue`, `metadata.description`
`variables`	No	Reusable computed values
`functions`	No	User-defined functions (rare)
`resources`	Yes	List of resources: `type`, `apiVersion`, `name`, `location`, `properties`, `dependsOn`, `condition`, `copy`, `tags`, `sku`, `kind`
`outputs`	No	Values returned on completion (IDs, FQDN, keys)

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "prefix": { "type": "string", "minLength": 3, "maxLength": 11 },
    "location": { "type": "string", "defaultValue": "[resourceGroup().location]" },
    "sku": { "type": "string", "defaultValue": "Standard_LRS", "allowedValues": [ "Standard_LRS", "Standard_GRS" ] }
  },
  "variables": { "stName": "[toLower(concat(parameters('prefix'), uniqueString(resourceGroup().id)))]" },
  "resources": [ {
    "type": "Microsoft.Storage/storageAccounts", "apiVersion": "2023-05-01",
    "name": "[variables('stName')]", "location": "[parameters('location')]",
    "sku": { "name": "[parameters('sku')]" }, "kind": "StorageV2",
    "properties": { "supportsHttpsTrafficOnly": true }
  } ],
  "outputs": { "stId": { "type": "string", "value": "[resourceId('Microsoft.Storage/storageAccounts', variables('stName'))]" } }
}
```

- **Expressions** go between square brackets `[ ]`. Functions that show up: `parameters()`, `variables()`, `resourceGroup().location`, `resourceGroup().id`, `subscription().subscriptionId`, `uniqueString()` (deterministic hash for unique names), `concat()` / `format()`, `toLower()`, `resourceId()`, `reference()` (runtime properties of a resource), `listKeys()` (storage keys), `copyIndex()`, `utcNow()` (only in a parameter's defaultValue), `deployment().name`.
- **dependsOn** sets the order when one resource needs another (NIC → VNet, VM → NIC); `reference()` and `listKeys()` **by name** to a resource in the same template create an implicit dependency; `resourceId()` does not (add `dependsOn`). **copy** deploys N instances (`"copy": { "name": "vms", "count": 3 }` + `copyIndex()`); **condition** deploys only if it is met (`"condition": "[equals(parameters('createPip'), 'yes')]"`).
- Secrets: `secureString` parameter and, in the **parameter file**, a reference to **Key Vault** (`"reference": { "keyVault": { "id": "..." }, "secretName": "adminPassword" }`); the vault needs **"Enabled for template deployment"** and the user needs the `Microsoft.KeyVault/vaults/deploy/action` permission.

2. Bicep: the same, shorter

```
targetScope = 'resourceGroup'                    // subscription | managementGroup | tenant

@minLength(3)
param prefix string
param location string = resourceGroup().location
@allowed([ 'Standard_LRS', 'Standard_GRS' ])
param sku string = 'Standard_LRS'
@secure()
param adminPassword string
var stName = toLower('${prefix}${uniqueString(resourceGroup().id)}')
resource st 'Microsoft.Storage/storageAccounts@2023-05-01' = {
  name: stName
  location: location
  sku: { name: sku }
  kind: 'StorageV2'
  properties: { supportsHttpsTrafficOnly: true }
}

resource vnet 'Microsoft.Network/virtualNetworks@2024-01-01' existing = { name: 'vnet-prod' }   // reference without creating
resource nic 'Microsoft.Network/networkInterfaces@2024-01-01' = [for i in range(0, 3): {
  name: 'nic-${i}'
  location: location
  properties: { ipConfigurations: [ { name: 'ipcfg', properties: { subnet: { id: vnet.properties.subnets[0].id } } } ] }
}]
module web './modules/webapp.bicep' = if (sku == 'Standard_GRS') { name: 'webDeploy', params: { location: location } }

output stId string = st.id
```

- **No explicit dependsOn**: Bicep infers dependencies from the **symbolic names** (`st.id`, `vnet.properties...`). Types and API versions with autocompletion in VS Code (Bicep extension).
- Keywords: `param`, `var`, `resource`, `module` (another .bicep or a template spec / registry), `output`, `existing`, `targetScope`, decorators `@secure()`, `@allowed()`, `@description()`, `@minLength`; loops `[for x in list: {...}]`, conditions `= if (...) {...}`, interpolation `'${a}-${b}'`, `?:` operator.
- Tools: `az bicep install` / `az bicep upgrade` / `az bicep version` · `az bicep build --file main.bicep` (Bicep → JSON) · `az bicep decompile --file template.json` (JSON → Bicep; review the result) · `bicep build` / `bicep decompile` if you have the Bicep CLI. Parameters in `main.bicepparam` (`using 'main.bicep'` + `param prefix = 'contoso'`) or in the classic `.parameters.json`.

3. Deploying

Scope	Azure CLI	PowerShell	Purpose
Resource group	`az deployment group create -g RG1 --template-file main.bicep --parameters main.bicepparam` · `--parameters @main.parameters.json` · `--parameters prefix=contoso`	`New-AzResourceGroupDeployment -ResourceGroupName RG1 -TemplateFile main.bicep -TemplateParameterFile main.parameters.json` · `-prefix contoso`	The usual: resources inside an RG
Subscription	`az deployment sub create -l westeurope --template-file main.bicep`	`New-AzSubscriptionDeployment -Location westeurope -TemplateFile main.bicep` (alias `New-AzDeployment`)	Create RGs, Policy, RBAC at subscription level
Management group	`az deployment mg create -m mg-prod -l westeurope --template-file main.bicep`	`New-AzManagementGroupDeployment -ManagementGroupId mg-prod -Location westeurope ...`	Policy / RBAC for many subscriptions
Tenant	`az deployment tenant create -l westeurope ...`	`New-AzTenantDeployment ...`	Management groups

Option	Detail
Incremental mode (default)	Creates or updates what is in the template; **does not touch** what already exists in the RG and is not listed
Complete mode	Additionally **deletes** the RG resources that are **not** in the template (RG scope only; `--mode Complete` / `-Mode Complete`). "Ensure the RG contains only what the template defines" → Complete. Dangerous: validate first with what-if
What-if	`az deployment group what-if ...` / `--confirm-with-what-if` · `New-AzResourceGroupDeployment -WhatIf` → lists Create / Modify / Delete / NoChange before applying
Validate	`az deployment group validate ...` · `Test-AzResourceGroupDeployment` → syntax and schema, not permissions or quotas
History	Resource group → **Deployments**: inputs, outputs, template, errors; you can **Redeploy**. Maximum **800** per RG (auto-deleted when approaching the limit; a CanNotDelete lock prevents it and deployments fail)
Permissions	Contributor at the scope (`Microsoft.Resources/deployments/write` + the resources); to assign roles or Policy from the template, Owner / UAA / Resource Policy Contributor
Deploying from the portal	"Deploy a custom template" → edit, upload a file or pick a **Quickstart template**; any resource has **Automation options** / "Download a template for automation" before creating it

- The same deployment (same name) can be repeated: it is **idempotent**. Repeated deployment names overwrite the history entry.

4. Export, reuse and organize

What	How	Watch out
Export template of an RG or resource	RG → Export template · resource → Export template · `az group export -n RG1` · `Export-AzResourceGroup -ResourceGroupName RG1 -Path .`	Reflects the **current state**, may miss some property or include hardcoded values: review and parameterize. Does not export secrets
Template of a deployment	RG → Deployments → deployment → **Template**	It is **exactly** the template and parameters that were used
Convert JSON → Bicep	`az bicep decompile --file template.json`	Best from a clean template; warns about what it cannot convert
Template specs	`az ts create --name tsStorage --version 1.0 -g RG1 -l westeurope --template-file main.bicep` · `New-AzTemplateSpec` · deploy: `az deployment group create --template-spec <id>` · `New-AzResourceGroupDeployment -TemplateSpecId <id>`	Template **versioned as an Azure resource**, shared with RBAC (Template Spec Reader) with no repository or SAS
Deployment stacks	`az stack group create --name s1 -g RG1 --template-file main.bicep --action-on-unmanage deleteAll --deny-settings-mode denyDelete`	Manage the lifecycle of what was deployed (delete what leaves the template, deny settings); they replace **Blueprints** (retired)
Nested / linked templates	Nested: template embedded in `Microsoft.Resources/deployments`. Linked: `templateLink.uri` to an **accessible** URL (storage + SAS, GitHub) or `relativePath` with template specs	For modularizing; in Bicep, **modules** are used
Quickstart templates	`azure/azure-quickstart-templates` gallery	Starting point; deployed from the portal with "Deploy to Azure"

5. Common errors and what they mean

Error	Cause	Fix
`InvalidTemplate`	Syntax, malformed expression, parameter without a value	Validate / what-if; check brackets and quotes
`ResourceNotFound` when deploying	Missing `dependsOn` or reference to something that does not exist yet	Add the dependency (in Bicep, use the symbolic name)
`DeploymentQuotaExceeded`	800 deployments in the history	Delete old entries (`az deployment group delete`) or remove the lock that prevents cleaning them up
`RequestDisallowedByPolicy`	An Azure Policy blocks the resource (location, SKU, tag)	Comply with the policy or request an exemption
`SkuNotAvailable` / `OperationNotAllowed` (quota)	SKU not available in the region or quota exhausted	Another region / size or increase the quota
`StorageAccountAlreadyTaken` / `Conflict`	Duplicate global name	`uniqueString()` in the name
`MissingSubscriptionRegistration`	Resource provider not registered	`az provider register --namespace Microsoft.X`
`NoRegisteredProviderFound` / invalid `apiVersion`	Nonexistent `apiVersion` for the type	Use a valid version (`az provider show -n Microsoft.Compute --query "resourceTypes[?resourceType=='virtualMachines'].apiVersions"`)
Resources deleted after deploying	**Complete mode** was used	Use Incremental or include everything in the template
Access denied to the Key Vault secret during deployment	Vault without "enabled for template deployment" or without the deploy/action permission	Enable the option and grant the permission

6. Commands that show up

Task	PowerShell (Az.Resources)	Azure CLI
Deploy to RG	`New-AzResourceGroupDeployment -ResourceGroupName RG1 -TemplateFile main.bicep -TemplateParameterFile main.parameters.json -Mode Incremental`	`az deployment group create -g RG1 --template-file main.bicep --parameters main.bicepparam --mode Incremental`
Inline parameter and URI	`-TemplateUri https://.../azuredeploy.json -TemplateParameterObject @{ prefix = "contoso" }`	`--template-uri https://.../azuredeploy.json --parameters prefix=contoso`
What-if / validate	`New-AzResourceGroupDeployment ... -WhatIf` · `Test-AzResourceGroupDeployment ...`	`az deployment group what-if ...` · `az deployment group validate ...`
Subscription	`New-AzSubscriptionDeployment -Location westeurope -TemplateFile main.bicep`	`az deployment sub create -l westeurope --template-file main.bicep`
History	`Get-AzResourceGroupDeployment -ResourceGroupName RG1` · `Get-AzResourceGroupDeploymentOperation` · `Remove-AzResourceGroupDeployment`	`az deployment group list -g RG1 -o table` · `az deployment operation group list` · `az deployment group delete`
Export	`Export-AzResourceGroup -ResourceGroupName RG1 -Path .` · `Save-AzResourceGroupDeploymentTemplate -ResourceGroupName RG1 -DeploymentName d1`	`az group export -n RG1 > rg.json` · `az deployment group export -g RG1 -n d1`
Bicep	`bicep build main.bicep` · `bicep decompile template.json`	`az bicep build --file main.bicep` · `az bicep decompile --file template.json` · `az bicep upgrade`
Template spec	`New-AzTemplateSpec -Name ts1 -Version 1.0 -ResourceGroupName RG1 -Location westeurope -TemplateFile main.bicep`	`az ts create --name ts1 --version 1.0 -g RG1 -l westeurope --template-file main.bicep`

7. Decision tree

```
What does the wording ask for?
├─ Deploy the same thing repeatedly without changes (idempotent) ..... ARM / Bicep in Incremental mode
├─ Leave the RG exactly as the template ............................... Complete mode (RG only)
├─ See what will change before applying ............................... What-if
├─ Reuse what already exists as a template ............................ Export template (RG / resource) or Deployments → Template
├─ Convert a JSON template to Bicep ................................... az bicep decompile
├─ Share versioned templates with the team ............................ Template specs (RBAC)
├─ Password in the template without exposing it ....................... secureString + Key Vault reference in parameters
├─ Unique name for a storage account .................................. uniqueString(resourceGroup().id)
├─ Three identical VMs ................................................ copy (ARM) / for (Bicep)
├─ Resource only in production ........................................ condition (ARM) / if (Bicep)
├─ Create resource groups or policies from the template ............... Subscription / management group scope deployment
├─ "DeploymentQuotaExceeded" .......................................... Delete history (800) or remove the CanNotDelete lock
└─ Replacement for Blueprints ......................................... Template specs + deployment stacks
```

8. Signals → answer

Wording	Answer
"deploy the template without affecting existing resources"	Incremental (default)
"resources not in the template must be deleted"	Complete
"preview changes before deploying"	`what-if` / `-WhatIf`
"template fails: resource not found"	Missing dependsOn (or reference to a resource not yet created)
"storage account name must be unique", "same template in several RGs"	`uniqueString()`
"password must not be stored in the template or parameter file"	secureString + Key Vault reference (enabled for template deployment)
"deploy N identical VMs"	copy / for loop
"VM deploys only when parameter is true"	condition / if
"reuse the exact template that created the resources"	RG → Deployments → Template
"export template is missing some properties"	Normal: the export reflects the state and not every property; use the deployment's template or complete it
"store templates in Azure and control access with RBAC"	Template specs
"convert Bicep to JSON for a tool that only reads ARM"	`az bicep build`
"deploy to a subscription scope", "create resource groups"	`az deployment sub create` / `New-AzSubscriptionDeployment`
"the deployment history is full"	Delete old deployments (800 limit)
"a resource lock prevents deployments"	CanNotDelete on the RG prevents cleaning up the history
"assign a tag / policy during deployment at subscription"	Subscription-level deployment with `Microsoft.Authorization/policyAssignments`

9. Mistakes to avoid

1. Using Complete mode in a shared RG: it deletes everything not in the template.
2. Forgetting `dependsOn` in JSON (in Bicep, symbolic names resolve it).
3. Putting passwords in clear text: `secureString` + Key Vault.
4. Hardcoding the region: `resourceGroup().location` or a parameter.
5. Expecting `validate` to detect quotas, policies or duplicate names: it only checks schema and syntax; `what-if` goes further.
6. Exporting an RG and deploying it as is without parameterizing names and reviewing properties.
7. Confusing template spec (versioned template) with deployment stack (lifecycle management) or with Blueprints (retired).
8. Deploying at RG level a template whose `$schema` is for subscription (or vice versa).

Cheatsheet AZ-104 — 08 · Virtual machines: creation, sizes, disks, encryption, moving and availability

1. Creating a VM: what each tab decides

Tab	Decisions	Exam details
Basics	Image (Marketplace, **Azure Compute Gallery**, your own VHD), **size**, **zone** or availability set, user and authentication (SSH / password), inbound ports	Gen 1 vs **Gen 2** (Gen 2: Trusted Launch, OS disks > 2 TiB, UEFI). **Trusted launch** (Secure Boot + vTPM) is the default for Gen 2
Disks	OS disk type, encryption, data disks, "Delete with VM"	Premium SSD requires a size with **"s"** (e.g. D4**s**_v5)
Networking	VNet, subnet, **public IP**, NSG (on the NIC), load balancer	An NSG on the NIC and another on the subnet are both evaluated. No public IP → Bastion
Management	Identity (managed identity), **Entra login**, auto-shutdown, **backup**, patch orchestration, hotpatch	"Sign in with Entra credentials" is enabled here (extension + Virtual Machine Administrator / User Login role)
Monitoring	**Boot diagnostics** (managed or your own account), guest OS metrics, alerts	Serial console needs boot diagnostics
Advanced	Extensions, **custom data / user data** (cloud-init), VM applications, dedicated host, proximity placement group, capacity reservation	cloud-init is Linux only and first boot only

- Creation generates **VM, OS disk, NIC, public IP, NSG** and, if none exists, a VNet. Deleting the VM **does not delete** the NIC, disks or PIP unless "Delete with VM" is checked. **VM names**: 15 characters on Windows, 64 on Linux; a VM cannot be **renamed** (recreate it).
- **Images:** Marketplace (some with a paid plan) · **Azure Compute Gallery** (image definitions with **versions**, generalized (sysprep / `waagent –deprovision`) or **specialized**, replicated to several regions and shareable with other tenants or the community) · managed image (one region, no versions) · **VM Image Builder** to build them. For "deploy 50 identical VMs in 3 regions from a custom image" → Compute Gallery.
- **Sizes:** family by letter: **B** burstable (credits; dev/test), **D** general purpose, **E** memory, **F** CPU, **L** local storage, **M** huge memory, **N** GPU, **H** HPC. Suffixes: **s** Premium Storage, **a** AMD, **p** ARM, **d** local temp disk, **l** low memory, **m** more memory, **i** isolated, **v5** generation. **vCPU quota per family and region** → "QuotaExceeded" is fixed by requesting an increase, not by changing subscription.
- **Resizing:** the VM **restarts**; if the new size is not listed it is not available in the current cluster → **Stop (deallocate)** and check again; in an **availability set** you may have to deallocate **all** the VMs in the set. Windows cannot go from a size with a temp disk to one without (Linux can). The **temp disk** contents are lost.

2. Disks

Type	Purpose	Max IOPS / MB/s	Max size	OS disk
Standard HDD	Backup, dev/test, low access	2,000 / 500	32 TiB	Yes (**to be retired** as OS disk in Sept. 2028)
Standard SSD	Web, light apps, consistency	6,000 / 750	32 TiB	Yes
Premium SSD	Production, low latency; fixed **P1-P80** sizes, bursting, performance tiers (bump P30 → P40 without resizing)	20,000 / 900	32 TiB	Yes (needs an "s" VM)
Premium SSD v2	Production with adjustable IOPS and throughput **without changing the size**, and cheaper	80,000 / 2,000	64 TiB	**No**; data only; **zonal**; no host caching; not in Compute Gallery
Ultra Disk	SAP HANA, top-tier databases; IOPS and MB/s configurable on the fly	400,000 / 10,000	64 TiB	**No**; data only; no caching; **not in availability sets**; LRS only

- Roles: **OS** (Windows 127 GiB by default, up to 4 TiB), **temporary** (D: / `/dev/sdb`, **lost** on deallocate, resize or redeploy: never data), **data** (maximum number depends on the VM size). **Host caching**: None / ReadOnly / ReadWrite (default OS ReadWrite, data ReadOnly; None for Ultra and v2).
- Operations: **attach** new or existing (a disk on only one VM except **shared disks** for clusters), **detach**, **expand** (never shrink; data disks online in many cases, OS with the VM deallocated), **change type** Standard ↔ Premium (VM **deallocated**; to v2 / Ultra only by creating a new one), **snapshot** (full or **incremental**, copied to another region; create a disk or VM from it), **swap OS disk**, **export** via SAS, **managed image**. A disk must be in the **same region and zone** as the VM.

Encryption	What it encrypts	Keys	Requirements
SSE with platform-managed keys (default, always)	OS and data at rest	Microsoft	None
SSE with customer-managed keys	OS and data	Your Key Vault via **Disk Encryption Set**	Key Vault with soft delete and purge protection; the DES needs access to the key
Encryption at host (recommended)	OS, data, **temp disk and caches**, and VM ↔ storage traffic	Platform or CMK (DES)	Register the `EncryptionAtHost` feature in the subscription, **VM deallocated** to enable it, compatible sizes, **incompatible with ADE** on the same VM
Azure Disk Encryption (ADE)	Inside the VM with **BitLocker** / **dm-crypt**; OS, data and temp	**Key Vault** (+ optional KEK), "Enabled for disk encryption"	Uses VM CPU; not on Basic/A-series or with < 2 GB RAM; no custom Linux images; **to be retired in Sept. 2028** → migrate to encryption at host
Confidential disk encryption	OS disk bound to the vTPM	DES	Confidential VMs only (DCasv5 / ECasv5)

- "Encrypt the temp disk and caches with minimal effort" → encryption at host. "Keys must be in our Key Vault" → CMK (DES) or ADE. "Defender says disk encryption unhealthy" → enable encryption at host or ADE.

3. Availability and SLA

Option	What it is	SLA	Rules
Single VM	No redundancy	99.9 % with Premium SSD / Ultra · 99.5 % Standard SSD · 95 % Standard HDD	The SLA depends on the **OS disk**
Availability set	Spreads VMs across **fault domains** (up to **3**: power and switch) and **update domains** (up to **20**, default 5: maintenance reboots) within a datacenter	**99.95 %** (≥ 2 VMs)	Chosen **when creating** the VM (cannot be added later); same region; aligned with managed disks; does not combine with zones; recommended when the region **has no zones**
Availability zones	**Zonal** VM (1, 2 or 3), separate datacenters with their own power and network	**99.99 %** (≥ 2 VMs in ≥ 2 zones)	**Standard** public IP and load balancer (zone-redundant); a VM is not "moved" to a zone: recreate from snapshot / Backup, zone-to-zone Site Recovery or Resource Mover
VMSS	Automatic scaling (cheatsheet 09)	99.95 % (FD) · 99.99 % (zones)	Flexible spreads across zones and FDs
Proximity placement group	VMs in the same datacenter for **minimum latency**	—	Reduces availability; a PPG can span only one availability zone
Dedicated host	A physical server just for you	99.99 % with ≥ 2 hosts	Compliance, per-host licensing, isolation
Capacity reservation	Guarantees capacity for a size in a region / zone	—	For DR or planned peaks
Spot VM	Spare capacity up to ~90 % cheaper, **evictable** with 30 s notice (deallocate or delete)	**No SLA**	Batch, test, rendering; never critical production

- "99.99 % for a two-tier app" → VMs of each tier in **2 or more zones** behind a Standard Load Balancer. "99.95 %, region without zones" → availability set. VMs in an availability set must be created with the set from the start: "add an existing VM to the set" = recreate it.

4. Moving a VM

Destination	How	Conditions
Another resource group	Move (portal / `Move-AzResource` / `az resource move`)	Move the VM with **NIC and disks**; the VNet can stay; resources not in Succeeded state block it; the resource's RBAC assignments are lost; no ReadOnly lock
Another subscription	Move with **all dependents in the same RG**: VM, disks, NIC, PIP, NSG and **VNet**	Same tenant; provider registered at the destination; quota; **stop backup** (Recovery Services) and ASR first
Another region	**Azure Resource Mover** (wizard: dependencies, prepare, initiate, commit) or **Site Recovery** (replicate and fail over)	There is no native cross-region "move"; copies are created and the source is deleted
A zone	Site Recovery (zone to zone), Resource Mover or snapshot → zonal disk → new VM	—
An availability set	Recreate the VM (same NIC and OS disk are fine)	An existing VM cannot be added
Another tenant	Transfer the **subscription** (RBAC is lost)	—

5. Day-to-day operations

Operation	What it does	Watch out
Stop (from the OS) vs **Stop (deallocate)** (portal / CLI)	OS shutdown: the VM keeps **billing** compute · Deallocate: releases the host, **no compute billing** (disks still billed)	Deallocate loses the temp disk and the **dynamic public IP** (use static if you need it fixed)
Restart	Reboot without changing host	—
Redeploy	Moves the VM to **another host** (RDP / host connectivity problems)	Loses the temp disk; the dynamic IP may change
Reapply	Reapplies the configuration when the state is "Failed"	—
Boot diagnostics	Screenshot and boot log	Managed (recommended) or your own storage account
Serial console	Console over the serial port, no network	Needs boot diagnostics; Windows requires SAC
Run command	Run PowerShell / shell **without RDP or network** (`az vm run-command invoke --command-id RunPowerShellScript --scripts "..."`)	Useful to repair firewall / RDP
Reset password / **Reset SSH key**	Help → Reset password (VMAccess extension)	Also "Reset configuration only" for RDP
Bastion	RDP / SSH through the browser without a public IP	Cheatsheets 09 and 13
Auto-shutdown	Shuts down at a set time (DevTest) with notification	Only shuts down; does not start (use Automation / Start-Stop v2 for that)
Azure Update Manager	OS patching; patch orchestration **AutomaticByPlatform**, hotpatch (Windows Server Azure Edition)	Replaces Automation's Update Management
Azure Hybrid Benefit	`--license-type Windows_Server` (or RHEL / SLES) to use your own licenses with Software Assurance	Up to 40 % on Windows; also SQL on VM
Hibernation	Saves RAM to the OS disk and deallocates	Specific sizes and OSes

6. Commands that show up

Task	PowerShell (Az.Compute)	Azure CLI
Quick VM create	`New-AzVM -ResourceGroupName RG1 -Name vm1 -Location westeurope -Image Win2022Datacenter -Size Standard_D2s_v5 -Credential (Get-Credential) -Zone 1 -OpenPorts 3389`	`az vm create -g RG1 -n vm1 --image Ubuntu2204 --size Standard_D2s_v5 --admin-username azureuser --generate-ssh-keys --zone 1 --nsg-rule SSH`
Create with detailed config	`New-AzVMConfig -VMName vm1 -VMSize ...` → `Set-AzVMOperatingSystem` → `Set-AzVMSourceImage` → `Add-AzVMNetworkInterface` → `Set-AzVMOSDisk` → `New-AzVM -VM $vm`	`az vm create ... --vnet-name vnet1 --subnet sub1 --public-ip-address "" --availability-set as1 --os-disk-size-gb 256 --data-disk-sizes-gb 128 --license-type Windows_Server`
Status and sizes	`Get-AzVM -Status` · `Get-AzVMSize -Location westeurope` · `Get-AzVMSize -ResourceGroupName RG1 -VMName vm1` (those in the cluster)	`az vm list -d -o table` · `az vm list-sizes -l westeurope` · `az vm list-vm-resize-options -g RG1 -n vm1`
Start, stop, redeploy	`Start-AzVM` · `Stop-AzVM -Force` (deallocates) · `Stop-AzVM -StayProvisioned` (shuts down only) · `Restart-AzVM` · `Set-AzVM -Redeploy`	`az vm start` · `az vm deallocate` · `az vm stop` (shuts down without releasing) · `az vm restart` · `az vm redeploy`
Resize	`$vm = Get-AzVM ...; $vm.HardwareProfile.VmSize = "Standard_D4s_v5"; Update-AzVM -VM $vm -ResourceGroupName RG1`	`az vm resize -g RG1 -n vm1 --size Standard_D4s_v5`
Disks	`Add-AzVMDataDisk -VM $vm -Name d1 -Lun 0 -CreateOption Empty -DiskSizeInGB 256 -StorageAccountType Premium_LRS; Update-AzVM` · `New-AzSnapshot` · `New-AzDiskUpdateConfig -DiskSizeGB 512` + `Update-AzDisk`	`az vm disk attach -g RG1 --vm-name vm1 --name d1 --new --size-gb 256 --sku Premium_LRS` · `az vm disk detach` · `az snapshot create --source d1` · `az disk update -n d1 --size-gb 512` · `az disk update --sku Premium_LRS`
Encryption	`Set-AzVMDiskEncryptionExtension -ResourceGroupName RG1 -VMName vm1 -DiskEncryptionKeyVaultUrl ... -DiskEncryptionKeyVaultId ...` (ADE) · `Update-AzVM ... -EncryptionAtHost $true`	`az vm encryption enable -g RG1 -n vm1 --disk-encryption-keyvault kv1` (ADE) · `az vm update -g RG1 -n vm1 --set securityProfile.encryptionAtHost=true`
Availability set	`New-AzAvailabilitySet -ResourceGroupName RG1 -Name as1 -Location westeurope -PlatformFaultDomainCount 2 -PlatformUpdateDomainCount 5 -Sku Aligned`	`az vm availability-set create -g RG1 -n as1 --platform-fault-domain-count 2 --platform-update-domain-count 5`
Image and gallery	`New-AzGallery` · `New-AzGalleryImageDefinition` · `New-AzGalleryImageVersion -TargetRegion ...` · generalize: `Set-AzVm -Generalized`	`az sig create` · `az sig image-definition create` · `az sig image-version create --target-regions westeurope northeurope` · `az vm generalize`
Move	`Move-AzResource -DestinationResourceGroupName RG2 -ResourceId <vm>,<nic>,<disk>`	`az resource move --destination-group RG2 --ids <vm> <nic> <disk>`
Run command and access	`Invoke-AzVMRunCommand -ResourceGroupName RG1 -VMName vm1 -CommandId RunPowerShellScript -ScriptPath fix.ps1` · `Set-AzVMAccessExtension` (reset password)	`az vm run-command invoke -g RG1 -n vm1 --command-id RunShellScript --scripts "sudo apt update"` · `az vm user update -g RG1 -n vm1 -u azureuser -p <pwd>`

7. Decision tree

```
What does the wording ask for?
├─ 99.99 % with VMs ..................................................... Availability zones (≥ 2 zones) + Standard LB
├─ 99.95 %, or region without zones ..................................... Availability set (≥ 2 VMs)
├─ Minimum latency between VMs .......................................... Proximity placement group
├─ Dedicated physical server / compliance ............................... Dedicated host
├─ Interruptible workload at minimum cost ............................... Spot VM
├─ Very fast OS disk .................................................... Premium SSD (size with "s")
├─ Adjustable IOPS without resizing, data ............................... Premium SSD v2 (or Ultra if extreme)
├─ Encrypt temp disk and caches ......................................... Encryption at host
├─ BitLocker / dm-crypt with Key Vault .................................. Azure Disk Encryption (retired 2028)
├─ Your own keys for the disks without touching the VM .................. SSE with CMK (Disk Encryption Set)
├─ Size not listed when resizing ........................................ Deallocate and retry
├─ Move to another region ............................................... Azure Resource Mover / Site Recovery
├─ Move to another subscription ......................................... Move with VM + disks + NIC + VNet in the same RG
├─ Put an existing VM into an availability set ........................... Recreate the VM
├─ Host problem, RDP not responding ..................................... Redeploy (or Run command / Serial console)
└─ No compute charges while stopped ..................................... Stop (deallocate), with a static IP if needed
```

8. Signals → answer

Wording	Answer
"VM stopped from the OS still incurs compute charges"	Stop (deallocate) from Azure
"public IP changes after each restart"	**Static** public IP (Standard)
"data on the temporary disk was lost"	Normal: D: / sdb is lost on deallocate, resize or redeploy
"the new size is not listed"	Deallocate first (cluster)
"resize a VM in an availability set"	May require deallocating all the VMs in the set
"attach Premium SSD fails"	The size is not "s" (Premium Storage capable)
"Ultra disk / Premium SSD v2 as OS disk"	Impossible: data only
"encrypt temp disk", "end-to-end encryption"	Encryption at host (feature registered, VM deallocated)
"ADE and encryption at host on the same VM"	Incompatible
"Key Vault for disk encryption not enabled"	Enable "Enabled for disk encryption" (ADE)

Wording	Answer
"deploy the same custom image to several regions"	Azure Compute Gallery with version replication
"generalize the VM before capturing"	sysprep /generalize (Windows) · `waagent -deprovision+user` (Linux) → `az vm generalize`
"specialized image keeps the computer name and accounts"	Specialized image in the gallery
"move VM to another subscription" fails	Dependents not moved (VNet, disks) or backup / ASR active
"move VM to another region"	Resource Mover (or Site Recovery)
"add VM to availability zone after creation"	Recreate (snapshot) or zone-to-zone Site Recovery
"Windows VM cannot change to a size without temp disk"	Limitation: create a new VM from a snapshot
"use existing Windows Server licenses"	Azure Hybrid Benefit (`--license-type Windows_Server`)
"connect without public IP"	Bastion (or Run command / Serial console to repair)
"boot problem, see the screen"	Boot diagnostics screenshot + Serial console

9. Mistakes to avoid

1. Storing data on the temp disk.
2. Choosing a size without "s" and expecting Premium disks.
3. Assuming a VM can be added to an availability set or a zone after creation.
4. Confusing the OS Stop with Stop (deallocate): only the latter stops billing compute.
5. Combining ADE with encryption at host, or choosing ADE for new projects (retired in 2028).
6. Moving to another subscription without the dependent resources in the same RG, or with backup active.
7. Expecting a native cross-region "move": it is Resource Mover or Site Recovery.
8. Using Spot for production, or Standard HDD as OS disk in new designs.
9. Forgetting that resizing restarts the VM and that a Premium SSD v2 or Ultra disk is zonal and has no cache.

Cheatsheet AZ-104 — 09 · VM Scale Sets, autoscale, extensions and VM access

1. VMSS: Flexible vs Uniform

	Flexible (recommended, default)	**Uniform**
VM type	**Standard** VMs (Microsoft.Compute/virtualMachines): RBAC, tags, Backup, Site Recovery, normal API	Child VMs of the scale set (own API); **no Backup or ASR**
Maximum instances	**1,000**	100 with FD guarantee (up to 3,000 with multiple placement groups)
Mixing	Sizes, **Linux and Windows**, **Spot + regular** in the same set; add existing VMs	All identical (same image, size and priority)
Zones and fault domains	Spreads across 1-3 zones; max FD 3 (1 if zonal); **no update domains**; zone / FD can be pinned per VM	Zones 1-3; FD 2, 3 or 5; **5 update domains**
Scaling and health	Autoscale, scale-in policy, instance protection, instance repair (with **Application Health extension**), upgrade policy Automatic / Rolling / Manual	Same; plus **automatic OS image upgrades** (still preview on Flexible) and health via **LB health probe**; **overprovisioning**
Networking	**Standard Load Balancer** / App Gateway only; **no outbound internet by default** (NAT Gateway, LB outbound rule or PIP)	Supported Basic LB (retired in Sept. 2025) and NAT pool
Identity	**User-assigned** managed identity only	System or user-assigned
SLA	99.95 % in FDs · **99.99 %** across several zones	Same

- The mode is chosen **at creation** and cannot be changed. "Mixed VM sizes", "Spot and regular", "attach an existing VM" → Flexible. "Service Fabric / AKS node pools", "automatic OS upgrades", "overprovisioning" → Uniform.
- New instances come from the model's **image** (Marketplace or **Compute Gallery**); initial configuration is done with **extensions** (Custom Script), cloud-init or a pre-built image.

2. Autoscale (VMSS and App Service)

Element	**What it is**	**Detail**
Profile (scale condition)	Up to **20** per resource: **Default** (one), **Recurrence** (days and hours), **Fixed date** (specific event)	Evaluation order: fixed date → recurrence → default
Capacity	**Minimum / Maximum / Default**	Default = what it falls back to if it **cannot read the metric** (never scales below it in that case)
Rule	Up to **10** per profile; metric (Percentage CPU, Available Memory Bytes, Network In/Out, Disk, Storage / Service Bus **queue length**, custom metrics), statistic, **duration** (window, e.g. 10 min), **time grain** (1 min), operator and threshold, action (**increase / decrease** count, percent or exact), **cooldown** (default **5 min**, per rule)	One scale-out rule and one scale-in rule; "> 75 % for 10 min → +2; < 25 % → −1"
Evaluation	Every 30-60 s; if several scale-out rules fire, the **largest** capacity wins; it only **scales in** if **all** scale-in rules are met; avoids **flapping** (does not scale in if that would trigger a scale-out)	Cooldown is counted per rule from the last change
Predictive autoscale	Scales ahead of time with ML on CPU (VMSS only)	"scale before the daily peak"
Scale-in policy	**Default** (balances zones, then FDs, deletes the highest ID), **NewestVM**, **OldestVM**	"remove the oldest instances first"
Instance protection	Protect an instance from scale-in or from all set operations	Keep a diagnostics VM
Notifications	Email to admins / co-admins / addresses, **webhook**	—

- Manual: set the instance count (`az vmss scale --new-capacity 5`). Scheduled: recurrence profile (business hours 10 instances, night 2). External metric: Storage queue → `dividePerInstance`.
- In **App Service** autoscale belongs to the **plan** (affects all its apps) and needs Standard or higher; Premium v2/v3 also has managed **automatic scaling** (cheatsheet 11).

3. Updating the model and keeping instances healthy

Piece	**What it does**
Upgrade policy	**Manual**: you change the model and apply it yourself instance by instance (`az vmss update-instances --instance-ids "*"`) · **Automatic**: all at once, no order (downtime) · **Rolling**: in batches (max %, max unhealthy %, pause) and requires a **health probe** (Uniform) or the **Application Health extension**
Automatic OS image upgrades	Uniform: new version of the Marketplace / gallery image → automatic rolling upgrade; requires health configured
Automatic VM guest patching	Flexible (and standalone VMs): patches inside the OS with platform orchestration
Instance repair	Replaces unhealthy instances (grace period); needs the Application Health extension or a probe
Reimage / Upgrade	Reimage reinstalls the instance from the image; Upgrade applies the **latest model** ("Latest model: No" = pending)
Terminate notification	Notice (up to 15 min) before deleting an instance so it can be drained

4. VM extensions

Extension	Purpose	Exam details
Custom Script Extension (Microsoft.Compute / CustomScriptExtension; Linux: Microsoft.Azure.Extensions / CustomScript)	Download and run a script **after** deployment (install IIS, configure)	`fileUris` + `commandToExecute`; downloads from Storage (key, SAS or **managed identity**) or GitHub; **90 min** maximum; runs **once** (to rerun: change `timestamp` or `forceUpdateTag`); **only one** per VM; secrets in `protectedSettings`; logs in `C:\WindowsAzure\Logs\Plugins\...` or `/var/log/azure/`; do not reboot inside the script; runs as SYSTEM / root
cloud-init (custom data)	Configuration on the **first boot** of Linux (packages, users, files)	Linux only, passed with `--custom-data`, runs **before** the extensions; **user data** (both OSes) is queried from IMDS but is not secret
Azure Monitor Agent (AMA)	Collect OS metrics and logs for Log Analytics / Monitor with **Data Collection Rules**	Replaces the Log Analytics agent (MMA, retired Aug. 2024); needs managed identity; VM insights = AMA + **Dependency agent**
Network Watcher agent	Packet capture, connection troubleshoot, Connection monitor	Installed automatically when using those tools
VMAccess	Reset password, user, SSH keys, RDP	Help → Reset password; `az vm user update`
AADLoginForWindows / AADSSHLoginForLinux	Sign in with **Entra ID**	Plus the Virtual Machine Administrator Login / User Login roles; MFA via Conditional Access
Microsoft Antimalware	Antivirus on Windows	Defender for Endpoint via Defender for Cloud is the modern option
Key Vault VM extension	Pull certificates from Key Vault and renew them	—
DSC / Machine Configuration	Desired state (DSC being retired → **Azure Machine Configuration** with Azure Policy)	"audit a setting inside the VM" → guest configuration policy
Azure Disk Encryption	BitLocker / dm-crypt (cheatsheet 08)	—

- States: Provisioning succeeded / **transitioning** (in progress) / **failed** (check the logs; cause no. 1: script error or URL unreachable because of an NSG / storage firewall). Extensions need the **VM Agent** (included in Marketplace images). In VMSS, better to deploy the extension with CLI / PowerShell / template and managed identity: the portal's SAS expires and breaks scale-out.

5. Automation around VMs

Tool	Purpose	Signal
Run command	One-off script without network or RDP (`az vm run-command invoke`)	"cannot RDP, fix the firewall"
Azure Automation	PowerShell / Python runbooks with **schedules**, webhooks, **Hybrid Runbook Worker** for on-prem; State Configuration (being retired)	"run a script every night", "stop VMs at 19:00"
Start/Stop VMs v2	Functions-based solution to start / stop by schedule or CPU	"start VMs at 8:00 and stop at 19:00"
Auto-shutdown	Only shuts down at a set time, with notification	"shut down dev VMs daily"
Azure Update Manager	Patching, maintenance schedules, patch orchestration	"install updates on a schedule"
Azure Policy (guest configuration)	Audit / enforce configuration inside the OS	"audit VMs where a service is disabled"
VM Image Builder	Build custom images in a pipeline	"build images automatically"

6. Accessing VMs

Method	How	Detail
RDP 3389 / SSH 22 over public IP	NSG with a rule from specific IPs; **Just-in-time** (Defender for Cloud) opens the port only on request and for a limited time	Avoid "Any" as the source
Azure Bastion	RDP / SSH **in the browser** (port 443) with no public IP on the VM; **AzureBastionSubnet /26 or larger** subnet, **Standard static** public IP; reaches VMs in **peered VNets**	SKUs: **Developer** (free, no subnet, limited) · **Basic** (2 instances) · **Standard** (host scaling 2-50, **native client** mstsc / ssh, **file transfer**, **shareable links**, custom ports, connection by IP) · **Premium** (+ **session recording**, private-only)
Serial console	Console without network	Boot diagnostics enabled
Entra ID login	Extension + role; MFA with Conditional Access	"no local admin accounts", "require MFA to RDP"
VPN / ExpressRoute	RDP to the **private** IP from on-prem	Cheatsheet 13
Run command / VMAccess	Repair without logging in	—

7. Commands that show up

Task	PowerShell	Azure CLI
Create VMSS	`New-AzVmss -ResourceGroupName RG1 -VMScaleSetName vmss1 -Location westeurope -OrchestrationMode Flexible -ImageName Ubuntu2204 -InstanceCount 2 -Zone 1,2,3 -UpgradePolicyMode Rolling`	`az vmss create -g RG1 -n vmss1 --image Ubuntu2204 --orchestration-mode Flexible --instance-count 2 --zones 1 2 3 --vm-sku Standard_D2s_v5 --upgrade-policy-mode Rolling --lb lb1 --admin-username azureuser --generate-ssh-keys`
Scale manually	`Update-AzVmss -ResourceGroupName RG1 -VMScaleSetName vmss1 -SkuCapacity 5`	`az vmss scale -g RG1 -n vmss1 --new-capacity 5`
Autoscale	`$r = New-AzAutoscaleScaleRuleObject -MetricTriggerMetricName "Percentage CPU" -MetricTriggerMetricResourceUri <vmss id> -MetricTriggerOperator GreaterThan -MetricTriggerStatistic Average -MetricTriggerThreshold 75 -MetricTriggerTimeAggregation Average -MetricTriggerTimeGrain (New-TimeSpan -Minutes 1) -MetricTriggerTimeWindow (New-TimeSpan -Minutes 10) -ScaleActionDirection Increase -ScaleActionType ChangeCount -ScaleActionValue 1 -ScaleActionCooldown (New-TimeSpan -Minutes 5) → $p = New-AzAutoscaleProfileObject -Name p1 -CapacityDefault 2 -CapacityMinimum 2 -CapacityMaximum 10 -Rule $r → New-AzAutoscaleSetting -Name as1 -ResourceGroupName RG1 -Location westeurope -TargetResourceId <vmss id> -Profile $p`	`az monitor autoscale create -g RG1 --resource vmss1 --resource-type Microsoft.Compute/virtualMachineScaleSets --min-count 2 --max-count 10 --count 2 · az monitor autoscale rule create -g RG1 --autoscale-name vmss1 --condition "Percentage CPU > 75 avg 10m" --scale out 2 · --condition "Percentage CPU < 25 avg 10m" --scale in 1`
Apply model	`Update-AzVmssInstance -ResourceGroupName RG1 -VMScaleSetName vmss1 -InstanceId "*"`	`az vmss update-instances -g RG1 -n vmss1 --instance-ids "*" · az vmss update -g RG1 -n vmss1 --set upgradePolicy.mode=Rolling`
Custom Script	`Set-AzVMCustomScriptExtension -ResourceGroupName RG1 -VMName vm1 -Location westeurope -FileUri https://.../install.ps1 -Run install.ps1 -Name cse · VMSS: Add-AzVmssExtension`	`az vm extension set -g RG1 --vm-name vm1 --name CustomScriptExtension --publisher Microsoft.Compute --settings '{"fileUris":["https://.../install.ps1"],"commandToExecute":"powershell -ExecutionPolicy Unrestricted -File install.ps1"}' · Linux: --name CustomScript --publisher Microsoft.Azure.Extensions · VMSS: az vmss extension set`
List extensions	`Get-AzVMExtension -ResourceGroupName RG1 -VMName vm1`	`az vm extension list -g RG1 --vm-name vm1 -o table · az vm get-instance-view`
Entra login	`Set-AzVMExtension -Publisher Microsoft.Azure.ActiveDirectory -ExtensionType AADLoginForWindows -Name AADLogin ... + New-AzRoleAssignment -RoleDefinitionName "Virtual Machine Administrator Login"`	`az vm extension set --publisher Microsoft.Azure.ActiveDirectory --name AADSSHLoginForLinux ... + az role assignment create --role "Virtual Machine User Login" --assignee ... --scope <vm id>`
Bastion	`New-AzBastion -ResourceGroupName RG1 -Name bas1 -PublicIpAddressRgName RG1 -PublicIpAddressName pip-bas -VirtualNetworkRgName RG1 -VirtualNetworkName vnet1 -Sku Standard`	`az network bastion create -g RG1 -n bas1 --public-ip-address pip-bas --vnet-name vnet1 --sku Standard -l westeurope · native client: az network bastion rdp --name bas1 -g RG1 --target-resource-id <vm id>`
Run command	`Invoke-AzVMRunCommand -ResourceGroupName RG1 -VMName vm1 -CommandId RunPowerShellScript -ScriptString "Set-NetFirewallRule ..."`	`az vm run-command invoke -g RG1 -n vm1 --command-id RunPowerShellScript --scripts "Get-Service"`

8. Decision tree

```
What does the wording ask for?
├─ Scale identical VMs automatically ..... VMSS (Flexible) + autoscale
├─ Mix sizes, Spot and regular, or add existing VMs ..... VMSS Flexible
├─ Automatic OS image upgrades or overprovisioning ..... VMSS Uniform
├─ Scale by schedule (office hours / night) ..... Autoscale profile Recurrence
├─ Scale by message queue ..... Rule with Storage Queue / Service Bus metric
├─ Prevent a specific instance from being deleted ..... Instance protection
├─ Update without downtime ..... Upgrade policy Rolling + Application Health extension
├─ Install software after deployment ..... Custom Script Extension (or cloud-init on Linux)
├─ Run the script again ..... Change timestamp / forceUpdateTag
├─ Collect OS logs into Log Analytics ..... Azure Monitor Agent + Data Collection Rule
├─ Reset the admin password ..... VMAccess (Help → Reset password)
├─ RDP without a public IP ..... Azure Bastion (native client → Standard)
├─ Record RDP sessions ..... Bastion Premium
├─ Sign in with Entra ID and MFA ..... AADLogin extension + VM Admin/User Login role + CA
├─ Shut down dev VMs every night ..... Auto-shutdown (or Start/Stop v2 if they must start too)
└─ Repair the VM without network ..... Run command / Serial console
```

9. Signals → answer

Wording	Answer
"scale out when CPU exceeds 70 % for 10 minutes"	Autoscale rule: Percentage CPU > 70, duration 10 min, increase
"instances keep scaling in and out"	Flapping: separate the thresholds, raise the cooldown
"autoscale uses default count"	It could not read the metric (agent / diagnostics)
"more instances on weekdays 8-18"	Recurrence profile
"mix Spot and regular instances"	Flexible
"scale set with Basic Load Balancer"	Uniform only (and Basic LB is retired: use Standard)

Wording	Answer
"new instances don't have the app installed"	Custom Script Extension in the VMSS model (or gallery image)
"script ran once, needs to run again"	timestamp / forceUpdateTag
"extension failed: cannot download the script"	NSG / storage firewall is blocking; use managed identity or open it
"script takes longer than 90 minutes"	Fails: split it or use Automation
"configure Linux VM at first boot"	cloud-init (custom data)
"collect Windows event logs to Log Analytics"	AMA + DCR
"Log Analytics agent", "MMA"	Retired: migrate to AMA
"RDP to VMs without public IPs from the portal"	Bastion
"Bastion subnet /27 fails"	AzureBastionSubnet minimum **/26**
"connect with mstsc through Bastion"	Standard SKU (native client)
"copy files through Bastion"	Standard (file transfer)
"VM in a peered VNet through Bastion"	Supported: one Bastion in the hub
"update the scale set model but instances unchanged"	Upgrade policy Manual: `update-instances`
"prevent the old instance from being removed"	Instance protection

10. Mistakes to avoid

1. Choosing Uniform when mixed sizes or Spot are required, or Flexible when automatic OS upgrades are required.
2. Setting a single autoscale rule (without the scale-in one) or overlapping thresholds (flapping).
3. Expecting a Custom Script Extension to run on every boot or twice without changing the timestamp.
4. Putting secrets in `settings` instead of `protectedSettings`.
5. Deploying the extension on a VMSS with a portal SAS that expires.
6. Creating AzureBastionSubnet with /27 or with another name, or expecting the native client on Basic.
7. Using the Log Analytics agent (MMA) instead of AMA.
8. Assuming auto-shutdown also starts the VMs.

Cheatsheet AZ-104 — 10 · Containers: Container Registry, Container Instances and Container Apps

1. Choosing the service

Service	What it is	When	Not suitable if
Azure Container Instances (ACI)	**Standalone** containers (container groups) without an orchestrator, billed per second	One-off tasks, batch, testing, **a quick container**, AKS bursting (virtual nodes)	You need **autoscale**, load balancing across replicas, revisions or managed HTTP ingress
Azure Container Apps (ACA)	**Serverless** microservices and HTTP / event-driven apps on managed Kubernetes (KEDA, Dapr, Envoy), **scales to zero**	APIs, containerized web apps, queue workers, jobs; no cluster to manage	You need access to the Kubernetes API, Windows containers or node control
AKS	Full Kubernetes	Full control, operators, complex workloads	You want something simple (AZ-104 barely asks about it)
App Service (Web App for Containers)	Web PaaS with your own image	Classic containerized web app with slots, domains, TLS	Complex multi-container, scale to zero
Azure Container Registry (ACR)	**Private** image registry (Docker / OCI)	Whenever you use the above with your own images	—

2. Azure Container Registry

Aspect	Basic	Standard	Premium
Included / maximum storage	10 GiB / 40 TiB	100 GiB / 40 TiB	**500 GiB** / 100 TiB
Webhooks	2	10	500
Geo-replication (one registry, several regions)	—	—	**Yes**
Private Link (private endpoints), IP rules, service endpoints	—	—	**Yes**
Content trust (signed images), **customer-managed keys**, **retention policy** (untagged manifests), dedicated data endpoints, connected registries	—	—	**Yes**
Anonymous pull, artifact cache	—	Yes	Yes
Zone redundancy, tokens and scope maps, Entra RBAC per repository	Yes	Yes	Yes
Performance (r/m, concurrency)	Low	Medium	High

- Name: 5-50 alphanumeric characters, **global**; login server `<name>.azurecr.io`. Changing SKU has no downtime (downgrading from Premium requires removing geo-replicas).
- **Authentication:** Entra ID with the **AcrPull** (pull only), **AcrPush** (push and pull), **AcrDelete**, **AcrImageSigner** roles; **managed identity** for AKS / ACA / App Service / VMs; **service principal** for CI/CD; **admin user** (two passwords; disabled by default; testing only); tokens with scope maps for specific repositories. "Least privilege for AKS to pull" → AcrPull to the cluster identity.
- **ACR Tasks:** `az acr build -r acr1 -t app:v1 .` builds in the cloud **without local Docker**; automatic tasks on Git **commit**, on **base image update** or on a schedule; multi-step tasks (YAML); `az acr run`. **Import** from Docker Hub or another registry: `az acr import -n acr1 --source docker.io/library/nginx:latest -t nginx:latest`.
- Operate: `az acr login -n acr1` → `docker tag app acr1.azurecr.io/app:v1` → `docker push acr1.azurecr.io/app:v1` · `az acr repository list -n acr1` · `az acr repository show-tags -n acr1 --repository app` · `az acr repository delete -n acr1 --image app:v1` · `az acr show-usage`. **Defender for Containers** scans for vulnerabilities. **Image lock** (`az acr repository update --write-enabled false`) prevents overwriting or deleting a tag.

3. Azure Container Instances

Aspect	Detail
Container group	Deployment unit: one or more containers (up to 60, **multi-container Linux only**) sharing lifecycle, network (same IP and ports) and volumes; sidecar pattern; **init containers**
Resources	**CPU and memory** are set per container (`--cpu 1 --memory 1.5`); the group can reach tens of vCPUs, but above 4 vCPU / 16 GB it depends on regional capacity; billed **per second** of vCPU and GB
Images	Linux (x64) and **Windows**; from ACR with `--registry-login-server --registry-username --registry-password` or **managed identity** (`--acr-identity`)
Networking	**Public IP** with DNS label (`--dns-name-label app1` → `app1.<region>.azurecontainer.io`) and ports (`--ports 80 443`, max. 5 per IP) **or** deployed in a **VNet** (`--vnet --subnet`, subnet delegated to `Microsoft.ContainerInstance/containerGroups`, **private** IP; not both)
Storage	**Azure Files** (`--azure-file-volume-account-name/-key --azure-file-volume-share-name --azure-file-volume-mount-path`; Linux only), emptyDir, gitRepo, secret volumes
Restart	`--restart-policy Always` (default; services) · `OnFailure` (tasks) · `Never` (run once)
Configuration	Variables `--environment-variables`, secure ones `--secure-environment-variables`, command `--command-line`; YAML (`az container create -f group.yaml`) for multi-container
Operate	`az container logs -n g1 -g RG1` · `az container attach` (live stdout) · `az container exec -n g1 --exec-command "/bin/sh"` · `az container show --query ipAddress.fqdn` · `az container restart / stop / start / delete`
Scaling	**No autoscale**: create more groups (or use ACA / AKS); **Spot containers** (preview) and **confidential containers**

Aspect	Detail
Quotas	By default 100 container groups and 100 vCPUs per region and subscription (can be raised); no built-in replicas or redundancy (HA = several groups behind your own load balancer)

- Typical case: "run a one-off job / build / batch without managing VMs, billed per second" → ACI with `--restart-policy Never`. "Persistent data across restarts" → Azure Files volume. "Accessible only inside the VNet" → deploy in the delegated subnet.

4. Azure Container Apps

Piece	What it is	Detail
Environment	Shared secure boundary (network, Log Analytics, Dapr) for several apps	**Consumption only** or **workload profiles** (Consumption + **Dedicated** D4/D8/E4... with more CPU and GPU); can live in your **VNet** (internal or external); apps in the same environment talk to each other by name
Container app	Image (Linux x64), resources, secrets, identity, **ingress**, scaling	Several containers per app (sidecar / init) only if they are tightly coupled; `az containerapp up` creates environment + app in one step
Revisions	Every change to the `template` (image, env, resources) creates an immutable **revision**; **single** mode (the new one replaces) or **multiple** (several active)	**Traffic splitting** by % between revisions (blue-green, canary), labels with their own URL; "roll back" = send traffic back to the previous one
Ingress	**External** (public, managed HTTPS, custom domain and certificate), **Internal** (environment / VNet only) or disabled; HTTP, TCP and gRPC; target port; IP restrictions; sticky sessions	"only reachable from other apps" → internal
Scaling	**Min / max replicas** (min 0 = **scale to zero**; max. 1,000 replicas); **HTTP** rules (concurrent requests), **TCP**, **custom** via **KEDA** scalers (Service Bus, Storage queue, Event Hubs, CPU, memory...). CPU / memory **do not scale to zero**	Without rules: 1 fixed replica between min and max; configurable cooldown and polling
Resources	Fixed combinations in Consumption: **0.25 vCPU / 0.5 Gi** ... up to **4 vCPU / 8 Gi** (memory = 2 × CPU); dedicated workload profiles allow more; in a *Consumption only* environment the maximum is 2 vCPU / 4 Gi	"0.5 CPU and 2 GiB" is not valid (it would be 0.5 / 1 Gi)
Jobs	Containers that **finish**: manual, **scheduled** (cron) or **event-triggered** (KEDA)	Replace ACI for tasks inside the environment
Secrets and identity	Own secrets or **Key Vault** references; **managed identity** (system / user) for ACR (`acrPull`) and other services	"no credentials in the app"
Dapr	Service invocation, pub/sub, state stores	Optional
Logs	Console and system in **Log Analytics** (`ContainerAppConsoleLogs_CL`, `ContainerAppSystemLogs_CL`) or Azure Monitor; `az containerapp logs show`	—

```
az containerapp env create -n env1 -g RG1 -l westeurope [--infrastructure-subnet-resource-id <subnet>] [--internal-only true]
az containerapp create -n app1 -g RG1 --environment env1 --image acr1.azurecr.io/app:v1 \
   --registry-server acr1.azurecr.io --registry-identity system --target-port 80 --ingress external \
   --min-replicas 0 --max-replicas 10 --cpu 0.5 --memory 1.0Gi --scale-rule-name http --scale-rule-http-concurrency 50
az containerapp update -n app1 -g RG1 --image acr1.azurecr.io/app:v2            # new revision
az containerapp revision list -n app1 -g RG1 -o table
az containerapp ingress traffic set -n app1 -g RG1 --revision-weight app1--v1=80 app1--v2=20
az containerapp job create -n job1 -g RG1 --environment env1 --trigger-type Schedule \
   --cron-expression "0 2 * * *" --image ... --replica-timeout 1800
```

5. Commands that show up (ACR and ACI)

Task	Azure CLI	PowerShell
Create registry	`az acr create -n acr1 -g RG1 --sku Premium --admin-enabled false`	`New-AzContainerRegistry -ResourceGroupName RG1 -Name acr1 -Sku Premium`
Build and push	`az acr build -r acr1 -t app:v1 .` · `az acr login -n acr1` + `docker push acr1.azurecr.io/app:v1`	`Connect-AzContainerRegistry -Name acr1` + docker push
Import and list	`az acr import -n acr1 --source docker.io/library/nginx:latest -t nginx:latest` · `az acr repository list -n acr1` · `az acr repository show-tags -n acr1 --repository app`	`Get-AzContainerRegistryRepository -RegistryName acr1` · `Get-AzContainerRegistryTag -RegistryName acr1 -RepositoryName app`
Geo-replicate	`az acr replication create -r acr1 -l northeurope`	`New-AzContainerRegistryReplication -RegistryName acr1 -Location northeurope`
Pull permission	`az role assignment create --assignee <mi id> --role AcrPull --scope <acr id>`	`New-AzRoleAssignment -ObjectId <mi> -RoleDefinitionName AcrPull -Scope <acr id>`
Create public ACI	`az container create -g RG1 -n c1 --image acr1.azurecr.io/app:v1 --cpu 1 --memory 1.5 --dns-name-label app1 --ports 80 --os-type Linux --restart-policy Always --acr-identity <id> --assign-identity <id>`	`$p = New-AzContainerInstancePortObject -Port 80 -Protocol TCP; $c = New-AzContainerInstanceObject -Name c1 -Image ... -RequestCpu 1 -RequestMemoryInGb 1.5 -Port $p; New-AzContainerGroup -ResourceGroupName RG1 -Name c1 -Location westeurope -Container $c -OsType Linux -IPAddressType Public -IPAddressDnsNameLabel app1 -RestartPolicy Always`
ACI in a VNet	`az container create ... --vnet vnet1 --subnet sub-aci` (without `--dns-name-label`)	`-SubnetId <id>`
ACI with Azure Files	`az container create ... --azure-file-volume-account-name st1 --azure-file-volume-account-key <key> --azure-file-volume-share-name share1 --azure-file-volume-mount-path /data`	`New-AzContainerGroupVolumeObject -Name v1 -AzureFileShareName share1 -AzureFileStorageAccountName st1 -AzureFileStorageAccountKey <key>`

Task	Azure CLI	PowerShell
Operate ACI	`az container logs / attach / exec / show / restart / delete -g RG1 -n c1`	`Get-AzContainerInstanceLog` · `Get-AzContainerGroup` · `Restart-AzContainerGroup` · `Remove-AzContainerGroup`

6. Decision tree

```
What does the wording ask for?
├─ Store your own images privately ..................................... Azure Container Registry
├─ Registry in several regions / private endpoint / signed images ..... ACR Premium
├─ Build the image without Docker on the local machine ................. ACR Tasks (az acr build)
├─ Let AKS / ACA / App Service pull images without credentials ......... Managed identity + AcrPull
├─ A quick container, task or batch, per-second billing ................ ACI (restart-policy Never / OnFailure)
├─ Container with a private IP in the VNet ............................. ACI in a delegated subnet (or ACA with VNet)
├─ Several containers sharing lifecycle and localhost .................. ACI container group (Linux) or ACA sidecar
├─ Persist a container's data .......................................... Azure Files volume
├─ HTTP API that scales with load and down to zero ..................... Container Apps (ingress + HTTP rules)
├─ Process messages from a queue, scale by queue length ................ Container Apps with a KEDA scaler
├─ Roll out v2 to 10 % of users and roll back .......................... Container Apps: multiple revisions + traffic split
├─ Scheduled task in a container ....................................... Container Apps Job (schedule)
├─ Only reachable from other apps in the environment ................... Ingress internal
└─ Full Kubernetes with kubectl access ................................. AKS
```

7. Signals → answer

Wording	Answer
"private registry with geo-replication"	ACR Premium
"registry must not be reachable from the internet"	ACR Premium + private endpoint (and public access disabled)
"least privilege for the container app to pull images"	AcrPull to its managed identity
"admin user credentials in the deployment"	Bad practice: use an identity / service principal
"build the image in Azure from the Dockerfile"	`az acr build`
"automatically rebuild when the base image updates"	ACR Task with a base image trigger
"delete untagged manifests automatically"	Retention policy (Premium)
"run a container for a few minutes and bill only for that"	ACI, restart policy Never
"container needs a public DNS name"	ACI with `--dns-name-label`
"container group with Windows and Linux containers"	Impossible: multi-container is Linux only
"ACI must access resources in the VNet only"	ACI in a delegated subnet (no public IP)
"data must survive container restarts"	Azure Files volume
"scale to zero when idle"	Container Apps (min replicas 0)
"scale on HTTP concurrent requests / queue length"	Container Apps scale rule HTTP / custom (KEDA)
"0.5 vCPU with 2 GiB" rejected	Fixed combinations: 0.5 vCPU → 1 Gi
"run two versions and split traffic"	Multiple revision mode + traffic weights
"container app only reachable inside the environment"	Ingress internal
"nightly job in a container inside the environment"	Scheduled Container Apps Job
"ACI needs more replicas automatically"	No autoscale in ACI → ACA

8. Mistakes to avoid

1. Choosing ACI when autoscale, revisions or scale to zero are required: that is Container Apps.
2. Expecting Private Link, geo-replication or content trust in ACR Basic / Standard.
3. Using the ACR admin user in production instead of managed identity + AcrPull.
4. Asking for a public IP and a VNet at the same time in a container group, or multi-container on Windows.
5. Forgetting the Azure Files volume and losing data on restart.
6. Setting CPU / memory outside the allowed combinations in Container Apps (memory = 2 × vCPU).
7. Assuming CPU or memory can scale to zero in Container Apps (only HTTP, TCP and event scalers).
8. Leaving ingress external when the wording asks for internal-only access.

Cheatsheet AZ-104 — 11 · App Service: plans, scaling, slots, domains, TLS, backup and networking

1. App Service plan: tiers and what each one unlocks

Tier	Max. instances	Slots	What it adds	Signal
Free (F1)	1 (shared)	0	60 min CPU/day, 1 GB; no custom domain or SLA	"test at no cost"
Shared (D1, Windows only)	1 (shared)	0	Custom domain; no custom TLS; CPU quota	Rare
Basic (B1-B3)	**3**	0	Dedicated compute, **domain + TLS**, **Always On**, **manual** scaling, **VNet integration**, **private endpoint**, hybrid connections, automatic and custom backups (production slot only)	"custom domain, minimal cost"
Standard (S1-S3)	**10**	**5**	**Autoscale**, **deployment slots**, Traffic Manager	"staging slot", "autoscale"
Premium v2 / v3 / v4 (P0v3-P5mv3, Pv4...)	**30**	**20**	More power, **zone redundancy**, automatic scaling, more slots	"99.99 %", "zone redundant", "faster"
Isolated v2 (I1v2-I6v2)	**100**	20	**App Service Environment v3**: isolated network inside your VNet, maximum scale	"fully isolated", "no shared infrastructure"

- The plan sets the **OS (Windows / Linux), region, tier, size and instances**; all apps in the plan share the instances and scale together (optional per-app scaling). To **isolate** a heavy app or scale it separately → **new plan**. An app can only be moved to a plan in the **same region, RG and OS**.
- **Scale up** = change tier (more CPU / RAM / features; the whole plan). **Scale out** = more instances (manual or autoscale). App Service autoscale = Azure Monitor autoscale (cheatsheet 09): rules on **CPU %, memory %, HTTP queue length, Data In/Out** or another resource's metric, schedule-based profiles, cooldown. **Automatic scaling** (Premium v2+): the platform scales per app with "maximum burst" and "always ready instances", no rules.
- Default domain: `https://<app>.azurewebsites.net` (new apps carry a unique suffix `<app>-<hash>.<region>-01.azurewebsites.net`). Globally unique name.

2. Deployment slots

Concept	Detail
What it is	A live app with its own hostname (`<app>-<slot>.azurewebsites.net`) in the **same plan**; Standard **5**, Premium / Isolated **20**; no extra cost
Swap	Exchanges **content and configuration** with production **without downtime** (warm-up first); **rollback** = swap again. Usually with **production as the target** (any two slots can be swapped)
Swap with preview	Two phases: applies the production configuration to staging for validation, then completes (or cancels)
Auto swap	Every deployment to the slot is swapped automatically (CI/CD); **not on Linux** or containers
Traffic routing	% of production traffic sent to a slot (canary, A/B); `x-ms-routing-name` cookie; `?x-ms-routing-name=staging` for manual opt-in
Swapped	Framework and version, 32/64-bit, WebSockets, **app settings and connection strings** (unless sticky), handler mappings, public certificates, WebJobs content, hybrid connections, service endpoints, path mappings
Not swapped (slot-specific)	**Custom domains**, private certificates and TLS bindings, **scale settings**, publishing endpoints, WebJobs schedulers, **IP restrictions**, Always On, diagnostic logs, CORS, HTTPS only / TLS version, **managed identities**, **VNet integration**, settings marked "Deployment slot setting"

- "Connection string must point to the test DB in staging and prod DB in production" → mark the setting as a **Deployment slot setting** (sticky). "Validate with the production settings before going live" → swap with preview. "Send 10 % of users to the new version" → traffic %.

3. Domains and certificates

Task	How	Detail
Custom domain	Custom domains → Add: prove ownership with **TXT** `asuid.<domain>` = Custom Domain Verification ID, and **CNAME** `www` → `<app>.azurewebsites.net` (subdomain) or **A** record to the inbound IP + TXT (root)	Basic tier or higher for HTTPS; added to **each slot** (not swapped); also for Traffic Manager (CNAME to the profile)
App Service domain	Buy a domain from the portal (managed in Azure DNS)	"purchase a domain in Azure"
Certificates	**App Service managed certificate** (free, automatic renewal, Basic+, no wildcard, not exportable) · **App Service certificate** (purchased, stored in Key Vault, wildcard supported) · **Import from Key Vault** (grant access to the "Microsoft Azure App Service" principal) · **Upload PFX** · public certificate (.cer) so the app trusts others	Import from Key Vault = centralized renewal
TLS binding	**SNI SSL** (free, several certificates per IP; the usual choice) or **IP-based SSL** (dedicated IP; cost; for clients without SNI)	"dedicated IP address for TLS" → IP-based
Protocols	**HTTPS Only**, minimum version **TLS 1.2** (1.3 available), client certificates (mTLS)	"redirect HTTP to HTTPS" → HTTPS Only

4. Backup and restore

Type	Tier	What it includes	Details
Automatic backups	Basic, Standard, Premium, Isolated	Content and configuration, up to **30 GB**	**Hourly**, **30-day** retention, nothing to configure and no storage account; cannot be downloaded; restore over the app, to another app or to a slot
Custom backups	Basic (production only), Standard, Premium, Isolated	Content + configuration up to **10 GB**; a **linked database** can no longer be added to new configurations (existing ones keep working until March 31, 2028)	Needs a **storage account** (SAS) and container; **on demand or scheduled** (min. every 2 h, 12 per day); retention 0-30 days or indefinite; **partial backups** with `_backup.filter`; backup to firewalled storage via VNet integration; for the database, use its **native backup**

- Restoring **stops** the app during the process → restore to a **slot** first and swap. "Backup must include the database" → the database's native backup (Azure SQL, flexible server), not the custom backup. "Backup fails: content exceeds 10 GB" → exclude folders with `_backup.filter` or move files to Storage.

5. Networking

Direction	Feature	What it does	Tier
Inbound	**Access restrictions**	List of allow / deny rules by **IP**, **service tag** or **subnet (Microsoft.Web service endpoint)** with priority; once the first one is created, everything else is denied; also for the **SCM** site	All
Inbound	**Private endpoint**	Private IP in your VNet; combined with "public network access disabled" the app is no longer reachable from the internet; DNS `privatelink.azurewebsites.net`	Basic+
Inbound	**App Service Environment v3**	The whole infrastructure inside your VNet (ILB or external)	Isolated v2
Outbound	**VNet integration** (regional)	The app egresses into your VNet through a subnet **delegated** to `Microsoft.Web/serverFarms` (/26 recommended; up to 2 integrations per plan, shared by its apps); reaches VMs, databases with private endpoint, on-prem via VPN / ER; **Route All** to force all traffic; NAT Gateway for a fixed outbound IP	Basic+
Outbound	**Hybrid Connections**	TCP to an on-prem host:port via Relay, no VPN, with the Hybrid Connection Manager installed there	Basic+
Outbound	**Outbound IPs**	Listed in Properties (they change when the tier changes); for whitelists in external firewalls (or NAT Gateway for a fixed one)	All

- "App must reach a SQL server on-premises without a VPN" → Hybrid Connections. "App must call a database with private endpoint" → VNet integration. "Only the corporate IP range can reach the app" → access restrictions. "App must not be reachable from the internet at all" → private endpoint + public access disabled (or ASE).

6. Configuration, deployment and monitoring

Area	What shows up
App settings / connection strings	Environment variables encrypted at rest; sticky per slot; Key Vault references `@Microsoft.KeyVault(SecretUri=...)` with managed identity
General settings	Stack and version, 32/64-bit, **Always On** (keeps the app from unloading when idle; Basic+; required for continuous WebJobs), **ARR affinity** (sticky sessions; disable in stateless apps), HTTP 2.0, WebSockets, FTP state (FTPS only / disabled), basic auth for publishing
Path mappings	Virtual directories; **mount Azure Files / Blob** as a path
Authentication (Easy Auth)	Managed login with Entra ID, Microsoft, Google, Facebook, Apple, OpenID Connect without code; "require authentication"
Managed identity	System or user-assigned for Key Vault, Storage, SQL, ACR
Deployment Center	GitHub Actions, Azure Repos, Bitbucket, **Local Git**, FTPS, **ZIP deploy** (`az webapp deploy --src-path app.zip`), `az webapp up`, container from ACR / Docker Hub (single or Compose), **Run From Package** (`WEBSITE_RUN_FROM_PACKAGE=1`)
Logs	App Service logs: application logging (Filesystem 12 h / Blob), web server logging, detailed errors, failed request tracing, deployment logs; **Log stream**; **Kudu** (`<app>.scm.azurewebsites.net`: console, processes, files); diagnostic settings to Log Analytics; **Application Insights**
Health check	Path the platform probes; unhealthy instances are taken out of rotation (≥ 2 instances) and replaced
Diagnose and solve problems	Automatic detectors for CPU, memory, 5xx, slowness

7. Commands that show up

Task	Azure CLI	PowerShell (Az.Websites)
Plan and app	`az appservice plan create -n plan1 -g RG1 --sku S1 --is-linux` · `az webapp create -n app1 -g RG1 --plan plan1 --runtime "DOTNETCORE:8.0"` (`az webapp list-runtimes`)	`New-AzAppServicePlan -Name plan1 -ResourceGroupName RG1 -Location westeurope -Tier Standard -NumberofWorkers 2 -WorkerSize Small` · `New-AzWebApp -Name app1 -ResourceGroupName RG1 -AppServicePlan plan1`
Scale	`az appservice plan update -n plan1 -g RG1 --sku P1V3 --number-of-workers 3` · autoscale: `az monitor autoscale create --resource plan1 --resource-type Microsoft.Web/serverfarms --min-count 2 --max-count 10 --count 2` + `az monitor autoscale rule create --condition "CpuPercentage > 70 avg 10m" --scale out 1`	`Set-AzAppServicePlan -Name plan1 -ResourceGroupName RG1 -Tier PremiumV3 -WorkerSize Medium -NumberofWorkers 3`
Settings	`az webapp config appsettings set -n app1 -g RG1 --settings KEY=value --slot-settings STICKY=1` · `az webapp config connection-string set --connection-string-type SQLAzure --settings Db="..."` · `az webapp config set --always-on true --min-tls-version 1.2 --ftps-state Disabled` · `az webapp update --https-only true`	`Set-AzWebApp -Name app1 -ResourceGroupName RG1 -AppSettings @{KEY="value"} -HttpsOnly $true -AlwaysOn $true` · `Set-AzWebAppSlotConfigName -AppSettingNames STICKY`

Task	Azure CLI	PowerShell (Az.Websites)
Slots	`az webapp deployment slot create -n app1 -g RG1 --slot staging --configuration-source app1` · `az webapp deployment slot swap -n app1 -g RG1 --slot staging --target-slot production [--action preview]` · `az webapp traffic-routing set --distribution staging=10`	`New-AzWebAppSlot -Name app1 -ResourceGroupName RG1 -Slot staging` · `Switch-AzWebAppSlot -SourceSlotName staging -DestinationSlotName production -Name app1 -ResourceGroupName RG1` · `Add-AzWebAppTrafficRouting`
Domain and TLS	`az webapp config hostname add --webapp-name app1 -g RG1 --hostname www.contoso.com` · `az webapp config ssl upload --certificate-file c.pfx --certificate-password ...` · `az webapp config ssl bind --certificate-thumbprint <t> --ssl-type SNI` · `az webapp config ssl create` (managed) · `az webapp config ssl import --key-vault kv1 --key-vault-certificate-name cert1`	`Set-AzWebApp -HostNames @("www.contoso.com","app1.azurewebsites.net")` · `New-AzWebAppSSLBinding -Name www.contoso.com -ResourceGroupName RG1 -Thumbprint <t> -SslState SniEnabled -WebAppName app1`
Backup	`az webapp config backup create -n app1 -g RG1 --container-url "<sas>" --backup-name b1` · `az webapp config backup update --frequency 1d --retain-one true --retention 30` · `az webapp config backup restore` · automatic: `az webapp config snapshot list / restore`	`New-AzWebAppBackup -Name app1 -ResourceGroupName RG1 -StorageAccountUrl "<sas>"` · `Edit-AzWebAppBackupConfiguration -FrequencyInterval 1 -FrequencyUnit Day -RetentionPeriodInDays 30` · `Restore-AzWebAppBackup`
Networking	`az webapp vnet-integration add -n app1 -g RG1 --vnet vnet1 --subnet sub-apps` · `az webapp config access-restriction add -n app1 -g RG1 --rule-name office --action Allow --ip-address 203.0.113.0/24 --priority 100` · `az network private-endpoint create ... --group-id sites` · `az webapp hybrid-connection add`	`Add-AzWebAppAccessRestrictionRule -ResourceGroupName RG1 -WebAppName app1 -Name office -Priority 100 -Action Allow -IpAddress 203.0.113.0/24`
Deploy and logs	`az webapp deploy -n app1 -g RG1 --src-path app.zip --type zip` · `az webapp up` · `az webapp deployment source config --repo-url ... --branch main` · `az webapp log config --application-logging filesystem --level information` · `az webapp log tail` · `az webapp restart`	`Publish-AzWebApp -ResourceGroupName RG1 -Name app1 -ArchivePath app.zip` · `Restart-AzWebApp`

8. Decision tree

```
What does the wording ask for?
├─ Custom domain and HTTPS at minimal cost ............................. Basic
├─ Staging slot / autoscale / scheduled backups ........................ Standard
├─ Zone redundancy, 20 slots, more power ............................... Premium v3 / v4
├─ Full network isolation .............................................. Isolated v2 (ASE v3)
├─ One app needs more CPU than the rest ................................ New plan (scale up or isolate)
├─ More capacity by schedule / load .................................... Scale out with autoscale (Standard+)
├─ Deploy without downtime and be able to roll back .................... Slot + swap (rollback = swap again)
├─ Different setting in staging and production ......................... Deployment slot setting (sticky)
├─ Test the new version with some of the users ......................... Traffic routing %
├─ Free, auto-renewed certificate ...................................... App Service managed certificate
├─ Centrally managed certificate ....................................... Import from Key Vault
├─ Legacy clients without SNI .......................................... IP-based SSL
├─ Downloadable or scheduled backup to a storage account ............... Custom backup (Basic+)
├─ Office IPs only ..................................................... Access restrictions
├─ Nothing from the internet ........................................... Private endpoint + public access disabled
├─ Reach VNet resources / on-prem via VPN .............................. VNet integration (delegated subnet)
├─ Reach an on-prem server without VPN ................................. Hybrid Connections
├─ Secrets without putting them in settings ............................ Key Vault references + managed identity
└─ The app "sleeps" and is slow to respond ............................. Always On (Basic+)
```

9. Signals → answer

Wording	Answer
"deployment slots" + "Basic plan"	Upgrade to Standard (5 slots)
"need 8 slots"	Premium (20)
"scale to 15 instances"	Premium (Standard tops out at 10)
"autoscale based on CPU"	Standard or higher + autoscale rule on the plan
"move the app to a plan in another region / OS"	Not possible: create a new app (same region, RG and OS to move)
"after the swap the app points to the test database"	The connection string was not sticky: mark it "Deployment slot setting"
"custom domain missing after swap"	Expected: domains are not swapped; add it to the slot
"validate the production configuration before completing the swap"	Swap with preview
"domain verification fails"	Missing the `asuid.<domain>` TXT or the CNAME / A record
"root domain (contoso.com) to the app"	A record to the IP + asuid TXT (or alias in Azure DNS)
"free certificate for www.contoso.com"	App Service managed certificate (no wildcard)
"wildcard certificate"	App Service certificate (purchased) or your own PFX
"restore the app without downtime"	Restore to a slot and swap
"backup every 6 hours to a storage account"	Scheduled custom backup (Basic+, every 2 h minimum); the database with its native backup
"automatic backups retained 30 days"	Included in Basic+ (hourly)
"allow traffic only from the Application Gateway"	Access restriction by service tag / subnet + service endpoint

Wording	Answer
"app cannot resolve the private endpoint of the database"	Missing VNet integration (and private DNS)
"app needs a static outbound IP"	NAT Gateway on the integration subnet
"WebJob stops running"	Enable Always On
"user sessions lost when scaling out"	ARR affinity (or better: external state)

10. Mistakes to avoid

1. Looking for slots or autoscale in Basic, or more than 10 instances in Standard.
2. Scaling an app "separately" without moving it out of the plan: scale out affects every app in the plan.
3. Forgetting to mark environment settings as sticky before the swap, or expecting the domain and certificate to travel with it.
4. Confusing VNet integration (outbound) with private endpoint (inbound) or with App Service Environment.
5. Using Hybrid Connections when a VPN and VNet integration already exist (or the reverse: asking for a VPN for a single TCP port).
6. Trying to move the app to a plan in another region, RG or OS.
7. Counting on the linked database backup: it can no longer be configured and goes away in 2028.
8. Leaving basic FTP and HTTP allowed when the ask is "secure": FTPS only, HTTPS only, TLS 1.2.

Cheatsheet AZ-104 — 12 · Virtual networks: VNet, subnets, public IPs, peering, routes and troubleshooting

1. VNet and subnets

Concept	Rules
VNet	Lives in **one region and one subscription**; one or more **address spaces** (RFC 1918 or public; they can be added, expanded or removed **even with peering**, syncing afterwards); limits: 65,536 private IPs, 3,000 subnets, **650 peerings** per VNet (previously 500), 1,000 VNets per region and subscription
Subnet	Cannot overlap or span two VNets; minimum **/29** (8 addresses, **3 usable**): Azure reserves **5** per subnet (network, gateway .1, two DNS .2 and .3, broadcast); a subnet can carry an **NSG**, **route table**, **service endpoints**, **delegation** to a service (App Service, ACI, PostgreSQL...), **NAT Gateway** and the **private subnet** setting (no implicit outbound access)
Special subnets	`GatewaySubnet` (VPN / ExpressRoute, **/27 or larger**), `AzureBastionSubnet` (**/26 or larger**), `AzureFirewallSubnet` (/26), `AzureFirewallManagementSubnet`, `RouteServerSubnet` (/27): exact name and no other resources
VNet DNS	Default: **Azure-provided** (168.63.129.16; resolves VM names in the **same VNet** and public names) · **Custom**: your own servers (DC, BIND); the change requires **restarting or renewing DHCP** on the VMs; can be overridden per NIC. Names across VNets → Private DNS zone (cheatsheet 14)
Inside the VNet	All subnets communicate by default (system route); filtered with NSG; a NIC can only be in a subnet of **its** VNet and **region**

2. IP addresses and NICs

Type	Options	Exam details
Private IP	**Dynamic** (default; kept as long as the NIC exists) or **static** (reserve for DCs, DNS, NVAs)	Assigned from the subnet; configured **on the NIC** (ipconfig), never inside the OS
Standard public IP	**Static** only; **zone-redundant** or zonal; **secure by default** (without an NSG allowing it, nothing gets in); 99.99 % SLA; required with Standard Load Balancer, Bastion, NAT Gateway, Firewall	The **Basic** IP (dynamic or static, no zones) **retired on September 30, 2025**: any new answer is Standard
Public IP prefix	Reserved contiguous block (/28...)	Whitelists at third parties
DNS label	`<label>.<region>.cloudapp.azure.com` on the public IP	"VM reachable by name without Azure DNS"
NIC	Several **ipconfigs** (one primary), several NICs per VM (depending on size; one primary), **IP forwarding** (mandatory on NVAs), **accelerated networking**, own NSG, own DNS	**Deallocating** a VM loses the dynamic public IP (no longer exists with Standard)
Internet egress	**Default outbound access** (implicit Microsoft IP) is going away: new VNets create **private** subnets by default (portal already; API from 2026). **Explicit** egress: **NAT Gateway** (recommended, fixed IP, avoids SNAT exhaustion), public IP on the NIC, **outbound rules** of a public Standard Load Balancer, or NVA / Firewall with UDR	Flexible VMSS and VMs in private subnets **have no egress** without one of these. "VM cannot reach Windows Update / activate" in a private subnet → NAT Gateway
IPv6	Dual stack in VNet, subnet, NIC, Standard LB	Rare questions

3. Peering

Setting	What it does	When
Allow access to remote virtual network	Allows traffic between the two VNets (default yes)	Disable to cut traffic without deleting the peering
Allow forwarded traffic	Accepts traffic that **did not originate** in the remote VNet (e.g. forwarded by an **NVA** in the hub)	Hub-spoke with a firewall / NVA in the hub
Allow gateway transit (on the hub)	The hub offers its **VPN / ExpressRoute gateway** to the spokes	Spokes reach on-prem through the hub gateway
Use remote gateways (on the spoke)	The spoke uses the hub gateway; **cannot have its own gateway** and can only use **one**	Same; also valid with **global** peering

- **Regional** (same region) or **global** (different regions); both over the backbone, no gateway or encryption, latency like inside a VNet; cost per GB. Global: the frontend of a **Basic** ILB cannot be reached and some services do not support it.
- **Not transitive**: A ↔ B and B ↔ C does not give A ↔ C → direct A ↔ C peering, or NVA / Azure Firewall in B with **UDRs** in A and C and *forwarded traffic*. Peering routes show up with next hop **Virtual network peering** in the effective routes.
- Requirements: **non-overlapping** address spaces, create the peering **on both VNets** (the portal does both at once), **Network Contributor** role (or `virtualNetworks/virtualNetworkPeerings/write` + `peer/action`) on both; works across subscriptions and **tenants** (with a guest user or permission on both). States: **Initiated** (the other side is missing) → **Connected**; **Disconnected** if one is deleted.
- Alternative **VNet-to-VNet VPN**: encrypted, transitive via gateways, bandwidth limited by the gateway SKU; only when encryption is required or peering is not possible. For hundreds of spokes → **Azure Virtual Network Manager** (connectivity configurations, up to 1,000 spokes).

4. Routing

Route	Prefix	Next hop	Who creates it
Inside the VNet	Each address space	**Virtual network**	System
Egress	0.0.0.0/0	**Internet** (Azure services do not leave the backbone)	System

Route	Prefix	Next hop	Who creates it
Private RFC 1918 / 100.64	10/8, 172.16/12, 192.168/16, 100.64/10	**None** (dropped) unless they are in the address space	System
Peering	Remote address space	**Virtual network peering**	When the peering is created
On-prem	BGP prefixes or local network gateway prefixes	**Virtual network gateway**	When the gateway is created
Service endpoint	Service IPs	**VirtualNetworkServiceEndpoint** (cannot be overridden)	When the endpoint is enabled on the subnet

- **UDR** (route table associated with one or more subnets; **one** table per subnet; **400** routes per table, 1,000 with Virtual Network Manager): next hops **Virtual appliance** (+ private IP of the NVA or of the ILB in front of the NVAs; the NVA needs **IP forwarding** on its NIC and must live in **another subnet**), **Virtual network gateway** (VPN only, not ExpressRoute), **Virtual network**, **Internet**, **None** (black hole). **Service tags** are also accepted as prefix (e.g. `Storage`).
- **Selection: longest** prefix wins; same prefix: **UDR > BGP > system**. VNet, peering and service endpoint routes take precedence even if there is a more specific BGP route.
- **Propagate gateway routes = No** on the table: the gateway's BGP / static routes **do not** enter that subnet (to force on-prem traffic through an NVA). **Never** disable it on `GatewaySubnet` or associate with it a table that contains 0.0.0.0/0.
- **Overriding 0.0.0.0/0** (forced tunneling) toward an NVA / Firewall / VPN: all traffic leaves that way, including Azure public services without a service endpoint; resources stop being directly reachable from the internet; in private subnets, routes with next hop **Internet** stop working without explicit egress.
- Diagnostics: NIC → **Effective routes** (the VM must be **running**) and Network Watcher → **Next hop**.

5. Connectivity troubleshooting

Tool (Network Watcher)	Answers	Detail
IP flow verify	Does the NSG allow or deny this flow (source, destination, port, protocol, direction)? And **which rule**	First step for "VM1 cannot RDP to VM2"
NSG diagnostics	Evaluation of **all** NSGs in the path for a flow	More complete than IP flow verify
Effective security rules / **Effective routes**	Rules and routes that actually apply to a NIC (subnet + NIC NSG; UDR + system)	On the NIC in the portal, `az network nic list-effective-nsg`, `az network nic show-effective-route-table`
Next hop	Which way traffic to a destination leaves from a NIC	Detects a misplaced UDR or a broken peering
Connection troubleshoot	**One-off** test VM → destination (IP, FQDN, URI, VM): reachable, latency, hops and where it breaks	Needs the Network Watcher extension on the source VM
Connection monitor	Same but **continuous**, with metrics and alerts (cheatsheet 15)	—
Packet capture	pcap from the VM to storage or local, with filters and time limit	Extension required
Topology	Diagram of the VNet and its resources	—
VPN troubleshoot	Diagnostics for VPN gateways and connections	—
VNet flow logs / Traffic analytics	Log of accepted and denied flows (**NSG flow logs** retire in Sept. 2027)	"which traffic is being blocked"

- Checklist when two VMs cannot see each other: **peering** status, **NSG** on NIC and subnet (both must allow), **UDR** (next hop None or NVA down), **OS firewall** (Windows blocks ICMP by default), **DNS** (name vs IP), VM powered off, and nobody blocking **168.63.129.16**.

6. Commands that show up

Task	PowerShell (Az.Network)	Azure CLI
Create VNet and subnet	`$sub = New-AzVirtualNetworkSubnetConfig -Name sub1 -AddressPrefix 10.0.1.0/24; New-AzVirtualNetwork -Name vnet1 -ResourceGroupName RG1 -Location westeurope -AddressPrefix 10.0.0.0/16 -Subnet $sub` · add: `Add-AzVirtualNetworkSubnetConfig ... ; Set-AzVirtualNetwork`	`az network vnet create -g RG1 -n vnet1 --address-prefixes 10.0.0.0/16 --subnet-name sub1 --subnet-prefixes 10.0.1.0/24` · `az network vnet subnet create -g RG1 --vnet-name vnet1 -n sub2 --address-prefixes 10.0.2.0/24`
Subnet: NSG, table, delegation, private	`Set-AzVirtualNetworkSubnetConfig -VirtualNetwork $vnet -Name sub1 -AddressPrefix ... -NetworkSecurityGroup $nsg -RouteTable $rt -Delegation $d; Set-AzVirtualNetwork`	`az network vnet subnet update -g RG1 --vnet-name vnet1 -n sub1 --network-security-group nsg1 --route-table rt1 --delegations Microsoft.Web/serverFarms --default-outbound false`
Public IP and NIC	`New-AzPublicIpAddress -Name pip1 -ResourceGroupName RG1 -Location westeurope -Sku Standard -AllocationMethod Static -Zone 1,2,3` · `New-AzNetworkInterface -Name nic1 ... -SubnetId <id> -PublicIpAddressId <id> -EnableIPForwarding` · static private IP: `$nic.IpConfigurations[0].PrivateIpAllocationMethod = "Static"; Set-AzNetworkInterface`	`az network public-ip create -g RG1 -n pip1 --sku Standard --allocation-method Static --zone 1 2 3 --dns-name app1` · `az network nic create -g RG1 -n nic1 --vnet-name vnet1 --subnet sub1 --public-ip-address pip1 --ip-forwarding true` · `az network nic ip-config update -g RG1 --nic-name nic1 -n ipconfig1 --private-ip-address 10.0.1.10`

Task	PowerShell (Az.Network)	Azure CLI
Peering	`Add-AzVirtualNetworkPeering -Name hub-to-spoke -VirtualNetwork $hub -RemoteVirtualNetworkId $spoke.Id -AllowGatewayTransit` + `Add-AzVirtualNetworkPeering -Name spoke-to-hub -VirtualNetwork $spoke -RemoteVirtualNetworkId $hub.Id -UseRemoteGateways -AllowForwardedTraffic` · `Get-AzVirtualNetworkPeering`	`az network vnet peering create -g RG1 -n hub-to-spoke --vnet-name hub --remote-vnet spoke --allow-vnet-access --allow-gateway-transit` + `az network vnet peering create -g RG1 -n spoke-to-hub --vnet-name spoke --remote-vnet hub --allow-vnet-access --allow-forwarded-traffic --use-remote-gateways` · `az network vnet peering list` · `az network vnet peering sync`
Routes	`New-AzRouteTable -Name rt1 -ResourceGroupName RG1 -Location westeurope -DisableBgpRoutePropagation` · `Add-AzRouteConfig -RouteTable $rt -Name toNVA -AddressPrefix 0.0.0.0/0 -NextHopType VirtualAppliance -NextHopIpAddress 10.0.100.4; Set-AzRouteTable`	`az network route-table create -g RG1 -n rt1 --disable-bgp-route-propagation true` · `az network route-table route create -g RG1 --route-table-name rt1 -n toNVA --address-prefix 0.0.0.0/0 --next-hop-type VirtualAppliance --next-hop-ip-address 10.0.100.4` · associate: `az network vnet subnet update ... --route-table rt1`
NAT Gateway	`New-AzNatGateway -Name nat1 -ResourceGroupName RG1 -Location westeurope -Sku Standard -PublicIpAddress $pip` + subnet `-NatGateway`	`az network nat gateway create -g RG1 -n nat1 --public-ip-addresses pip-nat --idle-timeout 10` · `az network vnet subnet update ... --nat-gateway nat1`
Diagnostics	`Get-AzEffectiveRouteTable -NetworkInterfaceName nic1 -ResourceGroupName RG1` · `Get-AzEffectiveNetworkSecurityGroup` · `Test-AzNetworkWatcherIPFlow -NetworkWatcher $nw -TargetVirtualMachineId <id> -Direction Inbound -Protocol TCP -LocalIPAddress 10.0.1.4 -LocalPort 3389 -RemoteIPAddress 203.0.113.5 -RemotePort 60000` · `Get-AzNetworkWatcherNextHop`	`az network nic show-effective-route-table -g RG1 -n nic1 -o table` · `az network nic list-effective-nsg -g RG1 -n nic1` · `az network watcher test-ip-flow --vm vm1 -g RG1 --direction Inbound --protocol TCP --local 10.0.1.4:3389 --remote 203.0.113.5:60000` · `az network watcher show-next-hop --vm vm1 -g RG1 --source-ip 10.0.1.4 --dest-ip 10.1.0.4`

7. Decision tree

```
What does the wording ask for?
├─ Subnet for N VMs ..... Prefix with N + 5 addresses (minimum /29 = 3 usable)
├─ Connect two VNets, low latency, no encryption ..... Peering (regional or global)
├─ Spokes using the hub VPN / ExpressRoute ..... Gateway transit (hub) + Use remote gateways (spoke)
├─ Spoke A → spoke C through the hub ..... UDR to the hub NVA + Allow forwarded traffic
├─ Encrypted connection between VNets ..... VNet-to-VNet VPN
├─ All outbound traffic through a firewall ..... UDR 0.0.0.0/0 → Virtual appliance (IP forwarding on the NVA)
├─ Keep on-prem routes out of a subnet ..... Route table with "Propagate gateway routes = No"
├─ Drop traffic to a range ..... Route with next hop None
├─ Find out why VM1 cannot reach VM2 ..... IP flow verify → Next hop → Effective routes / rules
├─ Find out which NSG blocks ..... NSG diagnostics / Effective security rules
├─ Fixed outbound IP for all VMs ..... NAT Gateway
├─ VMs with no internet egress in a new VNet ..... Private subnet by default → NAT Gateway / PIP / LB outbound
├─ Public IP that survives zone failures and has an SLA ..... Standard, static, zone-redundant
└─ Two VNets with the same range must talk ..... Impossible with peering: change ranges (or NAT on an NVA)
```

8. Signals → answer

Wording	Answer
"subnet /29, only 3 addresses available"	Correct: Azure reserves 5 per subnet
"address spaces overlap", "peering fails"	Change one VNet's range (it can be resized even with peerings)
"VNet1 peered to VNet2, VNet2 peered to VNet3, VNet1 cannot reach VNet3"	Peering is not transitive: peering 1-3 or NVA + UDR
"spoke VMs must reach on-premises through the hub gateway"	Allow gateway transit + Use remote gateways
"traffic from spoke through the firewall in the hub is dropped"	Allow forwarded traffic is missing on the peering (and the UDR)
"peering status Initiated"	The peering on the other VNet still has to be created
"Basic public IP", "dynamic public IP"	Retired: use Standard static
"VM cannot reach the internet after creating a new VNet"	Private subnet: add a NAT Gateway (or PIP / LB outbound)
"same outbound IP for all VMs", "SNAT exhaustion"	NAT Gateway
"route all traffic through the NVA"	UDR 0.0.0.0/0 → Virtual appliance + IP forwarding
"NVA receives traffic but does not forward it"	Enable IP forwarding on the NIC (and in the OS)
"on-premises routes should not apply to subnet X"	Disable BGP route propagation on its route table
"which rule blocks RDP"	IP flow verify
"traffic takes an unexpected path"	Next hop / Effective routes
"effective routes show nothing"	The VM is deallocated: start it
"VMs cannot resolve each other by name"	Different VNets → Private DNS zone linked to both; custom DNS → the server must forward to 168.63.129.16
"DNS server changed but VMs still use the old one"	Restart the VMs / renew DHCP
"NIC in a different region than the VNet"	Impossible: the NIC must be in the VNet's region

9. Mistakes to avoid

1. Calculating hosts without subtracting the 5 reserved addresses, or using /30 or /31 as a subnet.
2. Believing peering is transitive or that it includes the gateway without enabling transit / remote gateways.
3. Setting Use remote gateways on a spoke that already has its own gateway.
4. Forgetting IP forwarding on the NVA or deploying it in the same subnet it routes for (loop).
5. Associating a table with 0.0.0.0/0 to the GatewaySubnet or disabling its BGP propagation.
6. Assuming a UDR can override a service endpoint route.
7. Designing with Basic IPs or implicit internet egress: they no longer exist for new deployments.
8. Looking for the fault in the OS before checking NSG (NIC and subnet), UDR and peering status.

Cheatsheet AZ-104 — 13 · Network security: NSG and ASG, effective rules, Bastion, service endpoints and private endpoints

1. Network Security Groups

Rule property	Values
Priority	**100-4096**; evaluated from **lowest to highest** and the first match **stops** the evaluation; no two rules can have the same priority and direction
Source / Destination	Any, IP or CIDR, **several IPs** (augmented), **service tag** (VirtualNetwork, Internet, AzureLoadBalancer, AzureCloud, Storage, Storage.WestEurope, Sql, AzureKeyVault, AzureBastion, GatewayManager, AzureMonitor...) or **application security group**
Protocol	TCP, UDP, ICMP, ESP, AH, Any
Port	Single, range or list (`80,443,8000-8010`)
Direction	Inbound / Outbound
Action	Allow / Deny

Default rule	Priority	What it does
AllowVnetInBound	65000	Allows everything coming from the **VirtualNetwork** tag (the VNet, **peered VNets** and **on-prem** via gateway)
AllowAzureLoadBalancerInBound	65001	Allows the load balancer **probes** (168.63.129.16)
DenyAllInBound	65500	Denies the rest (including **internet**)
AllowVnetOutBound	65000	Outbound to the VNet
AllowInternetOutBound	65001	Outbound to the **internet** allowed by default
DenyAllOutBound	65500	Denies the rest

- They cannot be deleted, but they are **overridden** by lower-numbered rules. For "deny all outbound internet" → Deny rule to Internet with priority < 65001 (and Allow to the required service tags).
- **Association:** one NSG to **subnets** and/or **NICs** (a subnet or NIC has only **one** NSG; one NSG serves many). It must be in the **same region** (and subscription) as the VNet. Limit **2,000 rules** per NSG.
- **Evaluation** with an NSG on the subnet **and** on the NIC: inbound → **subnet first, then NIC**; outbound → **NIC first, then subnet**. Traffic must be allowed in **both**. The subnet NSG also filters traffic **between VMs in the same subnet**.
- **Stateful:** an Inbound rule allows the replies; no Outbound rule is needed (and vice versa). Changing rules only affects **new connections**. LB probes and platform services (DHCP, DNS, IMDS at 168.63.129.16 / 169.254.169.254) are not blocked except with specific tags. Outbound port **25** blocked except on EA / MCA-E subscriptions (use an SMTP relay). **ICMP** is not allowed from the internet by default.

2. Application Security Groups

- They group **NICs** by role (web, app, db) and are used as **source or destination** in NSG rules: "Allow TCP 1433 from asg-web to asg-db". A NIC can belong to **several** ASGs; all NICs in an ASG must be in the **same VNet**, and if a rule uses two ASGs, both in the same VNet. They allow stable rules even when IPs change or VMs are added.

3. Interpreting effective rules (what shows up with tables)

Scenario	Result	Why
NIC NSG allows RDP from the internet; subnet NSG has no rule	**Blocked**	DenyAllInBound wins on the subnet; both are needed
Subnet NSG allows RDP; the NIC has no NSG	Allowed	Only the subnet one is evaluated
Rule 100 Allow 3389 from Any; rule 110 Deny 3389 from Any	**Allowed**	The lower number wins
Rule 100 Deny 3389 from Internet; rule 200 Allow 3389 from 203.0.113.0/24	Blocked for everyone	100 matches first (Internet includes that IP): swap the priorities
VM1 and VM2 in different subnets of the same VNet, no custom rules	They see each other	AllowVnetInBound
LB backend marked unhealthy after a new NSG	AzureLoadBalancer / 168.63.129.16 was blocked	Add Allow from the AzureLoadBalancer tag to the probe port
Deny Any → Internet outbound rule with priority 100	VMs do not update and cannot reach Storage	Add Allow to the tags (Storage, AzureMonitor...) with a lower priority number, or service endpoints
VM in a peered VNet cannot get in	NSG with source limited to the local subnet	VirtualNetwork includes peered VNets; review custom rules

- **Effective security rules** (NIC → Networking) shows the sum of subnet + NIC; **IP flow verify** tells which rule allows or denies a specific flow; **NSG diagnostics** evaluates all NSGs in the path (cheatsheet 12). **Flow logs** (VNet flow logs; NSG flow logs retire in Sept. 2027) record accepted / denied per rule.

4. Azure Bastion

Aspect	Detail
What it is	RDP / SSH to VMs **through the portal (HTML5, port 443)** or with a native client, with no **public IP** on the VMs and without opening 3389 / 22 to the internet
Requirements	`AzureBastionSubnet` **/26 or larger** subnet (same VNet and RG), **Standard static** public IP; reaches VMs in the VNet and in **peered VNets** (one Bastion in the hub); the VM needs an NSG rule allowing 3389 / 22 from the Bastion subnet (the VirtualNetwork tag covers it)
NSG on AzureBastionSubnet	Inbound: 443 from **Internet**, 443 from **GatewayManager**, 443 from **AzureLoadBalancer**, 8080 and 5701 from VirtualNetwork · Outbound: 3389 and 22 to VirtualNetwork, 443 to **AzureCloud**, 8080 and 5701 to VirtualNetwork, 80 to Internet
Developer	Free, no dedicated subnet, one connection at a time, portal only; for dev/test
Basic	2 instances; portal; peered VNets; Kerberos
Standard	**Host scaling** (2-50 instances), **native client** (mstsc / ssh / `az network bastion rdp`), **file transfer**, **shareable links**, connection by **IP** (on-prem via VPN), custom ports
Premium	Plus **session recording** and **private-only** deployment (no public IP)

- "Users must connect with their own RDP client through Bastion" → Standard. "Record RDP sessions for audit" → Premium. "Bastion deployment fails on /27" → the subnet must be /26.

5. Service endpoints vs private endpoints

	Service endpoint	Private endpoint (Private Link)
What it does	Extends the **subnet identity** to the service; traffic leaves with a **private** source IP over the backbone toward the service's **public endpoint**	Creates a **NIC with a private IP** from your subnet that represents the resource (blob, sqlServer, vault, sites...); the service becomes reachable as if it were in the VNet
Configured on	The **subnet** (`Microsoft.Storage`, `Microsoft.Sql`, `Microsoft.KeyVault`, `Microsoft.Web`, `Microsoft.ServiceBus`, `Microsoft.EventHub`, `Microsoft.AzureCosmosDB`, `Microsoft.ContainerRegistry`...) + **VNet rule** in the resource firewall	The **resource** (sub-resource) + subnet where the NIC lives + **private DNS zone** `privatelink.<service>`
DNS	Unchanged (resolves to the public IP)	Must resolve to the **private IP**: `privatelink.*` zone linked to the VNets (and forwarding from on-prem with **DNS Private Resolver**)
From on-prem	**No** (on-prem traffic arrives with NAT public IPs → IP rule)	**Yes**, via VPN / ExpressRoute
From peered VNets	Each subnet enables its own endpoint	Yes (the peering reaches the private IP; link the DNS zone to that VNet)
Public endpoint	Still exists (restricted with the firewall)	Can be **disabled** entirely ("public network access disabled")
Cost	Free	Per hour and per GB
Granularity	Whole account / server; **endpoint policies** (Storage) to limit to specific accounts	A specific resource and sub-resource; automatic or manual approval
When	Quick and cheap for "only from my VNet"	"No public endpoint", "from on-premises", "private IP", "zero trust"

Service	Private zone
Blob / File / Queue / Table / DFS / Web (static)	`privatelink.blob.core.windows.net` · `privatelink.file.core.windows.net` · `privatelink.queue...` · `privatelink.table...` · `privatelink.dfs...` · `privatelink.web...`
Azure SQL Database	`privatelink.database.windows.net`
Key Vault	`privatelink.vaultcore.azure.net`
App Service	`privatelink.azurewebsites.net`
Container Registry	`privatelink.azurecr.io`

- When a service endpoint is enabled, existing connections are dropped and the public IP rules for that subnet stop working (it now arrives with a private IP). For **Azure SQL** the endpoint only works in the **same region**; Storage has `Microsoft.Storage.Global` for other regions.
- Private endpoint and NSG: the private endpoint **network policies** can be enabled on the subnet to apply NSG / UDR to traffic toward the endpoint NIC. To expose **your own** service behind a Standard ILB to other tenants → **Private Link service**.

6. Other layers

Service	Purpose	Versus NSG
Azure Firewall	Managed, stateful firewall centralized in the hub (`AzureFirewallSubnet` /26): **network** rules, **application rules by FQDN**, **DNAT**, threat intelligence, Premium with IDPS and TLS inspection; **Firewall Policy** and Firewall Manager	NSG filters by IP / port on subnet or NIC; Firewall inspects and centralizes (all traffic arrives via **UDR**)
DDoS Protection	Network Protection (whole VNet) or IP Protection (per public IP); automatic mitigation	—
Just-in-time VM access (Defender for Cloud)	Opens 3389 / 22 **only on request** and for a limited time, creating a temporary NSG rule	"reduce exposure of management ports"

Service	Purpose	Versus NSG
Virtual Network Manager: security admin rules	Global rules evaluated **before** NSGs (Allow, Always allow, Deny)	Central governance
Web Application Firewall	On Application Gateway / Front Door (layer 7, OWASP)	Cheatsheet 14

7. Commands that show up

Task	PowerShell (Az.Network)	Azure CLI
Create NSG and rule	`$r = New-AzNetworkSecurityRuleConfig -Name AllowRDP -Priority 100 -Direction Inbound -Access Allow -Protocol Tcp -SourceAddressPrefix 203.0.113.0/24 -SourcePortRange * -DestinationAddressPrefix * -DestinationPortRange 3389; New-AzNetworkSecurityGroup -Name nsg1 -ResourceGroupName RG1 -Location westeurope -SecurityRules $r`	`az network nsg create -g RG1 -n nsg1` · `az network nsg rule create -g RG1 --nsg-name nsg1 -n AllowRDP --priority 100 --direction Inbound --access Allow --protocol Tcp --source-address-prefixes 203.0.113.0/24 --destination-port-ranges 3389`
Rule with tag or ASG	`-SourceAddressPrefix AzureLoadBalancer` · `-SourceApplicationSecurityGroupId <asg-web> -DestinationApplicationSecurityGroupId <asg-db>`	`--source-address-prefixes AzureLoadBalancer` · `--source-asgs asg-web --destination-asgs asg-db`
Modify / list rules	`$nsg = Get-AzNetworkSecurityGroup -Name nsg1 -ResourceGroupName RG1; Set-AzNetworkSecurityRuleConfig -NetworkSecurityGroup $nsg -Name AllowRDP -Access Deny ...; Set-AzNetworkSecurityGroup -NetworkSecurityGroup $nsg` · `Get-AzNetworkSecurityRuleConfig -DefaultRules`	`az network nsg rule update -g RG1 --nsg-name nsg1 -n AllowRDP --access Deny` · `az network nsg rule list -g RG1 --nsg-name nsg1 --include-default -o table`
Associate	Subnet: `Set-AzVirtualNetworkSubnetConfig ... -NetworkSecurityGroup $nsg; Set-AzVirtualNetwork` · NIC: `$nic.NetworkSecurityGroup = $nsg; Set-AzNetworkInterface`	`az network vnet subnet update -g RG1 --vnet-name vnet1 -n sub1 --network-security-group nsg1` · `az network nic update -g RG1 -n nic1 --network-security-group nsg1` · remove: `--network-security-group ""`
ASG	`New-AzApplicationSecurityGroup -Name asg-web -ResourceGroupName RG1 -Location westeurope` · on the NIC: `$nic.IpConfigurations[0].ApplicationSecurityGroups = @($asg); Set-AzNetworkInterface`	`az network asg create -g RG1 -n asg-web` · `az network nic ip-config update -g RG1 --nic-name nic1 -n ipconfig1 --application-security-groups asg-web`
Effective rules	`Get-AzEffectiveNetworkSecurityGroup -NetworkInterfaceName nic1 -ResourceGroupName RG1`	`az network nic list-effective-nsg -g RG1 -n nic1`
Bastion	`New-AzBastion -ResourceGroupName RG1 -Name bas1 -PublicIpAddressRgName RG1 -PublicIpAddressName pip-bas -VirtualNetworkRgName RG1 -VirtualNetworkName vnet1 -Sku Standard -ScaleUnit 4`	`az network vnet subnet create -g RG1 --vnet-name vnet1 -n AzureBastionSubnet --address-prefixes 10.0.255.0/26` · `az network bastion create -g RG1 -n bas1 --vnet-name vnet1 --public-ip-address pip-bas --sku Standard --scale-units 4 --enable-tunneling true`
Service endpoint	`Set-AzVirtualNetworkSubnetConfig ... -ServiceEndpoint Microsoft.Storage,Microsoft.Sql; Set-AzVirtualNetwork` + `Add-AzStorageAccountNetworkRule -VirtualNetworkResourceId <subnet id>`	`az network vnet subnet update -g RG1 --vnet-name vnet1 -n sub1 --service-endpoints Microsoft.Storage Microsoft.Sql` + `az storage account network-rule add -n st1 --vnet-name vnet1 --subnet sub1`
Private endpoint	`New-AzPrivateEndpoint -Name pe-st1 -ResourceGroupName RG1 -Location westeurope -Subnet $subnet -PrivateLinkServiceConnection (New-AzPrivateLinkServiceConnection -Name c1 -PrivateLinkServiceId <st id> -GroupId blob)` + `New-AzPrivateDnsZone -Name privatelink.blob.core.windows.net` + `New-AzPrivateDnsVirtualNetworkLink` + `New-AzPrivateDnsZoneGroup`	`az network private-endpoint create -g RG1 -n pe-st1 --vnet-name vnet1 --subnet sub1 --private-connection-resource-id <st id> --group-id blob --connection-name c1` · `az network private-dns zone create -g RG1 -n privatelink.blob.core.windows.net` · `az network private-dns link vnet create -g RG1 -z privatelink.blob.core.windows.net -n link1 -v vnet1 -e false` · `az network private-endpoint dns-zone-group create -g RG1 --endpoint-name pe-st1 -n zg --private-dns-zone privatelink.blob.core.windows.net --zone-name blob`

8. Decision tree

```
What does the wording ask for?
├─ Filter by IP / port on a subnet or NIC ..... NSG (remember subnet and NIC must both allow)
├─ Rules by server role without IPs ..... Application security groups
├─ Find out which rule blocks a flow ..... IP flow verify / NSG diagnostics
├─ RDP / SSH without public IP or VPN ..... Azure Bastion (native client, files → Standard)
├─ Open RDP only on request and for a while ..... Just-in-time VM access
├─ PaaS reachable only from my VNet, free, quick ..... Service endpoint + VNet rule
├─ PaaS with no public endpoint, access from on-prem ..... Private endpoint + privatelink zone + DNS
├─ PaaS reachable from a peered VNet by private IP ..... Private endpoint + link the zone to that VNet
├─ FQDN filtering, inspection, centralized DNAT ..... Azure Firewall (hub + UDR)
├─ Protect public IPs from DDoS ..... DDoS Network / IP Protection
├─ Expose my service to another tenant by private IP ..... Private Link service (Standard ILB)
└─ See denied traffic ..... VNet flow logs + Traffic analytics
```

9. Signals → answer

Wording	Answer
"RDP allowed on the NIC NSG but still blocked"	The subnet NSG denies it: allow on both
"rule 100 allows, rule 200 denies"	100 wins (lower number)
"allow web tier to reach database tier on 1433, servers change often"	ASGs + NSG rule

Wording	Answer
"load balancer backend unhealthy after adding NSG"	Allow AzureLoadBalancer (168.63.129.16) to the probe port
"block all outbound internet but allow Azure Storage"	Deny Internet + Allow service tag Storage with a lower priority number (or service endpoint)
"VMs in peered VNet blocked"	VirtualNetwork covers the peering; review custom rules
"NSG can be applied to a VNet"	False: subnet or NIC only
"RDP without public IP from the portal"	Bastion
"Bastion needs native RDP client / file copy / shareable link"	Standard
"session recording"	Premium
"AzureBastionSubnet /27 error"	Minimum /26
"storage only from subnet, no cost"	Service endpoint Microsoft.Storage + VNet rule
"storage not reachable from on-premises after service endpoint"	Expected: add the on-prem NAT public IP or use a private endpoint
"SQL private endpoint resolves to public IP"	Missing `privatelink.database.windows.net` zone linked to the VNet (or custom DNS without forwarding)
"private endpoint from on-premises"	VPN / ExpressRoute + DNS Private Resolver inbound (or forwarder)
"disable public access to the storage account"	Public network access Disabled + private endpoint
"inspect outbound traffic by FQDN"	Azure Firewall (application rules)
"management ports exposed", "reduce attack surface"	JIT VM access

10. Mistakes to avoid

1. Forgetting that between the subnet NSG and the NIC NSG, a deny from either one wins.
2. Giving the Deny rule a lower number than the specific Allow (or the reverse) without thinking about the order.
3. Blocking 168.63.129.16 / AzureLoadBalancer and breaking probes, DHCP or DNS.
4. Associating an NSG to a VNet (not possible) or from another region.
5. Creating the Bastion subnet with another name or with /27.
6. Expecting a service endpoint to work from on-prem or to change DNS.
7. Creating the private endpoint without the private DNS zone linked to every VNet that uses it.
8. Choosing NSG when the ask is FQDN filtering or centralized inspection (Azure Firewall).

Cheatsheet AZ-104 — 14 · DNS and load balancing: Azure DNS, private zones, Load Balancer and its troubleshooting

1. Azure DNS (public zones)

Concept	Detail
Zone	Hosts the records of a domain you already **own** (Azure DNS **does not sell** domains: that is **App Service Domain**). On creation it receives 4 name servers (`ns1-xx.azure-dns.com / .net / .org / .info`): you must **delegate** at the registrar by changing the domain's **NS** records. Several zones with the same name can coexist (in different RGs); only the delegated one answers
Records	A, AAAA, **CNAME**, MX, NS, PTR, SOA, SRV, TXT, CAA; record set = same name and type; **TTL** per record set. A **CNAME** cannot coexist with other types under the same name nor sit at the **apex** (`@`)
Alias records	A / AAAA / CNAME pointing to an **Azure resource** (public IP, Traffic Manager profile, Front Door / CDN endpoint) or to another record set in the zone; they update automatically if the IP changes and **are valid at the apex** ("contoso.com → Traffic Manager")
Subdomains	Child zone (`dev.contoso.com`) delegated with **NS** records in the parent zone
Import / export	BIND zone files: `az network dns zone import / export`
Permissions	**DNS Zone Contributor**; RBAC per zone or per record set
DNSSEC	Zone signing (DS in the parent)
Verify	`nslookup -type=SOA contoso.com` · `nslookup www.contoso.com ns1-01.azure-dns.com` · `dig`

2. Private resolution

Mechanism	What it resolves	Detail
Azure-provided DNS (default, 168.63.129.16)	Names of VMs in the **same VNet** (`internal.cloudapp.net` suffix) and public names	Does not cross VNets; cannot be customized
Private DNS zone	Your own names (`contoso.internal`, `privatelink.*`) for one or several VNets	It is **linked** to each VNet: link with **autoregistration** (the VNet is a *registration* VNet: Azure creates and deletes the **A** records of its VMs; **only one registration zone per VNet**; up to 100 per zone) or **resolution** only (up to 1,000 per zone); a single link per zone-VNet pair; also **reverse** zones (PTR) and **split-horizon** (same zone public and private with different records)
Custom DNS (on the VNet or NIC)	Whatever your server decides (DC, BIND)	To resolve AD / on-prem; the server must **forward** to 168.63.129.16 for Azure names; VMs need a restart after changing the setting
Azure DNS Private Resolver	**VM-less** bridge between on-prem and Azure: **inbound endpoint** (IP in your VNet to which on-prem forwards the queries for private / privatelink zones) and **outbound endpoint + ruleset** (Azure forwards specific domains to the on-prem DNS servers)	Replaces VM-based forwarders; linked to VNets like a zone

- "VMs in VNet A must resolve VMs in VNet B by name" → a Private DNS zone linked to both (registration on those that create VMs) **or** custom DNS; peering only provides IP, not names.
- "On-premises clients must resolve the private endpoint of the storage account" → Private Resolver **inbound** + conditional forwarding of `privatelink.blob.core.windows.net` from the on-prem DNS servers.
- "Azure VMs must resolve contoso.local on-premises" → **outbound** endpoint with ruleset `contoso.local → on-prem DNS IP` (or custom DNS pointing to the DC).

3. Azure Load Balancer (layer 4)

Component	Detail
SKU	**Standard**: backend any VM / VMSS in a **single VNet** (by NIC or by IP), **zone-redundant**, **99.99 %** SLA, TCP / HTTP / **HTTPS** probes, **HA ports**, **outbound rules**, multiple frontends, TCP reset, **secure by default** (inbound traffic requires an **NSG** that allows it), global peering and Private Link; **Standard** public IP mandatory. **Basic**: **retired on September 30, 2025** (no SLA, availability set / VMSS only, open by default). **Gateway**: for NVA chains
Public vs internal	Public: frontend with a public IP (internet → VMs) · **Internal (ILB)**: frontend with a **private IP** from a subnet (web tier → app tier, on-prem → Azure)
Backend pool	VM / VMSS NICs or **IP addresses** in the same VNet; **no** availability set needed; VMs across several zones
Health probe	TCP (handshake) · HTTP / HTTPS (GET to a path; only **200** = healthy); default interval **5 s** in the portal (15 s via API), failure threshold; source **168.63.129.16** (AzureLoadBalancer tag); if **all** fail, Standard keeps the existing TCP flows but sends no new ones
Load balancing rule	Frontend IP:port → backend port, **TCP / UDP** protocol, probe, **session persistence** (None = 5-tuple hash · **Client IP** · Client IP and protocol), **idle timeout** 4-100 min (default 4), **floating IP** (Direct Server Return: for **SQL Always On listener** and NVAs), TCP reset
HA ports (ILB only)	One rule for **all ports** (and protocols): highly available NVAs
Inbound NAT rule	Port forwarding to **one** specific VM (e.g. frontend 50001 → VM1:3389, 50002 → VM2:3389); no probe required
Outbound rule	Explicit SNAT so the backend reaches the internet through the LB's public IP (**public** Standard LB only); a Standard ILB **provides no outbound access**: NAT Gateway (recommended), PIP or public LB with outbound rule
Global tier	Cross-region LB (public) that distributes across regional LBs

- It is **layer 4**: it does not terminate TLS, does not route by URL or host, does not inspect. For that, **Application Gateway**.

4. Choosing the load balancer

Service	Layer	Scope	Purpose	Signal
Load Balancer	4 (TCP / UDP)	Regional	VMs / VMSS, non-HTTP traffic, minimum latency, ILB between tiers, SQL listener	"TCP", "UDP", "internal", "SQL Always On", "RDP port forwarding"
Application Gateway	7 (HTTP / HTTPS)	Regional	**WAF**, TLS termination, **URL / host-based routing** (multi-site), cookie affinity, redirection, rewrite, autoscale v2, VM / VMSS / App Service / IP backends; dedicated subnet (/24 recommended)	"web application firewall", "/images to pool A", "multiple sites one IP", "SSL offload"
Front Door	7 global	Global (anycast)	Global entry point for multi-region web apps with **CDN**, WAF, TLS, fast failover, caching	"global", "lowest latency worldwide", "web app in two regions"
Traffic Manager	DNS	Global	Returns the endpoint IP according to the method: **Priority** (failover), **Weighted**, **Performance** (latency), **Geographic**, Multivalue, Subnet; Azure, **external** or nested endpoints; does not proxy traffic, DNS TTL	"DNS-based", "failover to another region", "route by user location", "any protocol"

- "HTTP + regional" → App Gateway · "HTTP + global" → Front Door · "non-HTTP + regional" → Load Balancer · "non-HTTP + global" or external endpoints → Traffic Manager. They combine: Traffic Manager / Front Door in front of regional App Gateways or LBs.

5. Load Balancer troubleshooting

Symptom	Usual cause	Fix
All backends **unhealthy**	The app is not listening on the probe port, the **NSG** blocks AzureLoadBalancer, the OS firewall, or the HTTP path does not return **200**	Check the listener, allow 168.63.129.16, adjust the probe (TCP vs HTTP and path)
Public LB unreachable from the internet	Standard LB is **secure by default**: NSG rule missing on subnet / NIC; public IP SKU does not match	Allow in NSG; Standard IP with Standard LB
The same VM always receives the traffic	**Session persistence** Client IP, or few clients (hash)	Set None if affinity is not needed
VM behind an ILB has no internet	Standard ILB provides no SNAT	NAT Gateway / outbound rule of a public LB / PIP
VM from another VNet cannot be added	Backend from **one VNet** only	Peering is not enough; move it or use App Gateway by IP
RDP to each VM on a different port	—	Inbound NAT rules
SQL listener does not respond	**Floating IP** missing on the rule	Enable floating IP and configure the listener
Asymmetric traffic, timeouts with NVA	UDRs without HA ports / without symmetric flows	HA ports + consistent routes
SNAT port exhaustion	Many outbound connections through the same IP	NAT Gateway (64,512 ports per IP, several IPs)
Metrics	**Health probe status**, **Data path availability**, SNAT connection count	Monitor → Metrics of the LB

6. Commands that show up

Task	PowerShell (Az.Network / Az.Dns)	Azure CLI
Zone and records	`New-AzDnsZone -Name contoso.com -ResourceGroupName RG1` · `New-AzDnsRecordSet -Name www -RecordType A -ZoneName contoso.com -ResourceGroupName RG1 -Ttl 3600 -DnsRecords (New-AzDnsRecordConfig -IPv4Address 203.0.113.10)` · alias: `-TargetResourceId <pip id>` · `Get-AzDnsRecordSet`	`az network dns zone create -g RG1 -n contoso.com` · `az network dns record-set a add-record -g RG1 -z contoso.com -n www -a 203.0.113.10` · `az network dns record-set a create -g RG1 -z contoso.com -n @ --target-resource <pip id>` · `az network dns record-set cname set-record -g RG1 -z contoso.com -n app -c app1.azurewebsites.net` · `az network dns zone show -g RG1 -n contoso.com --query nameServers`
Private zone	`New-AzPrivateDnsZone -Name contoso.internal -ResourceGroupName RG1` · `New-AzPrivateDnsVirtualNetworkLink -ZoneName contoso.internal -ResourceGroupName RG1 -Name link1 -VirtualNetworkId <id> -EnableRegistration` · `New-AzPrivateDnsRecordSet -Name db -RecordType A -ZoneName contoso.internal -ResourceGroupName RG1 -Ttl 300 -PrivateDnsRecords (New-AzPrivateDnsRecordConfig -IPv4Address 10.0.1.4)`	`az network private-dns zone create -g RG1 -n contoso.internal` · `az network private-dns link vnet create -g RG1 -z contoso.internal -n link1 -v vnet1 --registration-enabled true` · `az network private-dns record-set a add-record -g RG1 -z contoso.internal -n db -a 10.0.1.4`
VNet DNS	`$vnet.DhcpOptions.DnsServers = @("10.0.0.4"); Set-AzVirtualNetwork`	`az network vnet update -g RG1 -n vnet1 --dns-servers 10.0.0.4 10.0.0.5` (back to Azure: `--dns-servers null`)
Private Resolver	`New-AzDnsResolver` · `New-AzDnsResolverInboundEndpoint` · `New-AzDnsResolverOutboundEndpoint` · `New-AzDnsForwardingRuleset` · `New-AzDnsForwardingRulesetForwardingRule`	`az dns-resolver create` · `az dns-resolver inbound-endpoint create` · `az dns-resolver outbound-endpoint create` · `az dns-resolver forwarding-ruleset create` · `az dns-resolver forwarding-rule create --domain-name contoso.local. --target-dns-servers '[{"ip-address":"192.168.1.10"}]'`

Task	PowerShell (Az.Network / Az.Dns)	Azure CLI
Load Balancer	`New-AzLoadBalancer -Name lb1 -ResourceGroupName RG1 -Location westeurope -Sku Standard -FrontendIpConfiguration $fe -BackendAddressPool $be -Probe $probe -LoadBalancingRule $rule` with `New-AzLoadBalancerFrontendIpConfig`, `New-AzLoadBalancerBackendAddressPoolConfig`, `New-AzLoadBalancerProbeConfig -Protocol Http -Port 80 -RequestPath / -IntervalInSeconds 5 -ProbeCount 2`, `New-AzLoadBalancerRuleConfig -Protocol Tcp -FrontendPort 80 -BackendPort 80 -EnableFloatingIP -LoadDistribution SourceIP` · NAT: `Add-AzLoadBalancerInboundNatRuleConfig` · add a NIC: `$nic.IpConfigurations[0].LoadBalancerBackendAddressPools.Add($pool); Set-AzNetworkInterface`	`az network lb create -g RG1 -n lb1 --sku Standard --public-ip-address pip-lb --frontend-ip-name fe --backend-pool-name be` (internal: `--vnet-name vnet1 --subnet sub1 --private-ip-address 10.0.1.100`) · `az network lb probe create -g RG1 --lb-name lb1 -n p80 --protocol Http --port 80 --path / --interval 5 --threshold 2` · `az network lb rule create -g RG1 --lb-name lb1 -n r80 --protocol Tcp --frontend-port 80 --backend-port 80 --frontend-ip-name fe --backend-pool-name be --probe-name p80 --load-distribution SourceIP --floating-ip true --idle-timeout 15` · `az network lb inbound-nat-rule create -g RG1 --lb-name lb1 -n rdp1 --protocol Tcp --frontend-port 50001 --backend-port 3389 --frontend-ip-name fe` · `az network nic ip-config address-pool add -g RG1 --nic-name nic1 -n ipconfig1 --lb-name lb1 --address-pool be` · `az network lb outbound-rule create`
App Gateway / Traffic Manager (quick)	`New-AzApplicationGateway` · `New-AzTrafficManagerProfile -TrafficRoutingMethod Priority -RelativeDnsName contoso -Ttl 30 -MonitorProtocol HTTPS -MonitorPort 443 -MonitorPath /` + `New-AzTrafficManagerEndpoint`	`az network application-gateway create --sku WAF_v2 ...` · `az network traffic-manager profile create -n tm1 -g RG1 --routing-method Performance --unique-dns-name contoso` · `az network traffic-manager endpoint create --type azureEndpoints --target-resource-id <pip id>`

7. Decision tree

```
What does the wording ask for?
├─ Host a public domain I already own ..... Azure DNS zone + delegate NS at the registrar
├─ Buy a domain from Azure ..... App Service Domain
├─ contoso.com (apex) → Traffic Manager / public IP ..... Alias record (not CNAME)
├─ Private names across several VNets ..... Private DNS zone linked to all the VNets
├─ On-prem resolves private zones / private endpoints ..... DNS Private Resolver inbound + on-prem conditional forwarding
├─ Azure resolves on-prem domains ..... Private Resolver outbound + ruleset (or custom DNS)
├─ Distribute TCP / UDP across VMs in one VNet ..... Standard Load Balancer (public or internal)
├─ RDP to each VM on different ports of one IP ..... Inbound NAT rules
├─ SQL Always On listener ..... ILB + floating IP
├─ NVAs in HA ..... ILB + HA ports
├─ Internet egress for VMs behind an ILB ..... NAT Gateway (or public LB outbound rule)
├─ WAF, TLS offload, URL routing, multiple sites ..... Application Gateway
├─ Global multi-region web with CDN and WAF ..... Front Door
└─ DNS-based failover between regions, any protocol ..... Traffic Manager (Priority)
```

8. Signals → answer

Wording	Answer
"DNS zone created but the domain does not resolve"	NS delegation at the registrar is missing
"point the root domain to the web app / Traffic Manager"	Alias record A / CNAME-alias (CNAME not valid at the apex)
"CNAME and TXT with the same name"	Not allowed: CNAME cannot coexist with other types
"VMs register their names automatically"	Private zone with autoregistration link (one per VNet)
"VNet already has a registration zone"	Only one registration zone per VNet: the second one is resolution only
"private endpoint resolves to public IP from on-premises"	Private Resolver inbound + conditional forwarder
"replace the DNS forwarder VMs"	DNS Private Resolver
"Basic load balancer"	Retired: Standard
"load balancer across VMs in two VNets"	No: one VNet per backend pool
"VMs must be in an availability set to use the LB"	Only with Basic (retired); Standard does not require it
"backend shows unhealthy"	Probe: port, NSG (AzureLoadBalancer), HTTP 200
"requests from the same client must go to the same VM"	Session persistence Client IP
"connections drop after 4 minutes idle"	Raise the idle timeout (up to 100 min) or keep-alive
"VMs behind the internal LB cannot reach the internet"	NAT Gateway / outbound rule
"SSL termination and URL routing"	Application Gateway
"protect the web app with WAF"	Application Gateway WAF_v2 or Front Door Premium
"route users to the closest region"	Traffic Manager Performance (or Front Door for HTTP)
"failover to secondary region only when primary fails"	Traffic Manager Priority
"send 10 % of traffic to the new deployment"	Traffic Manager Weighted (DNS) or App Service slots

9. Mistakes to avoid

1. Forgetting the NS delegation or placing a CNAME at the apex.

2. Assuming peering resolves names: a Private DNS zone or custom DNS is needed.
3. Linking two zones with autoregistration to the same VNet.
4. Choosing Load Balancer for WAF, TLS or URL routing, or Application Gateway for UDP.
5. Designing with Basic LB or Basic IP (retired) or mixing IP and LB SKUs.
6. Blocking the probe (168.63.129.16) with the NSG or probing a port where nothing listens.
7. Expecting a Standard ILB to provide internet egress.
8. Forgetting floating IP for SQL Always On or HA ports for NVAs.

Cheatsheet AZ-104 — 15 · Azure Monitor: metrics, logs and KQL, alerts, Insights and Network Watcher

1. Azure Monitor map

Data	What it is	Where it comes from	Where to query it
Platform metrics	Numeric series (CPU %, Disk IOPS, transactions...) every **1 min**, retained **93 days**, nothing to configure	All resources	**Metrics explorer** (avg / min / max / sum / count aggregation, split by dimension, pin to dashboard), metric alerts
Guest OS metrics	Memory, logical disk, processes inside the VM	**Azure Monitor Agent** + Data Collection Rule (or Diagnostics extension to storage)	Metrics explorer, Log Analytics (`InsightsMetrics`, `Perf`)
Activity log	Control plane: who created, deleted, modified, started; Service Health, Policy, Autoscale, Security	Automatic per subscription, **90 days**	Activity log blade; to Log Analytics (`AzureActivity`) with a diagnostic setting to retain longer
Resource logs	Data plane of the resource (StorageBlobLogs, AppServiceHTTPLogs, NSG, Key Vault audit...)	**Diagnostic settings** per resource	Log Analytics, Storage, Event Hub
OS logs	Event logs, Syslog, counters, files	AMA + DCR	`Event`, `Syslog`, `Perf`
Application Insights	App telemetry: requests, dependencies, exceptions, **availability tests**, Live Metrics	SDK / auto-instrumentation	`AppRequests`, `AppExceptions`...; Application Map
Insights	Ready-made views: **VM insights** (Performance + Map; AMA + Dependency agent), **Storage insights**, **Network insights**, Container insights	Enabled per resource	Monitor → Insights

2. Diagnostic settings

Destination	Purpose	Signal
Log Analytics workspace	Query with **KQL**, log alerts, workbooks, Insights	"query", "correlate", "alert on log"
Storage account	**Cheap archive** and long retention (days per category or lifecycle)	"retain for 7 years", "audit archive"
Event Hub	Streaming to a **SIEM** or third parties (Splunk, QRadar)	"third-party SIEM", "stream"
Partner solution	Datadog, Elastic...	—

- Up to **5 diagnostic settings** per resource; each category only once per destination. Categories: logs (**allLogs** / **audit** groups or individual categories) and metrics. The **Activity log** is sent from Monitor → Activity log → Export (subscription-level diagnostic setting).
- VMs: the diagnostic setting only covers host metrics; anything **inside** the OS needs **AMA + DCR** (Log Analytics agent / MMA retired in 2024). The **DCR** defines what to collect (counters, event logs, Syslog, custom text / JSON, IIS) and to which workspace; it is associated with VMs, VMSS or Arc; needs a **managed identity** on the VM.

3. Log Analytics and KQL

Aspect	Detail
Workspace	Own region, collects data from **any subscription and region**; priced per GB ingested (PAYG or **commitment tiers**) and per retention; **daily cap** to limit cost
Retention	Default **30 days** (90 for `AzureActivity`, `Usage` and App Insights tables); analytics up to **730 days**; **total retention** (long-term, queryable with search jobs) up to **12 years**; per workspace or **per table**
Table plans	**Analytics** (everything), **Basic** (cheap, 30 days, limited queries), **Auxiliary** (very cheap, high-volume logs)
Access	**Log Analytics Reader / Contributor** roles; *workspace-context* mode (the whole workspace) or *resource-context* (only data of the resources you have access to)
Tables that show up	`Heartbeat`, `Perf`, `Event`, `Syslog`, `InsightsMetrics`, `AzureActivity`, `AzureDiagnostics`, `AzureMetrics`, `SecurityEvent`, `StorageBlobLogs`, `AppRequests`, `Update`

```
Heartbeat | where TimeGenerated > ago(1h) | summarize LastBeat = max(TimeGenerated) by Computer   // reporting VMs
Perf | where ObjectName == "Processor" and CounterName == "% Processor Time"
     | summarize avg(CounterValue) by Computer, bin(TimeGenerated, 5m) | render timechart
Event | where EventLevelName == "Error" and TimeGenerated > ago(24h) | summarize count() by Computer, Source | top 10 by count_
AzureActivity | where OperationNameValue has "delete" and ActivityStatusValue == "Success"
              | project TimeGenerated, Caller, ResourceGroup, OperationNameValue
InsightsMetrics | where Name == "AvailableMB" | summarize min(Val) by Computer                    // free memory (VM insights)
StorageBlobLogs | where StatusCode >= 400 | summarize count() by StatusText, bin(TimeGenerated, 1h)
search "vm-sql01" | where TimeGenerated > ago(1d) | distinct $table                              // which tables it appears in
```

- Operators: `where` (filters), `project` (columns), `extend` (computes), `summarize ... by` (groups; `count()`, `avg()`, `max()`, `dcount()`), `bin()` (time buckets), `sort by` / `order by`, `top N by`, `take` / `limit`, `distinct`, `join`, `parse`, `render`, `let`. Time: `ago(1h)`, `between`, `startofday()`. Text comparison: == exact, =~ case-insensitive, `has` (whole word, fast), `contains` (substring), `startswith`. Save: **queries**, **functions**, **workbooks**.

4. Alerts

Rule type	Signal	How it evaluates	When
Metric	Platform, guest or custom metric	Every **1 min**; aggregation window (e.g. 5 min), **static** or **dynamic** (ML) threshold; one rule for **several resources** of the same type and region; **stateful** (resolves after 3 OK evaluations)	"CPU > 80 % for 5 minutes", near real-time
Log search	**KQL** query on the workspace	Frequency 1 min to 1 day; condition by **number of results** or **measure** (metric from the query); **split by dimensions** (one alert per resource); Basic / Auxiliary tables are not supported	"event ID 4625 more than 5 times", "service stopped"
Activity log	Control operations (Administrative), **Service Health**, **Resource Health**, Policy, Autoscale, Security	**Stateless**: every event fires; filter by operation and status	"when a VM is deleted", "planned maintenance in West Europe", "resource became unavailable"
Smart detection / Prometheus	App Insights anomalies / Prometheus metrics	—	Apps / AKS

Piece	Detail
Rule	Scope (resource, RG, subscription) + signal and condition + **severity 0 (critical) - 4 (verbose)** + actions + name; **Monitoring Contributor** (create) / **Monitoring Reader** (view) roles; fired alerts are kept **30 days**
Action group	Reusable, **global**, up to **5 per rule**: **email**, **SMS**, **push** (Azure app), **voice** notifications, **Email Azure Resource Manager role** (Owner, Contributor, Reader, Monitoring Contributor / Reader of the subscription); **Automation runbook**, **Azure Function**, **Logic App**, **Event Hub**, **ITSM**, **webhook** / **secure webhook** (Entra ID) actions; **common alert schema**; rate limiting on SMS / email / voice
Alert processing rules	Applied to already fired alerts within a scope: **suppress notifications** (maintenance window, by schedule) or **add action groups** to a whole RG / subscription without touching each rule
States	*User response*: New → Acknowledged → Closed · *Monitor condition*: Fired / Resolved (stateful)

- "Notify the team when the VM is deleted" → activity log alert (Administrative, Delete Virtual Machine, Succeeded). "Get notified of Azure outages in my region" → **Service Health alert** with action group. "Run a runbook when CPU is high" → metric alert + action group with Automation runbook. "Stop emails during the maintenance window" → alert processing rule. "Alert on a KQL result" → log search alert. "The VM alert must also apply to new VMs in the RG" → metric alert scoped to the **resource group** (multi-resource).

5. Insights and Network Watcher

Tool	What it does	Requirements / detail
VM insights	Performance (CPU, memory, disk, network) and **Map** (processes and connections)	AMA + **Dependency agent**; `InsightsMetrics` table; a dedicated DCR
Storage insights	Transactions, latency, availability, capacity per account; errors	Free metrics; logs with diagnostic settings (`StorageBlobLogs`...)
Network insights	Topology, health of LB / App Gateway / gateways, **Connection monitor**, Traffic analytics	Network Watcher
Connection monitor	**Continuous** tests from VMs (Network Watcher extension) or Arc to a VM, IP, FQDN or URL over TCP / ICMP / HTTP; **% checks failed** and **round-trip time** metrics; alerts	Replaces Network performance monitor
IP flow verify / **NSG diagnostics** / **Next hop** / **Effective security rules**	One-off diagnosis of NSGs and routes	Cheatsheet 12
Connection troubleshoot	One-off source → destination test with hops and latency	Network Watcher extension on the source
Packet capture	pcap capture on a VM / VMSS with filters and limit (5 h), to storage or disk	Extension
VPN troubleshoot	Diagnosis of gateways and connections	Logs to storage
Flow logs + **Traffic analytics**	**VNet** flow logging (NSG flow logs are retired in Sept. 2027) and analysis in Log Analytics	Storage account + workspace
Limits	**1 Network Watcher per region and subscription** (RG `NetworkWatcherRG`), enabled automatically when a VNet is created; 100 connection monitors per region	—

6. Commands that show up

Task	PowerShell (Az.Monitor / Az.OperationalInsights / Az.Network)	Azure CLI
Workspace	`New-AzOperationalInsightsWorkspace -ResourceGroupName RG1 -Name law1 -Location westeurope -Sku PerGB2018 -RetentionInDays 90` · `Set-AzOperationalInsightsWorkspace -RetentionInDays 365` · `Update-AzOperationalInsightsTable -TableName Event -RetentionInDays 30 -TotalRetentionInDays 730`	`az monitor log-analytics workspace create -g RG1 -n law1 --retention-time 90` · `az monitor log-analytics workspace update --retention-time 365` · `az monitor log-analytics workspace table update -n Event --retention-time 30 --total-retention-time 730`
Query	`Invoke-AzOperationalInsightsQuery -WorkspaceId <id> -Query "<KQL query>"`	`az monitor log-analytics query -w <id> --analytics-query "<KQL query>"`

Task	PowerShell (Az.Monitor / Az.OperationalInsights / Az.Network)	Azure CLI
Diagnostic setting	`New-AzDiagnosticSetting -Name ds1 -ResourceId <id> -WorkspaceId <law id> -Log (New-AzDiagnosticSettingLogSettingsObject -CategoryGroup allLogs -Enabled $true) -Metric (New-AzDiagnosticSettingMetricSettingsObject -Category AllMetrics -Enabled $true)`	`az monitor diagnostic-settings create -n ds1 --resource <id> --workspace <law id> --logs '[{"categoryGroup":"allLogs","enabled":true}]' --metrics '[{"category":"AllMetrics","enabled":true}]'` · to storage: `--storage-account st1` · to Event Hub: `--event-hub eh1 --event-hub-rule <auth rule id>`
Activity log	`Get-AzActivityLog -ResourceGroupName RG1 -StartTime (Get-Date).AddDays(-1)` (formerly `Get-AzLog`)	`az monitor activity-log list -g RG1 --offset 1d --query "[].{op:operationName.value, who:caller}"`
Action group	`$email = New-AzActionGroupEmailReceiverObject -Name ops -EmailAddress ops@contoso.com; New-AzActionGroup -ResourceGroupName RG1 -Name ag1 -ShortName ag1 -Location global -EmailReceiver $email`	`az monitor action-group create -g RG1 -n ag1 --short-name ag1 --action email ops ops@contoso.com --action sms oncall 34 600000000`
Metric alert	`$cond = New-AzMetricAlertRuleV2Criteria -MetricName "Percentage CPU" -TimeAggregation Average -Operator GreaterThan -Threshold 80; Add-AzMetricAlertRuleV2 -Name cpu80 -ResourceGroupName RG1 -TargetResourceId <vm id> -Condition $cond -WindowSize 00:05:00 -Frequency 00:01:00 -Severity 2 -ActionGroupId <ag id>`	`az monitor metrics alert create -n cpu80 -g RG1 --scopes <vm id> --condition "avg Percentage CPU > 80" --window-size 5m --evaluation-frequency 1m --severity 2 --action <ag id>`
Log alert	`New-AzScheduledQueryRule -Name errs -ResourceGroupName RG1 -Location westeurope -Scope <law id> -CriterionAllOf (New-AzScheduledQueryRuleConditionObject -Query '<KQL query>' -TimeAggregation Count -Operator GreaterThan -Threshold 5) -WindowSize 00:15:00 -EvaluationFrequency 00:05:00 -Severity 2 -ActionGroupResourceId <ag id>`	`az monitor scheduled-query create -n errs -g RG1 --scopes <law id> --condition "count 'Placeholder_1' > 5" --condition-query Placeholder_1="<KQL query>" --window-size 15m --evaluation-frequency 5m --severity 2 --action-groups <ag id>`
Activity log alert	`New-AzActivityLogAlert -Name vmdel -ResourceGroupName RG1 -Scope /subscriptions/<id> -Condition (New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -Equal Administrative -Field category),(... -Equal Microsoft.Compute/virtualMachines/delete -Field operationName) -Action (New-AzActivityLogAlertActionGroupObject -Id <ag id>)`	`az monitor activity-log alert create -n vmdel -g RG1 --scope /subscriptions/<id> --condition category=Administrative and operationName=Microsoft.Compute/virtualMachines/delete --action-group <ag id>`
AMA and DCR	`Set-AzVMExtension -Publisher Microsoft.Azure.Monitor -ExtensionType AzureMonitorWindowsAgent -Name AMA -EnableAutomaticUpgrade $true ...` · `New-AzDataCollectionRule` · `New-AzDataCollectionRuleAssociation -TargetResourceId <vm id> -DataCollectionRuleId <dcr id> -AssociationName a1`	`az vm extension set -n AzureMonitorWindowsAgent --publisher Microsoft.Azure.Monitor --vm-name vm1 -g RG1 --enable-auto-upgrade true` · `az monitor data-collection rule create ... --rule-file dcr.json` · `az monitor data-collection rule association create -n a1 --rule-id <dcr id> --resource <vm id>`
Network Watcher	`Get-AzNetworkWatcher` · `New-AzNetworkWatcherConnectionMonitor` · `New-AzNetworkWatcherPacketCapture -NetworkWatcher $nw -TargetVirtualMachineId <id> -PacketCaptureName pc1 -StorageAccountId <id> -TimeLimitInSeconds 300` · `Test-AzNetworkWatcherIPFlow` · `Get-AzNetworkWatcherNextHop`	`az network watcher configure -l westeurope --enabled true` · `az network watcher connection-monitor create` · `az network watcher packet-capture create -g RG1 --vm vm1 -n pc1 --storage-account st1 --time-limit 300` · `az network watcher test-connectivity --source-resource vm1 --dest-address www.contoso.com --dest-port 443` · `az network watcher flow-log create --vnet vnet1 --storage-account st1 --workspace law1 --traffic-analytics true`

7. Decision tree

```
What does the wording ask for?
├─ See CPU / IOPS of a resource right now ..... Metrics explorer (platform metrics)
├─ Free memory or logical disk of a VM ..... AMA + DCR (guest metrics) or VM insights
├─ Who deleted a resource ..... Activity log (90 days) → to Log Analytics for more
├─ Retain audit logs cheaply for years ..... Diagnostic setting → Storage account
├─ Send logs to an external SIEM ..... Diagnostic setting → Event Hub
├─ Query and correlate with KQL ..... Diagnostic setting → Log Analytics
├─ 2 years queryable / 7 years total retention ..... Workspace or table: analytics 730 d + total retention
├─ Alert "CPU > 80 % 5 min" on all VMs in the RG ..... Metric alert scoped to the RG
├─ Alert on text in the event log ..... Log search alert (KQL)
├─ Alert when a resource is deleted / modified ..... Activity log alert (Administrative)
├─ Notify about Azure incidents in my region ..... Service Health alert
├─ Run a runbook / function when it fires ..... Action group (Automation runbook / Function)
├─ Silence alerts during the maintenance window ..... Alert processing rule (suppress)
├─ Process and dependency map of a VM ..... VM insights (Map)
├─ Watch VM → service latency continuously ..... Connection monitor
└─ Capture packets from a VM ..... Network Watcher packet capture
```

8. Signals → answer

Wording	Answer
"metrics older than 93 days"	Send them to Log Analytics (diagnostic setting, AllMetrics)
"activity log for 1 year"	Activity log diagnostic setting to Log Analytics (retention) or Storage
"memory usage of the VM in Metrics"	Guest metrics: AMA + DCR (or VM insights)
"Log Analytics agent / MMA"	Retired: migrate to Azure Monitor Agent + DCR
"collect IIS logs / custom log file"	DCR with custom text logs
"query logs from VMs in several subscriptions"	One central workspace (collects from any subscription)
"users should see only logs of their resources"	Resource-context access (permission on the resource)

Wording	Answer
"reduce ingestion cost for verbose logs"	Basic / Auxiliary table, daily cap, commitment tier
"keep audit logs for 10 years"	Total retention (up to 12 years) or archive in Storage
"alert when VM CPU > 80 % for 10 minutes"	Metric alert, aggregation granularity 10 min, average > 80
"alert that also applies to VMs created later"	Scope at RG / subscription (multi-resource)
"alert when Windows service stops"	Log search alert on `Event` (ID 7036)
"alert when a resource group is deleted"	Activity log alert, delete operation
"notify Owners of the subscription"	Email Azure Resource Manager role = Owner in the action group
"same action group for 50 alert rules without editing each"	Alert processing rule that adds the action group
"no alerts during the weekly maintenance"	Suppression alert processing rule with recurrence
"alert severity for a critical outage"	Sev 0
"test connectivity continuously from VM to on-prem SQL"	Connection monitor (TCP 1433)
"capture traffic on the VM for support"	Packet capture (Network Watcher extension)
"Network Watcher not available in the region"	Enable it (one per region and subscription)

9. Mistakes to avoid

1. Looking for memory or logical disk in a VM's platform metrics: an agent is needed.
2. Sending to Storage what later has to be queried with KQL (or to Log Analytics what only needs cheap archiving).
3. Sticking with the Log Analytics agent (MMA) or the diagnostics extension for OS logs.
4. Assuming the Activity log lasts more than 90 days without exporting it.
5. Creating one metric alert per VM when a single one scoped to the RG covers the new ones.
6. Using a log alert for "VM deleted" (that is Activity log) or a metric alert for event log text (that is log search).
7. Editing 50 rules to add an action group instead of an alert processing rule.
8. Forgetting the Network Watcher extension for packet capture / connection troubleshoot / Connection monitor.

Cheatsheet AZ-104 — 16 · Backup and recovery: vaults, policies, restore and Site Recovery

1. The two vaults

	Recovery Services vault	Backup vault
What it protects	**Azure VMs**, **Azure Files** (SMB), **SQL Server and SAP HANA in VMs**, **MARS** (Windows files, folders and system state), **MABS / DPM** (on-prem), and **Azure Site Recovery**	**Managed disks**, **Blobs** (operational and vaulted), **Azure Database for PostgreSQL / MySQL**, **AKS**
Region	For **backup**, the vault must be in the **same region** as what it protects; for **Site Recovery**, in **any region except the source one**	Same region as the resource
Redundancy	**LRS / ZRS / GRS** (with optional **Cross Region Restore**); set **before the first backup**	LRS / ZRS / GRS
Security	**Soft delete** (14 days by default, free, **always on**, extendable to 180), **MUA** (multi-user authorization with **Resource Guard** for critical operations), vault **immutability**, CMK, private endpoints	Same
Management	**Business Continuity Center** (formerly Backup center): single view of vaults, jobs, policies, alerts and reports across all subscriptions	Same

- "Vault in another region to protect against regional disaster" → that is **GRS + Cross Region Restore** (backup) or **Site Recovery** (DR); VM backup requires the vault in its region.

2. Azure VM backup

Aspect	Detail
How it works	The **VMSnapshot** extension (Windows: **VSS**, app-consistent; Linux: file-consistent, app-consistent with **pre/post scripts**) takes a **snapshot** (instant tier, in the RG `AzureBackupRG_<region>_1`) and then transfers it **incrementally** to the vault (can take up to 8 h); there is also the **agentless crash-consistent** mode
Standard policy	Once a **day or a week**; snapshot retained **1-5 days** (instant restore); daily / weekly / monthly / yearly retention (up to 9,999 days)
Enhanced policy	**Several per day** (every 4, 6, 8 or 12 h), snapshot up to **30 days**, **zone-redundant** snapshots; **mandatory** for **Premium SSD v2** / **Ultra** disks and confidential VMs; for **Trusted Launch** it is the portal's option (Standard only via recent CLI / PowerShell / REST)
Restore	**Create new VM** (new, same or different RG / VNet) · **Restore disks** (to a storage account / RG; then the VM is created manually or with a template; to change settings) · **Replace existing** (replaces the disks of the original VM; keeps NIC and IP) · **Cross Region Restore** (in the paired region; requires **GRS + CRR** enabled) · **Cross subscription restore** · **File recovery** (script that mounts the recovery point as an iSCSI disk for 12 h to copy individual files) · restore to a specific **zone**; **selective disk** backup / restore
Stop	**Stop backup**: *retain backup data* (no more copies are taken, restores still possible) or *delete backup data* (goes into **soft delete** for 14 days)
Requirements	VM and vault in the **same region**; **VM agent** installed; disks up to **64 TB** (32 disks, 512 TB in total); VMs with **ADE** need Backup to read the keys from the Key Vault (BEK / KEK copy); no **CanNotDelete locks** on the snapshot RG; port 443 to the Storage / Backup service tags
Cost	Per **protected instance** (actual data size, excluding the temporary disk) + vault storage; **vault-archive tier** for long retention

- "Backup every 4 hours" → Enhanced. "Restore to a different VNet / change size" → Restore disks + create VM. "Restore a single file" → File recovery. "Backup fails after enabling encryption" → Backup permissions on Key Vault. "Backup fails: lock" → remove the lock from `AzureBackupRG`.

3. Other workloads

Workload	How	Exam details
Azure Files	Share **snapshots** orchestrated by Backup, stored in the **same storage account**; daily or **hourly** policy (several per day); GFS retention	Restore of the **whole share** or **item-level** (files / folders) in place or to another location; Backup enables soft delete and places a **lock** on the account; needs the account and the vault in the same region
SQL Server / SAP HANA in VM	Workload extension in the VM; full, differential and **log every 15 min**	Restores **individual databases** to a point in time
MARS agent (Microsoft Azure Recovery Services)	Agent on **Windows** (on-prem or VM) for **files, folders and system state**; registered with the **vault credentials** and a **passphrase** (if lost, no restore)	Maximum **3 backups per day**; **no Linux**, no full VM, not app-aware; bandwidth throttling; restores to the same machine or another one
MABS / DPM	On-prem backup server (Hyper-V, VMware, SQL, Exchange, SharePoint) with disk → vault copy	"back up on-premises VMware VMs to Azure" → MABS
Disks (Backup vault)	Operational incremental snapshots, agentless	For disks without a protected VM
Blobs (Backup vault)	**Operational**: continuous, uses versioning + change feed + soft delete + **point-in-time restore** in the account itself · **Vaulted**: copy in the vault	"restore the container to yesterday" → operational

4. Monitoring backup

Feature	Detail
Jobs	Status of backups and restores per vault (and centralized in Business Continuity Center)
Azure Monitor alerts for backup	**Built-in** alerts (backup / restore failure, security events such as deletions) already active in the vault; routed with **action groups** (email, SMS, runbook); shown in Monitor → Alerts
Backup reports	Vault diagnostic setting → **Log Analytics** (`AddonAzureBackup*` categories); workbooks with compliance, consumption, jobs; email reports

Feature	Detail
Metrics	Backup Health Events, Restore Health Events (metric alerts)

5. Azure Site Recovery (disaster recovery)

Piece	Detail
Scenarios	**Azure** → **Azure** (another region), **zone** → **zone**, **VMware / physical** → **Azure** (replication appliance), **Hyper-V** → **Azure** (provider + agent, with or without VMM)
A2A components	Vault in a **different region** from the VMs; **cache storage account** in the **source** region; **Mobility service** extension (installs itself); target resources created with the **asr** suffix (RG, VNet, availability set); `-ASRReplica` replica disks
Replication	**Continuous**: every write goes to the cache and from there to the target; **crash-consistent points every 5 min** (fixed); **app-consistent** per the policy (default **disabled**, 0 h); **default recovery point retention 1 day** (extendable, e.g. up to 15); **multi-VM consistency** (replication group) for shared points
Network mapping	Which target VNet / subnet corresponds to each source VNet; target IPs and NSG configurable per VM
Test failover	Brings up the VMs in an **isolated** test network **without touching** production or the replication; for drills ("perform a DR drill without impact")
Failover	Choose the point: *latest* (processes pending data), *latest processed* (minimum RTO), *latest app-consistent*, or custom; creates the VMs at the target; then **Commit** (deletes the other points)
Re-protect	Reverses the replication (target → source) to be able to go back
Failback	Failover back to the source once re-protected
Recovery plan	Groups VMs, defines **boot order** by groups, **scripts** (Automation runbooks) and **manual actions**; one click for the whole tier
Requirements	Managed disks; outbound 443 to the Storage, AzureSiteRecovery, EventHub, AzureActiveDirectory service tags; capacity at the target; target VM size configurable
Cost	Per protected instance (first 31 days free) + cache and replica storage

- **Backup vs Site Recovery:** backup = point-in-time copies to recover data or lost VMs (RPO of hours, high RTO); ASR = continuous replica to keep operating in another region (RPO of minutes, RTO of minutes). "RPO 1 hour, RTO 2 hours" → ASR. "Keep 7 years of recovery points" → backup. Both can coexist, but when **moving** a VM both must be disabled.

6. Commands that show up

Task	PowerShell (Az.RecoveryServices)	Azure CLI
Vault	`New-AzRecoveryServicesVault -Name rsv1 -ResourceGroupName RG1 -Location westeurope` · `Set-AzRecoveryServicesVaultContext -Vault $v` (or `-VaultId` on each cmdlet) · `Set-AzRecoveryServicesBackupProperty -Vault $v -BackupStorageRedundancy GeoRedundant` · CRR: `Set-AzRecoveryServicesBackupProperty -EnableCrossRegionRestore`	`az backup vault create -n rsv1 -g RG1 -l westeurope` · `az backup vault backup-properties set -n rsv1 -g RG1 --backup-storage-redundancy GeoRedundant --cross-region-restore-flag true` · `az backup vault backup-properties set --soft-delete-feature-state Enable`
Policy	`Get-AzRecoveryServicesBackupProtectionPolicy -Name DefaultPolicy` · `$sched = Get-AzRecoveryServicesBackupSchedulePolicyObject -WorkloadType AzureVM -PolicySubType Enhanced; $ret = Get-AzRecoveryServicesBackupRetentionPolicyObject -WorkloadType AzureVM; New-AzRecoveryServicesBackupProtectionPolicy -Name p1 -WorkloadType AzureVM -SchedulePolicy $sched -RetentionPolicy $ret`	`az backup policy list -v rsv1 -g RG1` · `az backup policy create -n p1 -v rsv1 -g RG1 --backup-management-type AzureIaasVM --policy @policy.json` · `az backup policy set`
Protect VM	`Enable-AzRecoveryServicesBackupProtection -Policy $pol -Name vm1 -ResourceGroupName RG1` · `Backup-AzRecoveryServicesBackupItem -Item $item` (now)	`az backup protection enable-for-vm -v rsv1 -g RG1 --vm vm1 --policy-name DefaultPolicy` · `az backup protection backup-now -v rsv1 -g RG1 -c vm1 -i vm1 --backup-management-type AzureIaasVM --retain-until 31-12-2026`
Restore	`$rp = Get-AzRecoveryServicesBackupRecoveryPoint -Item $item; Restore-AzRecoveryServicesBackupItem -RecoveryPoint $rp[0] -StorageAccountName st1 -StorageAccountResourceGroupName RG1 -TargetResourceGroupName RG-restore` (restore disks) · `-RestoreAsUnmanagedDisks` · cross region: `-RestoreToSecondaryRegion`	`az backup recoverypoint list -v rsv1 -g RG1 -c vm1 -i vm1 --backup-management-type AzureIaasVM` · `az backup restore restore-disks -v rsv1 -g RG1 -c vm1 -i vm1 --rp-name <rp> --storage-account st1 --target-resource-group RG-restore` · `--restore-mode OriginalLocation` (replace existing) · `az backup restore files mount-rp` (file recovery)
Stop / jobs	`Disable-AzRecoveryServicesBackupProtection -Item $item [-RemoveRecoveryPoints -Force]` · `Get-AzRecoveryServicesBackupJob -Status InProgress` · `Undo-AzRecoveryServicesBackupItemDeletion` (soft delete)	`az backup protection disable -v rsv1 -g RG1 -c vm1 -i vm1 [--delete-backup-data true]` · `az backup job list -v rsv1 -g RG1 -o table` · `az backup protection undelete`
Files and MARS	`Enable-AzRecoveryServicesBackupProtection -WorkloadType AzureFiles -Name share1 -StorageAccountName st1 -Policy $pol` · MARS: download the agent + `Get-AzRecoveryServicesVaultSettingsFile -Backup` (vault credentials)	`az backup protection enable-for-azurefileshare -v rsv1 -g RG1 --storage-account st1 --azure-file-share share1 --policy-name p1`
Site Recovery	`New-AzRecoveryServicesAsrFabric`, `New-AzRecoveryServicesAsrProtectionContainerMapping`, `New-AzRecoveryServicesAsrPolicy -AzureToAzure -RecoveryPointRetentionInHours 24 -ApplicationConsistentSnapshotFrequencyInHours 4`, `New-AzRecoveryServicesAsrReplicationProtectedItem`, `Start-AzRecoveryServicesAsrTestFailoverJob`, `Start-AzRecoveryServicesAsrUnplannedFailoverJob -Direction PrimaryToRecovery`, `Start-AzRecoveryServicesAsrCommitFailoverJob`, `Update-AzRecoveryServicesAsrProtectionDirection` (re-protect)	No native ASR commands in Azure CLI (portal / PowerShell / REST)

7. Decision tree

```
What does the wording ask for?
├─ Protect Azure VMs against deletion / corruption ..... Recovery Services vault + policy (Standard / Enhanced)
├─ Several backups a day, Premium SSD v2 or Trusted Launch ..... Enhanced policy
├─ Restore a file from a VM ..... File recovery (iSCSI mount)
├─ Restore the VM with another size / network ..... Restore disks + create VM
├─ Restore in the paired region ..... GRS + Cross Region Restore
├─ Protect against backup deletion / ransomware ..... Soft delete (14–180 days) + MUA + immutability
├─ Files and folders of an on-prem Windows server ..... MARS agent (3 per day)
├─ On-prem Hyper-V / VMware VMs to Azure (backup) ..... MABS / DPM
├─ Azure Files share with per-file restore ..... Azure Files backup (snapshots)
├─ SQL databases in a VM with 15 min RPO ..... SQL in VM backup (workload)
├─ Disks / blobs / PostgreSQL ..... Backup vault
├─ Keep operating in another region if the primary fails (RTO minutes) ... Site Recovery A2A + recovery plan
├─ Test DR without affecting production ..... Test failover
├─ Return to the original region after DR ..... Re-protect + failover (failback)
├─ Notify when a backup fails ..... Built-in Backup alerts + action group
└─ Backup compliance report ..... Backup reports (Log Analytics)
```

8. Signals → answer

Wording	Answer
"vault must be in a different region than the VMs"	For **backup** it is impossible (same region); for **ASR** yes
"change vault redundancy to GRS" fails	Items are already protected: it is set before the first backup
"restore the VM in the paired region"	Cross Region Restore (GRS + CRR enabled beforehand)
"backup twice a day"	Enhanced policy
"Trusted Launch VM, backup from the portal"	Enhanced policy (Standard only via recent CLI / PowerShell / REST)
"recover a deleted backup"	Soft delete: Undelete within 14 days (up to 180 if extended)
"prevent admins from deleting backups"	MUA with Resource Guard + immutability
"backup fails: CanNotDelete lock"	Remove the lock from the `AzureBackupRG_*` RG
"VM with Azure Disk Encryption backup fails"	Grant Backup permissions on Key Vault (Backup Management Service)
"restore only one disk"	Selective disk restore
"restore files without restoring the VM"	File recovery script (iSCSI, 12 h)
"MARS on Linux"	Does not exist: MARS is Windows only (use VM backup or MABS)
"MARS backs up 5 times a day"	Maximum 3
"restore a single file from the Azure file share"	Item-level restore from the snapshot
"on-premises VMware VMs replicated to Azure"	Site Recovery (appliance)
"RPO of minutes, RTO of minutes"	Site Recovery
"test the DR plan without downtime"	Test failover (isolated network)
"VMs must start in order (DB before web)"	Recovery plan with groups
"after failover, replicate back"	Re-protect
"Mobility service not installed"	Installs itself when A2A replication is enabled (check outbound 443)
"app-consistent recovery points every 4 hours"	Replication policy (app-consistent frequency; disabled by default)

9. Mistakes to avoid

1. Creating the backup vault in a different region from the VMs (the opposite of Site Recovery).
2. Changing the vault redundancy after the first backup: it cannot be done.
3. Using a Standard policy with Premium SSD v2 / Ultra (or with Trusted Launch from the portal).
4. Confusing backup (points in time, retention) with Site Recovery (continuous replica, failover).
5. Performing a real failover to "test" DR: that is Test failover.
6. Expecting MARS on Linux or more than 3 daily copies; or using MARS for a full VM.
7. Forgetting soft delete when working out when the data disappears (14 more days).
8. Placing locks on the Backup snapshot resource group.
9. Moving a VM to another subscription / RG without stopping backup and replication first.

Cheatsheet AZ-104 — Commands: equivalent Azure CLI and PowerShell, Cloud Shell and exam tricks

1. Tools and session

Tool	What it is	Exam details
Azure portal	GUI; every resource has **Automation options** (export template) and built-in **Cloud Shell**	The answer when "minimize administrative effort" refers to a one-off action
Azure CLI (`az`)	Cross-platform (Windows, macOS, Linux, Docker); syntax `az <group> <subgroup> <verb> --options`; output `-o json` (default), `-o table`, `-o tsv`; **JMESPath** filter with `--query`	`az login` (`--use-device-code`, `--identity` on a VM) · `az account list -o table` · `az account set -s "Prod"` · `az account show` · `az configure --defaults group=RG1 location=westeurope` · `az upgrade` · `az find "vm create"` · `az interactive` · `az <cmd> --help`
Azure PowerShell (**Az** module)	`Verb-AzNoun` cmdlets; objects through the pipeline; PowerShell 7 on any OS	`Install-Module Az -Scope CurrentUser` · `Update-Module Az` · `Connect-AzAccount` (`-UseDeviceAuthentication`, `-Identity`, `-Tenant`) · `Get-AzSubscription` · `Set-AzContext -Subscription "Prod"` · `Get-AzContext` · `Get-Command -Module Az.Compute -Verb Get` · `Get-Help New-AzVM -Examples`
Cloud Shell	Bash or PowerShell in the browser (portal, `shell.azure.com`, mobile app, Windows Terminal, VS Code) with Az CLI, Az PowerShell, Bicep, Terraform, kubectl, git, editor (`code`); already authenticated	Needs a **storage account and a file share** (`clouddrive`, 5 GB) to persist `$HOME` (**ephemeral** option without storage); closes after **20 min** idle; 20 concurrent users per tenant; share region is fixed
Retired modules	**AzureRM** (→ Az), **AzureAD** and **MSOnline** (→ **Microsoft Graph PowerShell**: `Connect-MgGraph`, `New-MgUser`, `Get-MgGroup...`)	Older questions show `New-AzureRmVM`, `New-AzureADUser`, `New-MsolUser`: the modern answer is Az / Graph
Others	`az rest` / `Invoke-AzRestMethod` (REST API without SDK), SDKs, Azure Mobile app, Terraform, Bicep	"automate with code" → ARM / Bicep (cheatsheet 07)

2. Syntax that shows up: JMESPath, pipeline and parameters

```
az vm list -o table                                          # basic columns
az vm list -d --query "[?powerState=='VM running'].{Name:name, RG:resourceGroup}" -o table
az vm list --query "[?location=='westeurope'].name" -o tsv    # flat list
az storage account list --query "[?sku.name=='Standard_LRS'] | length(@)"
az network nic show -g RG1 -n nic1 --query "ipConfigurations[0].privateIPAddress" -o tsv
az group list --query "[].{name:name, loc:location}" --output table
az vm create ... --no-wait                                   # do not wait; az vm wait --created
```

```
Get-AzVM -Status | Where-Object { $_.PowerState -eq "VM running" } | Select-Object Name, ResourceGroupName
Get-AzVM | Where-Object Location -eq "westeurope" |
    ForEach-Object { Stop-AzVM -Name $_.Name -ResourceGroupName $_.ResourceGroupName -Force -AsJob }
Get-AzResource -Tag @{Env="Prod"} | Select-Object Name, ResourceType
(Get-AzStorageAccount -ResourceGroupName RG1 -Name st1).PrimaryEndpoints.Blob
Get-AzRoleDefinition Contributor | ConvertTo-Json -Depth 5
Get-AzVM -ResourceGroupName RG1 -Name vm1 | Format-List *        # show all properties
```

- Conventions: CLI `-g` = `--resource-group`, `-n` = `--name`, `-l` = `--location`, `--ids` accepts resource IDs, `--yes` / `-y` no confirmation, `--no-wait` asynchronous, inline JSON in **single quotes** in Bash (and escaped in PowerShell / cmd) or `@file.json` · PowerShell `-ResourceGroupName`, `-Name`, `-Location`, `-Force` no confirmation, `-AsJob` in the background (`Get-Job`, `Receive-Job`), `-WhatIf` simulates, `-Confirm`, `@params` splatting.
- `New-` creates · `Set-` replaces / configures · `Update-` changes specific properties (`Update-AzVM`, `Update-AzTag`) · `Get-` reads · `Remove-` deletes · `Add-` adds to a collection (`Add-AzVMDataDisk`) · `Start-` / `Stop-` / `Restart-`.

3. Identity, RBAC and governance

Task	Azure CLI	PowerShell
Entra users and groups	`az ad user create --display-name Ana --user-principal-name ana@contoso.com --password ...` · `az ad group create --display-name g1 --mail-nickname g1` · `az ad group member add --group g1 --member-id <id>` · `az ad user list --filter "startswith(displayName,'A')"`	`New-MgUser` · `New-MgGroup -SecurityEnabled -MailEnabled:$false -MailNickname g1 -DisplayName g1` · `New-MgGroupMember -GroupId <id> -DirectoryObjectId <id>` · `New-MgInvitation` · Az: `New-AzADUser`, `New-AzADGroup`, `Add-AzADGroupMember`
Licenses and managed identities	`az rest` (licenses) · `az identity create -g RG1 -n mi1` · `az vm identity assign -g RG1 -n vm1 [--identities mi1]`	`Set-MgUserLicense` · `New-AzUserAssignedIdentity` · `Update-AzVM -IdentityType SystemAssigned`
RBAC	`az role assignment create --assignee ana@contoso.com --role Contributor --resource-group RG1` · `az role assignment list --assignee ana@contoso.com --all` · `az role definition create --role-definition role.json`	`New-AzRoleAssignment -SignInName ana@contoso.com -RoleDefinitionName Contributor -ResourceGroupName RG1` · `Get-AzRoleAssignment -SignInName ... -ExpandPrincipalGroups` · `New-AzRoleDefinition -InputFile role.json`
Subscriptions and MGs	`az account management-group create -n mg1 --display-name "MG 1"` · `az account management-group subscription add -n mg1 --subscription <id>` · `az provider register --namespace Microsoft.Compute`	`New-AzManagementGroup -GroupName mg1` · `New-AzManagementGroupSubscription -GroupName mg1 -SubscriptionId <id>` · `Register-AzResourceProvider -ProviderNamespace Microsoft.Compute`

Task	Azure CLI	PowerShell
Resource groups	`az group create -n RG1 -l westeurope --tags Env=Prod` · `az group delete -n RG1 --yes --no-wait` · `az group list --tag Env=Prod` · `az group exists -n RG1`	`New-AzResourceGroup -Name RG1 -Location westeurope -Tag @{Env="Prod"}` · `Remove-AzResourceGroup -Name RG1 -Force -AsJob` · `Get-AzResourceGroup -Tag @{Env="Prod"}`
Tags	`az tag update --resource-id <id> --operation Merge --tags Owner=Ana` · `az resource tag --tags Env=Prod --ids <id>` · `az resource list --tag Env=Prod`	`Update-AzTag -ResourceId <id> -Tag @{Owner="Ana"} -Operation Merge` · `Set-AzResourceGroup -Name RG1 -Tag @{...}` · `Get-AzResource -Tag @{Env="Prod"}`
Locks	`az lock create -n L1 --lock-type CanNotDelete -g RG1 [--resource-name vm1 --resource-type Microsoft.Compute/virtualMachines]` · `az lock list` · `az lock delete`	`New-AzResourceLock -LockLevel CanNotDelete -LockName L1 -ResourceGroupName RG1 [-ResourceName vm1 -ResourceType ...]` · `Get-AzResourceLock` · `Remove-AzResourceLock -LockId <id>`
Policy	`az policy assignment create -n loc --policy <id or name> --scope <id> --params '{...}'` · `az policy state list --filter "complianceState eq 'NonCompliant'"` · `az policy remediation create`	`New-AzPolicyAssignment -Name loc -PolicyDefinition $def -Scope <id> -PolicyParameterObject @{...}` · `Get-AzPolicyState -Filter "ComplianceState eq 'NonCompliant'"` · `Start-AzPolicyRemediation`
Move resources	`az resource move --destination-group RG2 --ids <id1> <id2> [--destination-subscription-id <id>]`	`Move-AzResource -DestinationResourceGroupName RG2 -ResourceId <id1>,<id2>`
Budget	`az consumption budget create --amount 500 --budget-name b1 --category cost --time-grain monthly --start-date ... --end-date ...`	`New-AzConsumptionBudget -Amount 500 -Name b1 -Category Cost -TimeGrain Monthly -StartDate ...`

4. Templates and deployments

Task	Azure CLI	PowerShell
Deploy to RG	`az deployment group create -g RG1 --template-file main.bicep --parameters main.bicepparam [--mode Complete] [--what-if]`	`New-AzResourceGroupDeployment -ResourceGroupName RG1 -TemplateFile main.bicep -TemplateParameterFile main.parameters.json [-Mode Complete] [-WhatIf]`
Subscription / MG	`az deployment sub create -l westeurope --template-file main.bicep` · `az deployment mg create -m mg1 -l westeurope ...`	`New-AzSubscriptionDeployment -Location westeurope -TemplateFile main.bicep` · `New-AzManagementGroupDeployment`
Validate, history, export	`az deployment group validate` · `az deployment group list -g RG1 -o table` · `az group export -n RG1` · `az deployment group export -g RG1 -n d1`	`Test-AzResourceGroupDeployment` · `Get-AzResourceGroupDeployment` · `Export-AzResourceGroup -ResourceGroupName RG1 -Path .` · `Save-AzResourceGroupDeploymentTemplate`
Bicep and template specs	`az bicep build / decompile / upgrade` · `az ts create -n ts1 --version 1.0 -g RG1 -l westeurope --template-file main.bicep`	`bicep build` · `New-AzTemplateSpec -Name ts1 -Version 1.0 -ResourceGroupName RG1 -Location westeurope -TemplateFile main.bicep`

5. Storage

Task	Azure CLI	PowerShell
Account	`az storage account create -n st1 -g RG1 -l westeurope --sku Standard_GRS --kind StorageV2 --access-tier Hot --min-tls-version TLS1_2 --allow-blob-public-access false` · `az storage account update --sku Standard_ZRS` · `az storage account show -n st1 --query primaryEndpoints`	`New-AzStorageAccount -ResourceGroupName RG1 -Name st1 -Location westeurope -SkuName Standard_GRS -Kind StorageV2 -AccessTier Hot -MinimumTlsVersion TLS1_2` · `Set-AzStorageAccount ... -SkuName Standard_ZRS`
Keys and SAS	`az storage account keys list -n st1` · `az storage account keys renew --key primary` · `az storage account generate-sas --services b --resource-types sco --permissions rl --expiry ... --https-only` · `az storage container generate-sas -n c1 --permissions r --expiry ... --auth-mode login --as-user`	`Get-AzStorageAccountKey` · `New-AzStorageAccountKey -KeyName key1` · `New-AzStorageAccountSASToken -Service Blob -ResourceType Service,Container,Object -Permission rl -ExpiryTime ...` · `New-AzStorageContainerSASToken` · `New-AzStorageContainerStoredAccessPolicy`
Firewall	`az storage account update --default-action Deny --bypass AzureServices` · `az storage account network-rule add --vnet-name vnet1 --subnet sub1` · `--ip-address 203.0.113.0/24`	`Update-AzStorageAccountNetworkRuleSet -DefaultAction Deny -Bypass AzureServices` · `Add-AzStorageAccountNetworkRule -VirtualNetworkResourceId <id>` · `-IPAddressOrRange ...`
Blobs	`az storage container create -n c1 --account-name st1 --auth-mode login [--public-access blob]` · `az storage blob upload -f a.pdf -c c1 -n a.pdf --tier Cool` · `az storage blob set-tier --tier Archive` · `az storage blob undelete` · `az storage blob list` · `az storage blob download` · `az storage blob copy start`	`$ctx = New-AzStorageContext -StorageAccountName st1 -UseConnectedAccount` · `New-AzStorageContainer -Name c1 -Context $ctx -Permission Blob` · `Set-AzStorageBlobContent -File a.pdf -Container c1 -StandardBlobTier Cool -Context $ctx` · `Get-AzStorageBlob` · `Get-AzStorageBlobContent` · `Start-AzStorageBlobCopy`
Protection and lifecycle	`az storage account blob-service-properties update --enable-delete-retention true --delete-retention-days 14 --enable-versioning true --enable-container-delete-retention true` · `az storage account management-policy create --policy @policy.json`	`Enable-AzStorageBlobDeleteRetentionPolicy -RetentionDays 14` · `Enable-AzStorageContainerDeleteRetentionPolicy` · `Update-AzStorageBlobServiceProperty -IsVersioningEnabled $true` · `Set-AzStorageAccountManagementPolicy`
Files	`az storage share-rm create --storage-account st1 -n share1 --quota 1024 --enabled-protocols SMB` · `az storage file upload -s share1 --source f.txt` · `az storage share-rm snapshot` · `az storage account file-service-properties update --enable-delete-retention true --delete-retention-days 14`	`New-AzRmStorageShare -ResourceGroupName RG1 -StorageAccountName st1 -Name share1 -QuotaGiB 1024` · `Set-AzStorageFileContent` · `New-AzRmStorageShare -Snapshot` · `Update-AzStorageFileServiceProperty -EnableShareDeleteRetentionPolicy $true -ShareRetentionDays 14`
AzCopy	`azcopy login` · `azcopy copy "C:\d" "https://st1.blob.core.windows.net/c1?<SAS>" --recursive` · `azcopy sync ... --delete-destination=true` · `azcopy jobs resume <id>`	Same (it is a separate tool)

6. Compute

Task	Azure CLI	PowerShell
VM	`az vm create -g RG1 -n vm1 --image Ubuntu2204 --size Standard_D2s_v5 --admin-username azureuser --generate-ssh-keys --zone 1 --vnet-name vnet1 --subnet sub1 --public-ip-address "" --nsg-rule NONE` · `az vm list -d -o table` · `az vm show -g RG1 -n vm1 -d --query publicIps`	`New-AzVM -ResourceGroupName RG1 -Name vm1 -Location westeurope -Image Win2022Datacenter -Size Standard_D2s_v5 -Credential (Get-Credential) -Zone 1 -VirtualNetworkName vnet1 -SubnetName sub1` · `Get-AzVM -Status` · `Get-AzPublicIpAddress`
Lifecycle	`az vm start / stop / deallocate / restart / redeploy / delete -g RG1 -n vm1` · `az vm resize --size Standard_D4s_v5` · `az vm list-vm-resize-options`	`Start-AzVM` · `Stop-AzVM -Force` (deallocates) · `Stop-AzVM -StayProvisioned` · `Restart-AzVM` · `Set-AzVM -Redeploy` · `$vm.HardwareProfile.VmSize = ...; Update-AzVM`
Disks	`az vm disk attach -g RG1 --vm-name vm1 --name d1 --new --size-gb 256 --sku Premium_LRS` · `az vm disk detach` · `az disk update -n d1 --size-gb 512` · `az snapshot create -g RG1 -n snap1 --source d1` · `az disk create --source snap1`	`Add-AzVMDataDisk -VM $vm -Name d1 -Lun 0 -CreateOption Empty -DiskSizeInGB 256 -StorageAccountType Premium_LRS; Update-AzVM` · `Remove-AzVMDataDisk` · `Update-AzDisk -ResourceGroupName RG1 -DiskName d1 -DiskUpdate (New-AzDiskUpdateConfig -DiskSizeGB 512)` · `New-AzSnapshot` · `New-AzDisk`
Extensions and commands	`az vm extension set -g RG1 --vm-name vm1 --name CustomScriptExtension --publisher Microsoft.Compute --settings '{...}'` · `az vm run-command invoke --command-id RunPowerShellScript --scripts "..."` · `az vm user update -u azureuser -p <pwd>` · `az vm boot-diagnostics enable`	`Set-AzVMCustomScriptExtension -FileUri ... -Run install.ps1 -Name cse ...` · `Invoke-AzVMRunCommand -CommandId RunPowerShellScript -ScriptPath fix.ps1` · `Set-AzVMAccessExtension` · `Set-AzVMBootDiagnostic -Enable`
Availability set, images	`az vm availability-set create -n as1 -g RG1 --platform-fault-domain-count 2 --platform-update-domain-count 5` · `az vm generalize` · `az sig image-version create`	`New-AzAvailabilitySet -Sku Aligned -PlatformFaultDomainCount 2 -PlatformUpdateDomainCount 5` · `Set-AzVm -Generalized` · `New-AzGalleryImageVersion`
VMSS and autoscale	`az vmss create -g RG1 -n vmss1 --image Ubuntu2204 --orchestration-mode Flexible --instance-count 2 --zones 1 2 3 --lb lb1` · `az vmss scale --new-capacity 5` · `az vmss update-instances --instance-ids "*"` · `az monitor autoscale create --resource vmss1 --resource-type Microsoft.Compute/virtualMachineScaleSets --min-count 2 --max-count 10 --count 2` · `az monitor autoscale rule create --condition "Percentage CPU > 75 avg 10m" --scale out 2`	`New-AzVmss -OrchestrationMode Flexible -InstanceCount 2 -Zone 1,2,3 ...` · `Update-AzVmss -SkuCapacity 5` · `Update-AzVmssInstance -InstanceId "*"` · `New-AzAutoscaleScaleRuleObject` + `New-AzAutoscaleProfileObject` + `New-AzAutoscaleSetting`
App Service	`az appservice plan create -n plan1 -g RG1 --sku S1 --is-linux` · `az webapp create -n app1 -g RG1 --plan plan1 --runtime "DOTNETCORE:8.0"` · `az webapp deployment slot create --slot staging` · `az webapp deployment slot swap --slot staging --target-slot production` · `az webapp config appsettings set --settings K=v --slot-settings K` · `az webapp config hostname add --hostname www.contoso.com` · `az webapp config ssl bind --certificate-thumbprint <t> --ssl-type SNI` · `az webapp deploy --src-path app.zip` · `az webapp log tail`	`New-AzAppServicePlan -Tier Standard -NumberofWorkers 2 -WorkerSize Small ...` · `New-AzWebApp` · `New-AzWebAppSlot` · `Switch-AzWebAppSlot -SourceSlotName staging -DestinationSlotName production` · `Set-AzWebApp -AppSettings @{K="v"}` · `Set-AzWebAppSlotConfigName -AppSettingNames K` · `New-AzWebAppSSLBinding` · `Publish-AzWebApp -ArchivePath app.zip`
Containers	`az acr create -n acr1 -g RG1 --sku Premium` · `az acr build -r acr1 -t app:v1 .` · `az acr login -n acr1` · `az container create -g RG1 -n c1 --image acr1.azurecr.io/app:v1 --cpu 1 --memory 1.5 --dns-name-label app1 --ports 80 --restart-policy Never` · `az container logs` · `az containerapp env create` · `az containerapp create --min-replicas 0 --max-replicas 10 --ingress external --target-port 80`	`New-AzContainerRegistry -Sku Premium` · `Connect-AzContainerRegistry` · `New-AzContainerGroup -Container (New-AzContainerInstanceObject -Name c1 -Image ... -Port (New-AzContainerInstancePortObject -Port 80 -Protocol TCP)) -OsType Linux -IPAddressType Public -IPAddressDnsNameLabel app1 -Location westeurope` · `Get-AzContainerInstanceLog` · Container Apps: `New-AzContainerApp` (Az.App module)

7. Networking

Task	Azure CLI	PowerShell
VNet and subnet	`az network vnet create -g RG1 -n vnet1 --address-prefixes 10.0.0.0/16 --subnet-name sub1 --subnet-prefixes 10.0.1.0/24` · `az network vnet subnet create / update` (`--network-security-group`, `--route-table`, `--service-endpoints`, `--delegations`, `--nat-gateway`, `--default-outbound false`) · `az network vnet update --dns-servers 10.0.0.4`	`New-AzVirtualNetwork -AddressPrefix 10.0.0.0/16 -Subnet (New-AzVirtualNetworkSubnetConfig -Name sub1 -AddressPrefix 10.0.1.0/24)` · `Add-AzVirtualNetworkSubnetConfig` / `Set-AzVirtualNetworkSubnetConfig` + `Set-AzVirtualNetwork` · `$vnet.DhcpOptions.DnsServers = @("10.0.0.4")`
IP, NIC	`az network public-ip create -n pip1 --sku Standard --allocation-method Static --zone 1 2 3 --dns-name app1` · `az network nic create --vnet-name vnet1 --subnet sub1 --public-ip-address pip1 --ip-forwarding true` · `az network nic ip-config update --private-ip-address 10.0.1.10`	`New-AzPublicIpAddress -Sku Standard -AllocationMethod Static -Zone 1,2,3` · `New-AzNetworkInterface -EnableIPForwarding` · `Set-AzNetworkInterfaceIpConfig -PrivateIpAddress 10.0.1.10; Set-AzNetworkInterface`
Peering	`az network vnet peering create -n hub-spoke --vnet-name hub --remote-vnet spoke --allow-vnet-access --allow-gateway-transit` (+ the reverse one with `--use-remote-gateways --allow-forwarded-traffic`)	`Add-AzVirtualNetworkPeering -Name hub-spoke -VirtualNetwork $hub -RemoteVirtualNetworkId $spoke.Id -AllowGatewayTransit` (+ reverse `-UseRemoteGateways -AllowForwardedTraffic`)
NSG and ASG	`az network nsg create -n nsg1` · `az network nsg rule create --nsg-name nsg1 -n AllowRDP --priority 100 --direction Inbound --access Allow --protocol Tcp --source-address-prefixes 203.0.113.0/24 --destination-port-ranges 3389` · `az network nsg rule list --include-default` · `az network asg create` · `az network nic ip-config update --application-security-groups asg-web` · `az network nic list-effective-nsg`	`New-AzNetworkSecurityGroup -SecurityRules (New-AzNetworkSecurityRuleConfig -Name AllowRDP -Priority 100 -Direction Inbound -Access Allow -Protocol Tcp -SourceAddressPrefix 203.0.113.0/24 -SourcePortRange * -DestinationAddressPrefix * -DestinationPortRange 3389)` · `Add-AzNetworkSecurityRuleConfig; Set-AzNetworkSecurityGroup` · `New-AzApplicationSecurityGroup` · `Get-AzEffectiveNetworkSecurityGroup`
Routes	`az network route-table create -n rt1 --disable-bgp-route-propagation true` · `az network route-table route create --route-table-name rt1 -n toNVA --address-prefix 0.0.0.0/0 --next-hop-type VirtualAppliance --next-hop-ip-address 10.0.100.4` · `az network nic show-effective-route-table`	`New-AzRouteTable -DisableBgpRoutePropagation` · `Add-AzRouteConfig -Name toNVA -AddressPrefix 0.0.0.0/0 -NextHopType VirtualAppliance -NextHopIpAddress 10.0.100.4; Set-AzRouteTable` · `Get-AzEffectiveRouteTable`

Task	Azure CLI	PowerShell
Bastion, NAT, endpoints	`az network bastion create -n bas1 --vnet-name vnet1 --public-ip-address pip-bas --sku Standard` · `az network nat gateway create -n nat1 --public-ip-addresses pip-nat` · `az network private-endpoint create --private-connection-resource-id <id> --group-id blob --connection-name c1 --vnet-name vnet1 --subnet sub1` · `az network private-dns zone create -n privatelink.blob.core.windows.net` · `az network private-dns link vnet create -z ... -n link1 -v vnet1 -e false`	`New-AzBastion -Sku Standard ...` · `New-AzNatGateway` · `New-AzPrivateEndpoint -PrivateLinkServiceConnection (New-AzPrivateLinkServiceConnection -PrivateLinkServiceId <id> -GroupId blob ...)` · `New-AzPrivateDnsZone` · `New-AzPrivateDnsVirtualNetworkLink`
DNS	`az network dns zone create -n contoso.com` · `az network dns record-set a add-record -z contoso.com -n www -a 203.0.113.10` · `az network dns record-set cname set-record -n app -c app1.azurewebsites.net` · `az network private-dns record-set a add-record`	`New-AzDnsZone` · `New-AzDnsRecordSet -Name www -RecordType A -Ttl 3600 -DnsRecords (New-AzDnsRecordConfig -IPv4Address 203.0.113.10)` · `New-AzPrivateDnsRecordSet`
Load Balancer	`az network lb create -n lb1 --sku Standard --public-ip-address pip-lb --frontend-ip-name fe --backend-pool-name be` · `az network lb probe create --protocol Http --port 80 --path /` · `az network lb rule create --protocol Tcp --frontend-port 80 --backend-port 80 --probe-name p80 --load-distribution SourceIP --floating-ip true` · `az network lb inbound-nat-rule create --frontend-port 50001 --backend-port 3389` · `az network nic ip-config address-pool add --lb-name lb1 --address-pool be`	`New-AzLoadBalancer -Sku Standard -FrontendIpConfiguration ... -BackendAddressPool ... -Probe (New-AzLoadBalancerProbeConfig -Protocol Http -Port 80 -RequestPath /) -LoadBalancingRule (New-AzLoadBalancerRuleConfig -Protocol Tcp -FrontendPort 80 -BackendPort 80 -LoadDistribution SourceIP -EnableFloatingIP)` · `Add-AzLoadBalancerInboundNatRuleConfig`
Diagnostics	`az network watcher test-ip-flow --vm vm1 --direction Inbound --protocol TCP --local 10.0.1.4:3389 --remote 203.0.113.5:60000` · `az network watcher show-next-hop` · `az network watcher test-connectivity` · `az network watcher packet-capture create`	`Test-AzNetworkWatcherIPFlow` · `Get-AzNetworkWatcherNextHop` · `Test-AzNetworkWatcherConnectivity` · `New-AzNetworkWatcherPacketCapture`

8. Monitor and backup

Task	Azure CLI	PowerShell
Log Analytics	`az monitor log-analytics workspace create -n law1 -g RG1 --retention-time 90` · `az monitor log-analytics query -w <id> --analytics-query "..."` · `az monitor log-analytics workspace table update -n Event --retention-time 30 --total-retention-time 730`	`New-AzOperationalInsightsWorkspace -RetentionInDays 90` · `Invoke-AzOperationalInsightsQuery -WorkspaceId <id> -Query "..."` · `Update-AzOperationalInsightsTable`
Diagnostic settings and AMA	`az monitor diagnostic-settings create -n ds1 --resource <id> --workspace <id> --logs '[{"categoryGroup":"allLogs","enabled":true}]' --metrics '[{"category":"AllMetrics","enabled":true}]'` · `az vm extension set -n AzureMonitorWindowsAgent --publisher Microsoft.Azure.Monitor ...` · `az monitor data-collection rule association create`	`New-AzDiagnosticSetting -ResourceId <id> -WorkspaceId <id> -Log ... -Metric ...` · `Set-AzVMExtension -Publisher Microsoft.Azure.Monitor -ExtensionType AzureMonitorWindowsAgent ...` · `New-AzDataCollectionRuleAssociation`
Alerts	`az monitor action-group create -n ag1 --short-name ag1 --action email ops ops@contoso.com` · `az monitor metrics alert create -n cpu80 --scopes <id> --condition "avg Percentage CPU > 80" --window-size 5m --evaluation-frequency 1m --action <ag id>` · `az monitor scheduled-query create` · `az monitor activity-log alert create --condition category=Administrative and operationName=Microsoft.Compute/virtualMachines/delete`	`New-AzActionGroup -EmailReceiver (New-AzActionGroupEmailReceiverObject ...)` · `Add-AzMetricAlertRuleV2 -Condition (New-AzMetricAlertRuleV2Criteria ...)` · `New-AzScheduledQueryRule` · `New-AzActivityLogAlert`
Backup	`az backup vault create -n rsv1 -g RG1 -l westeurope` · `az backup vault backup-properties set --backup-storage-redundancy GeoRedundant --cross-region-restore-flag true` · `az backup protection enable-for-vm -v rsv1 -g RG1 --vm vm1 --policy-name DefaultPolicy` · `az backup protection backup-now` · `az backup recoverypoint list` · `az backup restore restore-disks --rp-name <rp> --storage-account st1 --target-resource-group RG2` · `az backup protection disable [--delete-backup-data true]` · `az backup job list`	`New-AzRecoveryServicesVault` · `Set-AzRecoveryServicesVaultContext` · `Set-AzRecoveryServicesBackupProperty -BackupStorageRedundancy GeoRedundant` · `Enable-AzRecoveryServicesBackupProtection -Policy $pol -Name vm1 -ResourceGroupName RG1` · `Backup-AzRecoveryServicesBackupItem` · `Get-AzRecoveryServicesBackupRecoveryPoint` · `Restore-AzRecoveryServicesBackupItem` · `Disable-AzRecoveryServicesBackupProtection [-RemoveRecoveryPoints]` · `Get-AzRecoveryServicesBackupJob`
Site Recovery	No native commands (portal / PowerShell / REST)	`New-AzRecoveryServicesAsrPolicy -AzureToAzure -RecoveryPointRetentionInHours 24` · `New-AzRecoveryServicesAsrReplicationProtectedItem` · `Start-AzRecoveryServicesAsrTestFailoverJob` · `Start-AzRecoveryServicesAsrUnplannedFailoverJob` · `Start-AzRecoveryServicesAsrCommitFailoverJob`

9. Signals → answer

Wording	Answer
"run commands from any OS without installing anything"	Cloud Shell (needs storage account + clouddrive, or ephemeral mode)
"Cloud Shell asks to create storage"	First time: `clouddrive` file share to persist `$HOME`
"Cloud Shell session ended"	20 min of inactivity
"macOS admin must manage Azure with PowerShell"	PowerShell 7 + Az module (or Cloud Shell)
"New-AzureRmVM", "AzureRM"	Retired: Az module (`New-AzVM`)
"New-AzureADUser", "New-MsolUser"	Retired: Microsoft Graph PowerShell (`New-MgUser`)
"list VMs that are running"	`az vm list -d --query "[?powerState=='VM running']"` · `Get-AzVM -Status` + `Where-Object`

Wording	Answer
"only the private IP of the NIC"	`--query "...privateIPAddress" -o tsv` · `(Get-AzNetworkInterface ...).IpConfigurations[0].PrivateIpAddress`
"run the command without waiting"	`--no-wait` · `-AsJob`
"delete without confirmation"	`--yes` · `-Force`
"simulate before applying"	`--what-if` (deployments) · `-WhatIf`
"which command creates a role assignment"	`az role assignment create` · `New-AzRoleAssignment`
"deploy a Bicep file to a resource group"	`az deployment group create` · `New-AzResourceGroupDeployment`
"upload files to a container with identity"	`az storage blob upload --auth-mode login` · `Set-AzStorageBlobContent` with `New-AzStorageContext -UseConnectedAccount`
"stop the VM and stop billing"	`az vm deallocate` · `Stop-AzVM -Force`
"stop the VM but keep compute billing"	`az vm stop` · `Stop-AzVM -StayProvisioned`
"swap staging into production"	`az webapp deployment slot swap` · `Switch-AzWebAppSlot`
"enable backup for the VM"	`az backup protection enable-for-vm` · `Enable-AzRecoveryServicesBackupProtection`
"which NSG rule blocks the traffic"	`az network watcher test-ip-flow` · `Test-AzNetworkWatcherIPFlow`
"create the alert from the command line"	`az monitor metrics alert create` · `Add-AzMetricAlertRuleV2`

10. Mistakes to avoid

1. Confusing `az vm stop` (still billed) with `az vm deallocate`, or `Stop-AzVM -StayProvisioned` with `Stop-AzVM -Force`.
2. Choosing AzureRM, AzureAD or MSOnline cmdlets in a new question: Az and Microsoft Graph.
3. Forgetting `--auth-mode login` (or `-UseConnectedAccount`) and expecting the Entra identity to work with the storage data commands.
4. Using double quotes for inline JSON in Bash (`--settings '{"a":"b"}'` goes with single quotes) or not escaping them in PowerShell.
5. Believing Cloud Shell keeps files without the `clouddrive` file share (ephemeral mode loses them).
6. Passing names where the command expects IDs (`--ids`, `-ResourceId`) or the other way around.
7. Assuming `az deployment group validate` or `Test-AzResourceGroupDeployment` check quotas and policies.
8. Forgetting `-Force` / `--yes` in unattended scripts (they sit waiting for confirmation).